Supplementary appendix

Examining the bidirectional relationship between sleep and physical activity using pooled individual data

Table of contents

Methods 1
Data screening 1
Protocol deviations 1
Results 2
Study characteristics 2
Model diagnostics 8
Fixed Effect Model 18
Log Transforming PA Volume 26
All Eligible Participants Regardless of Valid Wear Days 33
Participants With at Least Seven Valid Days 40
Moderation 47
BMI z-score 48
SES 57
sex 68
weekday 79
season 93
region 104
daylight hours 118
wear location 127
most active hour 138
ethnicity 147

Methods

Data screening

We screened all data for implausibly low or high measurements, as these likely indicate measurement device failures. Screening had four stages. First, we excluded values falling outside fixed physiological limits. For average acceleration this was 200 mg, a value above any sustainable 24-hour average. Second, because a miscalibrated accelerometer corrupts every variable derived from acceleration magnitude rather than isolated days, we excluded all physical activity variables for files whose mean average acceleration exceeded that limit (76 participants). Third, we excluded remaining values more than 4 SD from the mean of their own study, rather than of the pooled sample, so that the threshold reflected implausible measurement rather than between-study differences in device, protocol, and population. Fourth, for participants under 18 years, we excluded heights, weights, and body mass index values that the CDC classifies as biologically implausible for the child's age and sex against its 2000 growth reference (height-for-age $z < -5$ or > 4, weight-for-age $z < -5$ or > 8, and BMI-for-age $z < -4$ or > 8). Values excluded at any stage were treated as missing data.

Protocol deviations

We made the following changes to our protocol while working with the holdout data.

Using study ID as a fixed effect

We initially planned to use Study ID as a fixed effect, given that our study selection was not random. But, as we aimed for our findings to generalise to other studies and populations, we instead opted to include random intercepts for Study ID which can better model generalisable patterns. This approach allows us to account for variability between studies without fully partialling out study effects, thereby better modelling common phenomena. While this change lessened the strength of model intercepts, it did not significantly alter model slopes. The results using Study ID as a fixed effect (i.e., those that align with the protocol) are in the Fixed Effect Model section below and are qualitatively similar to the main results.

Sleep indicator variable

Our research examines various metrics of sleep: onset, duration, regularity, and efficiency. We initially planned to merge these variables into a single indicator to simplify analysis. However, no validated indicator existed which could be estimated from our measured variables. We decided against creating one as we could not validate it with our data. Consequently, our models illustrate the relationship between each aspect of sleep and physical activity, instead of describing a straightforward link between overall sleep and physical activity.

Log transformation

We aimed to improve model diagnostics using log transformations of skewed variables. Pilot analysis using holdout data revealed that while diagnostics improved on the log scale, model predictions followed extreme exponential patterns near the edge of the data range when back-transformed. Accordingly, we do not use log transformations in the main paper. These models are in the Log Transforming PA Volume section below.

Results

Study characteristics

Table 1 describes each study in the primary analytic sample, and Table 2 shows how each study's measure of socioeconomic status was harmonised.

Valid observations were not uniform across the days of the week ($\chi^2_\{\(6\)\}$ = 343.32, p < .001). Weekends and Fridays were over-represented (Saturday: z = 11.24; Sunday: z = 7.84; Friday: z = 5.57), while Wednesdays (z = −9.30) and Tuesdays (z = −8.22) were under-represented.

Table 1: Characteristics of the included studies (primary analytic sample)

Study ID	Study	Reference	Country	Device	Wear location	N	Age, mean (range)	Female, %	Valid days, median
100			12 countries	GT3X	Hip	6,360	10.4 (9-11)	54.5	7
101			Switzerland	GENEActiv	Wrist	2,928	61.9 (45-86)	54.0	13
102			Denmark	GT3X	Hip	691	10.0 (8-11)	48.2	7
103			New Zealand	Actigraph	Wrist	651	19.3 (8-58)	60.3	6
104			Brazil	GT3X+	Wrist	737	16.4 (14-22)	50.9	6
105			Czech Republic	GT3X+, GT9X Link, wGT3X-BT, wGT3X+	Wrist	1,170	14.0 (8-19)	54.5	6
106			Australia	GENEActiv	Wrist	301	42.0 (18-65)	72.4	8
107			Spain	GT3X+	Wrist	2,423	72.6 (64-94)	53.1	6
108			Australia	GENEActiv	Wrist	1,038	8.8 (7-11)	51.6	8
110			Finland	Actisleep	Wrist	801	62.6 (58-64)	84.8	7
111			United Kingdom	Axivity AX3	Wrist	692	44.7 (20-70)	72.8	6
112			Chile	GENEActiv	Wrist	144	9.5 (8-12)	38.9	6
113			Spain	Actisleep	Wrist	353	8.4 (8-9)	47.3	6
114			United States	WGT3X-BT	Wrist	154	3.5 (2-4)	56.5	5
115			Ecuador	wGT3X-BT	Hip	1,019	8.8 (7-13)	48.0	6
117			Spain	GENEActiv	Wrist	175	8.8 (4-11)	48.6	6
118			Australia	Actigraph	Wrist	304	16.6 (15-18)	51.0	8

Study ID	Study	Reference	Country	Device	Wear location	N	Age, mean (range)	Female, %	Valid days, median
212			Spain	WGT3X	Hip	285	4.7 (2-6)	46.5	7
221			United Kingdom	GENEActiv	Wrist	4,071	69.4 (59-82)	26.0	8
222			Brazil	GT3X	Wrist	2,939	3.8 (3-4)	49.4	6

Table 2: Harmonisation of socioeconomic status

Study ID	Original category	Harmonised category
100	1	Low
100	2	Medium
100	4	High
100	3	High
101	High school	Medium
101	Apprenticeship	Low
101	Mandatory education	Low
101	University education	High
102	Vocational basic training/ vocational internship and main course	Medium
102	Short higher education	High
102	Long higher education/ Research degree	High
102	Regular secondary education and vocational secondary education	Medium
102	Medium higher education	High
102	Primary school	Low
102	Unanswered	Missing
104	Incomplete elementary school	Low
104	Complete Bachelor degree	High
104	Incomplete bachelor degree	Medium
104	Complete high school	Medium
104	Complete elementary school	Low
104	Unknown	Missing
104	Has not studied	Low
105	vocational education	Medium
105	high school education	Medium
105	master's degree or higher	High
105	bachelor's degree or equivalent	High
105	10th grade or less	Low
105	short tertiary education	Medium
105	26-100 books	Medium
105	11-25 books	Low

Study ID	Original category	Harmonised category
105	>200 books	High
105	101-200 books	High
105	0-10 books	Low
106	High school or less	Medium
106	Technical or further education institutions (e.g., TAFE)	Medium
106	University degree	High
107	Secondary education 2nd stage (i.e., High shool graduate, complete 12th grade or equivalent	Medium
107	Incomplete primary education (i.e., less than 6th grade or equivalent)	Low
107	Complete primary education (i.e., complete 6th grade or equivalent)	Low
107	Medium university studies (i.e., university diplomature or equivalent)	High
107	Higher university studies (i.e., Bachelor degree, Master degree, Doctoral degree or equivalent)	High
107	Secondary education 1st stage (i.e., complete 10th grade or equivalent)	Medium
107	Illiterate (i.e., not able to read or write)	Low
108	Enough to fill two bookcases (101-200 Books)	High
108	Enough to fill one shelf (11-25 Books)	Low
108	Enough to fill 3 or more bookcases (more than 200 Books)	High
108	Enough to fill one bookcases (26-100 books)	Medium
108	None or very few (0-10 Books)	Low
110	lower-level non-manual occupation	Medium

Study ID	Original category	Harmonised category
110	higher-level non-manual occupation	High
110	manual occupation	Low
110	service occupation	Medium
111	10	Medium
113	primary school	Low
113	university degree	High
113	bachelor's degree	High
113	high school	Medium
113	master's degree or higher	High
113	vocational training	Medium
114	poor	Low
114	O level	Low
114	A/S level	Low
114	ba/bsc	Low
114	higher degree	Low
114	No academic quals	Low
115	Medium	Medium
115	High	High
115	Low	Low
117	secondary education	Medium
117	higher education	High
117	primary education	Low
118	2	Low
118	7	Medium
118	5	Medium
118	3	Low
118	10	High
118	8	High
118	9	High
118	6	Medium
118	4	Medium
118	1	Low
221	O level	Medium

Study ID	Original category	Harmonised category
221	A/S level	Medium
221	ba/bsc	High
221	higher degree	High
221	No academic quals	Low
222	q5-wealthiest	High
222	q4	Medium
222	q3	Medium
222	q2	Medium
222	q1-poorest	Low

Model diagnostics

The model diagnostics for the main analysis, and for all models presented in this document are in Table 3

Table 3: Model diagnostics

Model name	Skewness	Kurtosis	Converged (%)	Singular (%)
Models moderated by age				
Scale sleep duration by PA volume	−0.05	2.17	100.00%	0.00%
Scale sleep efficiency by PA volume	−0.91	3.66	100.00%	0.00%
Scale sleep onset by PA volume	0.16	9.38	100.00%	0.00%
Scale sleep regularity by PA volume	−0.71	2.11	100.00%	0.00%
Scale sleep duration by PA intensity	−0.05	2.14	100.00%	0.00%
Scale sleep efficiency by PA intensity	−0.91	3.65	100.00%	0.00%
Scale sleep onset by PA intensity	0.18	9.40	100.00%	0.00%
Scale sleep regularity by PA intensity	−0.72	2.09	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.68	4.99	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.24	1.65	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.67	5.00	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.24	1.64	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.67	5.00	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.24	1.64	100.00%	0.00%

Model name	Skewness	Kurtosis	Converged (%)	Singular (%)
Scale PA volume by sleep regularity(lagged)	0.76	5.14	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.22	1.56	100.00%	0.00%
Models moderated by age fixedef				
Scale sleep duration by PA volume	−0.05	2.17	100.00%	0.00%
Scale sleep efficiency by PA volume	−0.91	3.66	100.00%	0.00%
Scale sleep onset by PA volume	0.16	9.38	100.00%	0.00%
Scale sleep regularity by PA volume	−0.71	2.11	100.00%	0.00%
Scale sleep duration by PA intensity	−0.05	2.14	100.00%	0.00%
Scale sleep efficiency by PA intensity	−0.91	3.65	100.00%	0.00%
Scale sleep onset by PA intensity	0.18	9.40	100.00%	0.00%
Scale sleep regularity by PA intensity	−0.72	2.09	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.68	4.99	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.24	1.65	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.67	5.00	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.24	1.64	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.67	4.99	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.24	1.64	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.76	5.14	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.22	1.56	100.00%	0.00%
Models moderated by age log				
Scale sleep duration by log PA volume	−0.05	2.17	100.00%	0.00%
Scale sleep efficiency by log PA volume	−0.91	3.67	100.00%	0.00%
Scale sleep onset by log PA volume	0.16	9.40	100.00%	0.00%
Scale sleep regularity by log PA volume	−0.72	2.13	100.00%	0.00%

Model name	Skewness	Kurtosis	Converged (%)	Singular (%)
Scale sleep duration by PA intensity	−0.05	2.14	100.00%	0.00%
Scale sleep efficiency by PA intensity	−0.91	3.65	100.00%	0.00%
Scale sleep onset by PA intensity	0.18	9.40	100.00%	0.00%
Scale sleep regularity by PA intensity	−0.72	2.09	100.00%	0.00%
Log PA volume by sleep duration(lagged)	−0.46	2.16	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.24	1.65	100.00%	0.00%
Log PA volume by sleep efficiency(lagged)	−0.46	2.18	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.24	1.64	100.00%	0.00%
Log PA volume by sleep onset(lagged)	−0.46	2.18	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.24	1.64	100.00%	0.00%
Log PA volume by sleep regularity(lagged)	−0.33	1.71	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.22	1.56	100.00%	0.00%
Models moderated by age allwear				
Scale sleep duration by PA volume	−0.05	2.17	100.00%	0.00%
Scale sleep efficiency by PA volume	−0.92	3.73	100.00%	0.00%
Scale sleep onset by PA volume	0.15	9.37	100.00%	0.00%
Scale sleep regularity by PA volume	−0.71	2.10	100.00%	0.00%
Scale sleep duration by PA intensity	−0.05	2.14	100.00%	0.00%
Scale sleep efficiency by PA intensity	−0.92	3.73	100.00%	0.00%
Scale sleep onset by PA intensity	0.18	9.39	100.00%	0.00%
Scale sleep regularity by PA intensity	−0.72	2.09	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.68	4.99	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.24	1.66	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.67	5.00	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.24	1.64	100.00%	0.00%

Model name	Skewness	Kurtosis	Converged (%)	Singular (%)
Scale PA volume by sleep onset(lagged)	0.67	4.99	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.24	1.64	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.76	5.14	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.22	1.56	100.00%	0.00%
Models moderated by age 7days				
Scale sleep duration by PA volume	0.02	2.79	100.00%	0.00%
Scale sleep efficiency by PA volume	−0.83	3.00	100.00%	0.00%
Scale sleep onset by PA volume	0.08	12.15	100.00%	0.00%
Scale sleep regularity by PA volume	−0.80	2.66	100.00%	0.00%
Scale sleep duration by PA intensity	0.02	2.79	100.00%	0.00%
Scale sleep efficiency by PA intensity	−0.83	2.99	100.00%	0.00%
Scale sleep onset by PA intensity	0.10	12.21	100.00%	0.00%
Scale sleep regularity by PA intensity	−0.80	2.61	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.53	5.08	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.22	1.54	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.53	5.16	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.22	1.52	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.53	5.11	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.22	1.52	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.60	5.81	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.18	1.33	100.00%	0.00%
Models moderated by bmi				
Scale sleep duration by PA volume	−0.12	2.48	100.00%	0.00%
Scale sleep efficiency by PA volume	−0.92	3.84	100.00%	0.00%
Scale sleep onset by PA volume	0.13	11.82	100.00%	0.00%

Model name	Skewness	Kurtosis	Converged (%)	Singular (%)
Scale sleep regularity by PA volume	−0.71	2.12	100.00%	0.00%
Scale sleep duration by PA intensity	−0.12	2.47	100.00%	0.00%
Scale sleep efficiency by PA intensity	−0.92	3.84	100.00%	0.00%
Scale sleep onset by PA intensity	0.15	11.85	100.00%	0.00%
Scale sleep regularity by PA intensity	−0.71	2.09	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.69	5.00	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.17	1.52	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.68	5.02	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.16	1.52	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.69	5.01	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.16	1.51	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.76	5.14	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.21	1.54	100.00%	0.00%
Models moderated by ses				
Scale sleep duration by PA volume	−0.03	2.15	100.00%	0.00%
Scale sleep efficiency by PA volume	−0.90	3.55	100.00%	0.00%
Scale sleep onset by PA volume	0.16	9.60	100.00%	0.00%
Scale sleep regularity by PA volume	−0.72	2.20	100.00%	0.00%
Scale sleep duration by PA intensity	−0.03	2.12	100.00%	0.00%
Scale sleep efficiency by PA intensity	−0.90	3.55	100.00%	0.00%
Scale sleep onset by PA intensity	0.18	9.61	100.00%	0.00%
Scale sleep regularity by PA intensity	−0.72	2.16	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.67	4.99	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.26	1.67	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.67	5.02	100.00%	0.00%

Model name	Skewness	Kurtosis	Converged (%)	Singular (%)
Scale PA intensity by sleep efficiency(lagged)	0.26	1.66	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.67	5.02	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.26	1.66	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.76	5.22	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.23	1.57	100.00%	0.00%
Models moderated by weekday				
Scale sleep duration by PA volume	−0.08	2.18	100.00%	0.00%
Scale sleep efficiency by PA volume	−0.91	3.69	100.00%	0.00%
Scale sleep onset by PA volume	0.09	9.83	100.00%	0.00%
Scale sleep regularity by PA volume	−0.79	2.64	100.00%	0.00%
Scale sleep duration by PA intensity	−0.08	2.16	100.00%	0.00%
Scale sleep efficiency by PA intensity	−0.90	3.69	100.00%	0.00%
Scale sleep onset by PA intensity	0.13	9.73	100.00%	0.00%
Scale sleep regularity by PA intensity	−0.80	2.69	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.73	5.07	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.29	1.71	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.73	5.10	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.29	1.69	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.73	5.09	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.31	1.72	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.79	5.16	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.26	1.63	100.00%	0.00%
Models moderated by season				
Scale sleep duration by PA volume	−0.05	2.17	100.00%	0.00%

Model name	Skewness	Kurtosis	Converged (%)	Singular (%)
Scale sleep efficiency by PA volume	−0.91	3.69	100.00%	0.00%
Scale sleep onset by PA volume	0.17	9.34	100.00%	0.00%
Scale sleep regularity by PA volume	−0.71	2.09	100.00%	0.00%
Scale sleep duration by PA intensity	−0.06	2.14	100.00%	0.00%
Scale sleep efficiency by PA intensity	−0.91	3.69	100.00%	0.00%
Scale sleep onset by PA intensity	0.19	9.37	100.00%	0.00%
Scale sleep regularity by PA intensity	−0.71	2.06	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.67	4.97	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.24	1.65	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.67	5.00	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.24	1.64	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.67	4.97	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.23	1.64	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.76	5.11	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.21	1.56	100.00%	0.00%
Models moderated by region				
Scale sleep duration by PA volume	−0.05	2.18	100.00%	0.00%
Scale sleep efficiency by PA volume	−0.91	3.71	100.00%	0.00%
Scale sleep onset by PA volume	0.16	9.39	100.00%	0.00%
Scale sleep regularity by PA volume	−0.72	2.09	100.00%	0.00%
Scale sleep duration by PA intensity	−0.06	2.14	100.00%	0.00%
Scale sleep efficiency by PA intensity	−0.91	3.70	100.00%	0.00%
Scale sleep onset by PA intensity	0.19	9.39	100.00%	0.00%
Scale sleep regularity by PA intensity	−0.72	2.07	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.68	4.97	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.24	1.66	100.00%	0.00%

Model name	Skewness	Kurtosis	Converged (%)	Singular (%)
Scale PA volume by sleep efficiency(lagged)	0.67	5.01	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.24	1.65	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.68	4.95	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.24	1.64	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.77	5.12	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.21	1.56	100.00%	0.00%
Models moderated by daylight				
Scale sleep duration by PA volume	−0.05	2.17	100.00%	0.00%
Scale sleep efficiency by PA volume	−0.91	3.69	100.00%	0.00%
Scale sleep onset by PA volume	0.17	9.34	100.00%	0.00%
Scale sleep regularity by PA volume	−0.71	2.10	100.00%	0.00%
Scale sleep duration by PA intensity	−0.06	2.14	100.00%	0.00%
Scale sleep efficiency by PA intensity	−0.91	3.69	100.00%	0.00%
Scale sleep onset by PA intensity	0.19	9.37	100.00%	0.00%
Scale sleep regularity by PA intensity	−0.71	2.07	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.67	4.97	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.24	1.65	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.67	5.00	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.24	1.64	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.67	4.99	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.24	1.64	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.76	5.11	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.21	1.56	100.00%	0.00%

Model name	Skewness	Kurtosis	Converged (%)	Singular (%)
Models moderated by wear location				
Scale sleep duration by PA volume	−0.05	2.17	100.00%	0.00%
Scale sleep efficiency by PA volume	−0.91	3.70	100.00%	0.00%
Scale sleep onset by PA volume	0.16	9.37	100.00%	0.00%
Scale sleep regularity by PA volume	−0.72	2.11	100.00%	0.00%
Scale sleep duration by PA intensity	−0.06	2.14	100.00%	0.00%
Scale sleep efficiency by PA intensity	−0.91	3.68	100.00%	0.00%
Scale sleep onset by PA intensity	0.19	9.39	100.00%	0.00%
Scale sleep regularity by PA intensity	−0.72	2.09	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.68	4.99	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.24	1.65	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.67	4.99	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.24	1.64	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.67	4.99	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.24	1.64	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.76	5.11	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.23	1.57	100.00%	0.00%
Models moderated by PA mostactivehr				
Scale sleep duration by PA volume	−0.05	2.17	100.00%	0.00%
Scale sleep efficiency by PA volume	−0.91	3.69	100.00%	0.00%
Scale sleep onset by PA volume	0.18	9.34	100.00%	0.00%
Scale sleep regularity by PA volume	−0.71	2.10	100.00%	0.00%
Scale sleep duration by PA intensity	−0.06	2.14	100.00%	0.00%
Scale sleep efficiency by PA intensity	−0.91	3.69	100.00%	0.00%
Scale sleep onset by PA intensity	0.20	9.38	100.00%	0.00%
Scale sleep regularity by PA intensity	−0.71	2.06	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.67	4.98	100.00%	0.00%

Model name	Skewness	Kurtosis	Converged (%)	Singular (%)
Scale PA intensity by sleep duration(lagged)	0.24	1.65	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.66	5.01	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.24	1.64	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.67	5.00	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.23	1.64	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.75	5.12	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.21	1.56	100.00%	0.00%
Models moderated by sex				
Scale sleep duration by PA volume	−0.05	2.17	100.00%	0.00%
Scale sleep efficiency by PA volume	−0.91	3.68	100.00%	0.00%
Scale sleep onset by PA volume	0.17	9.36	100.00%	0.00%
Scale sleep regularity by PA volume	−0.71	2.10	100.00%	0.00%
Scale sleep duration by PA intensity	−0.06	2.14	100.00%	0.00%
Scale sleep efficiency by PA intensity	−0.91	3.69	100.00%	0.00%
Scale sleep onset by PA intensity	0.19	9.39	100.00%	0.00%
Scale sleep regularity by PA intensity	−0.71	2.07	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.67	4.97	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.24	1.66	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.67	5.00	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.24	1.64	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.67	4.99	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.24	1.64	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.76	5.11	100.00%	0.00%

Model name	Skewness	Kurtosis	Converged (%)	Singular (%)
Scale PA intensity by sleep regularity(lagged)	0.21	1.56	100.00%	0.00%
Models moderated by ethnicity				
Scale sleep duration by PA volume	−0.05	2.18	100.00%	0.00%
Scale sleep efficiency by PA volume	−0.91	3.69	100.00%	0.00%
Scale sleep onset by PA volume	0.17	9.37	100.00%	0.00%
Scale sleep regularity by PA volume	−0.71	2.09	100.00%	0.00%
Scale sleep duration by PA intensity	−0.05	2.14	100.00%	0.00%
Scale sleep efficiency by PA intensity	−0.91	3.69	100.00%	0.00%
Scale sleep onset by PA intensity	0.19	9.37	100.00%	0.00%
Scale sleep regularity by PA intensity	−0.71	2.05	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.67	4.98	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.24	1.66	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.67	5.00	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.24	1.64	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.67	4.97	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.23	1.64	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.76	5.11	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.21	1.56	100.00%	0.00%

Fixed Effect Model

In this document we show the results using another analytic technique. Namely treating study ID as a fixed effect. This is what we wrote in the protocol originally.

The association between physical activity volume and sleep

We estimated the associations between physical activity and sleep (RQ1) using fixed-effects models. We estimated the association between physical activity volume and sleep by age, and the results are presented in Table 4 and Figure 1.

Table 4: Physical activity predicting sleep controlling for SES, sex, BMI z-score, and the fixed effects of study IDs.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.72 [0.68, 0.76]	< .001	0.61 [0.57, 0.65]	< .001
Physical activity	0.10 [0.09, 0.11]	< .001	0.05 [0.04, 0.06]	< .001
Age	−0.02 [−0.03, 0.00]	.024	−0.02 [−0.03, 0.00]	.013
Physical activity²	−0.02 [−0.03, −0.01]	< .001	0.00 [0.00, 0.01]	.591
Physical activity × Age	−0.01 [−0.02, −0.01]	< .001	−0.01 [−0.01, −0.01]	< .001
Age × Physical activity²	0.00 [−0.01, 0.00]	.001	−0.01 [−0.01, 0.00]	< .001
Sleep efficiency (z)				
(Intercept)	−0.51 [−0.55, −0.46]	< .001	−0.58 [−0.62, −0.54]	< .001
Physical activity	0.03 [0.03, 0.04]	< .001	0.02 [0.02, 0.03]	< .001
Age	−0.01 [−0.03, 0.00]	.146	−0.02 [−0.03, 0.00]	.024
Physical activity²	−0.02 [−0.02, −0.01]	< .001	−0.02 [−0.02, −0.01]	< .001
Physical activity × Age	−0.02 [−0.03, −0.02]	< .001	−0.01 [−0.02, −0.01]	< .001
Age × Physical activity²	0.00 [0.00, 0.00]	.094	0.01 [0.00, 0.01]	< .001
Sleep onset (z)				
(Intercept)	−1.14 [−1.18, −1.11]	< .001	−1.04 [−1.07, −1.01]	< .001
Physical activity	−0.12 [−0.12, −0.11]	< .001	−0.06 [−0.06, −0.05]	< .001
Age	−0.01 [−0.03, 0.00]	.025	0.00 [−0.02, 0.01]	.436
Physical activity²	0.02 [0.02, 0.03]	< .001	−0.01 [−0.02, −0.01]	< .001
Physical activity × Age	−0.01 [−0.01, −0.01]	< .001	0.01 [0.00, 0.01]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Age × Physical activity²	0.01 [0.01, 0.01]	< .001	0.01 [0.01, 0.01]	< .001
Sleep regularity (z)				
(Intercept)	−0.06 [−0.11, 0.00]	.045	−0.49 [−0.54, −0.43]	< .001
Physical activity	0.29 [0.27, 0.30]	< .001	0.14 [0.13, 0.15]	< .001
Age	0.00 [−0.02, 0.02]	.939	−0.09 [−0.11, −0.07]	< .001
Physical activity²	−0.15 [−0.16, −0.14]	< .001	−0.09 [−0.09, −0.08]	< .001
Physical activity × Age	0.00 [−0.01, 0.00]	.044	−0.05 [−0.05, −0.05]	< .001
Age × Physical activity²	−0.03 [−0.04, −0.03]	< .001	0.01 [0.01, 0.01]	< .001

Adjusted for SES, sex, BMI z-score, and the fixed effects of study IDs. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction. Age was centred at 35.6 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 35.6 and age interactions are the change per 10 years.

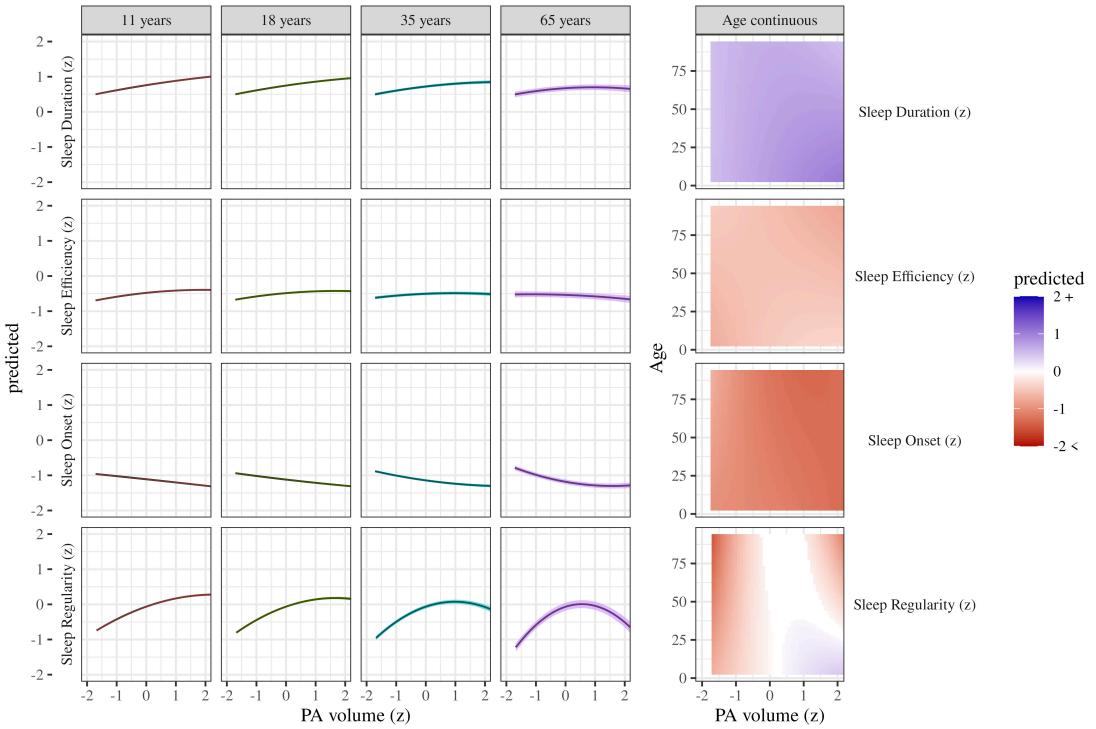

Figure 1: Sleep metrics on Physical activity volume

The association between physical activity intensity and sleep

We estimated how physical activity intensity is associated with sleep across different age groups. We present the results controlling for SES, sex, BMI z-score, and the fixed effects of study ID, in Table 4 and Figure 2.

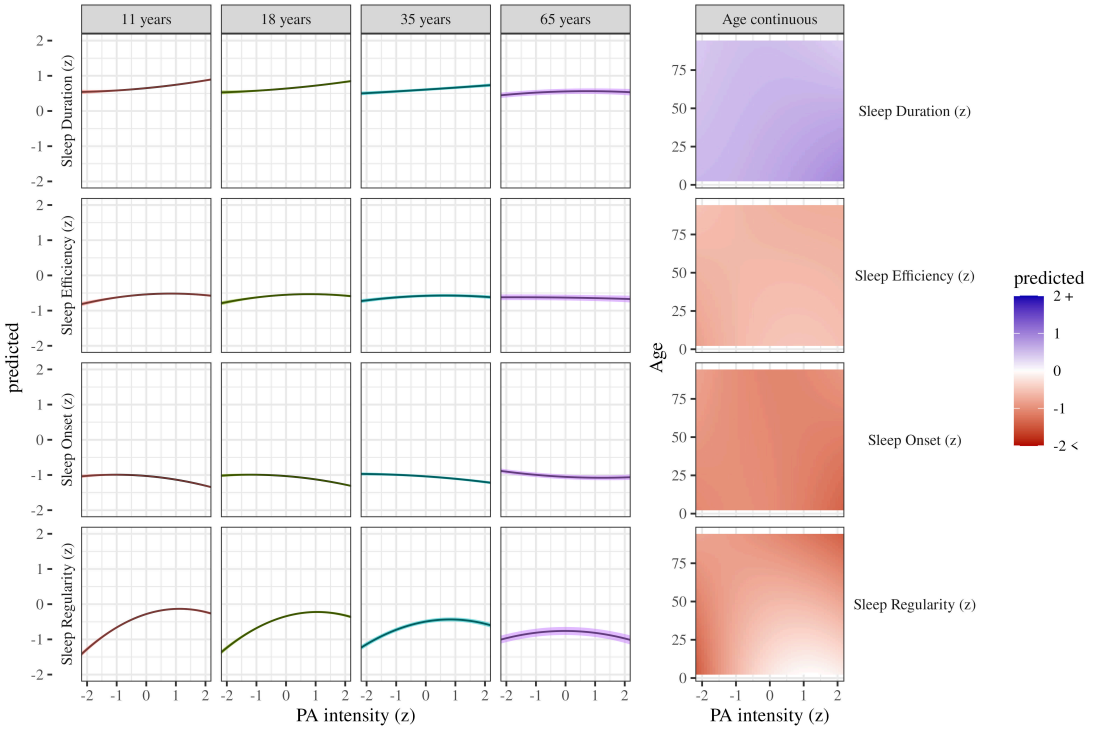

Figure 2: Sleep metrics on Physical activity intensity

The association between sleep duration and next-day physical activity

We estimated the association between sleep duration and next-day physical activity by age. Results, controlling for SES, sex, BMI z-score, and the fixed effects of study ID are presented in Table 5 and Figure 3.

Table 5: Sleep predicting physical activity controlling for SES, sex, BMI z-score, and the fixed effects of study IDs

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	−1.09 [−1.12, −1.05]	< .001	−0.41 [−0.44, −0.37]	< .001
Sleep duration	0.02 [0.01, 0.02]	< .001	0.03 [0.02, 0.03]	< .001
Age	−0.23 [−0.24, −0.22]	< .001	−0.27 [−0.28, −0.26]	< .001
Sleep duration²	−0.02 [−0.03, −0.02]	< .001	−0.01 [−0.01, −0.01]	< .001
Sleep duration × Age	−0.01 [−0.01, −0.01]	< .001	0.00 [0.00, 0.00]	.050
Age × Sleep duration²	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	.330
Sleep efficiency (z)				
(Intercept)	−1.07 [−1.10, −1.03]	< .001	−0.38 [−0.41, −0.35]	< .001
Sleep efficiency	0.00 [0.00, 0.01]	.283	0.01 [0.00, 0.01]	.010
Age	−0.23 [−0.24, −0.21]	< .001	−0.27 [−0.28, −0.26]	< .001
Sleep efficiency²	−0.01 [−0.01, 0.00]	< .001	−0.01 [−0.01, 0.00]	< .001
Sleep efficiency × Age	0.00 [0.00, 0.01]	< .001	0.00 [0.00, 0.00]	.204
Age × Sleep efficiency²	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	.227
Sleep onset (z)				
(Intercept)	−1.10 [−1.14, −1.07]	< .001	−0.44 [−0.47, −0.40]	< .001
Sleep onset	−0.02 [−0.03, −0.02]	< .001	−0.01 [−0.02, −0.01]	< .001
Age	−0.22 [−0.24, −0.21]	< .001	−0.27 [−0.28, −0.26]	< .001
Sleep onset²	0.00 [−0.01, 0.00]	.001	0.00 [0.00, 0.00]	.796
Sleep onset × Age	0.00 [0.00, 0.00]	.011	0.01 [0.01, 0.01]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Age × Sleep onset²	0.00 [0.00, 0.00]	.002	0.00 [0.00, 0.00]	.020
Sleep regularity (z)				
(Intercept)	−1.11 [−1.15, −1.06]	< .001	−0.52 [−0.56, −0.48]	< .001
Sleep regularity	0.12 [0.11, 0.12]	< .001	0.06 [0.06, 0.07]	< .001
Age	−0.24 [−0.26, −0.22]	< .001	−0.33 [−0.34, −0.31]	< .001
Sleep regularity²	0.00 [0.00, 0.00]	.445	0.00 [0.00, 0.00]	.408
Sleep regularity × Age	−0.01 [−0.01, −0.01]	< .001	−0.01 [−0.01, −0.01]	< .001
Age × Sleep regularity²	0.00 [0.00, 0.00]	.146	0.00 [0.00, 0.00]	.024

Adjusted for SES, sex, BMI z-score, and the fixed effects of study IDs. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction. Age was centred at 35.6 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 35.6 and age interactions are the change per 10 years.

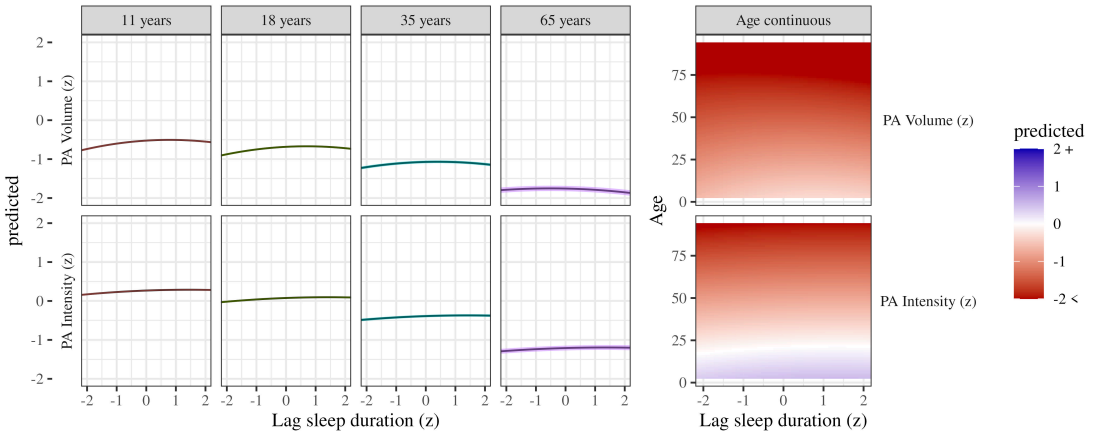

Figure 3: Physical activity by sleep duration

The association between sleep efficiency and next-day physical activity

We estimated the association between sleep efficiency and next-day physical activity by age. Results, controlling for SES, sex, BMI z-score, and the fixed effects of study ID are presented in Table 5 and Figure 4.

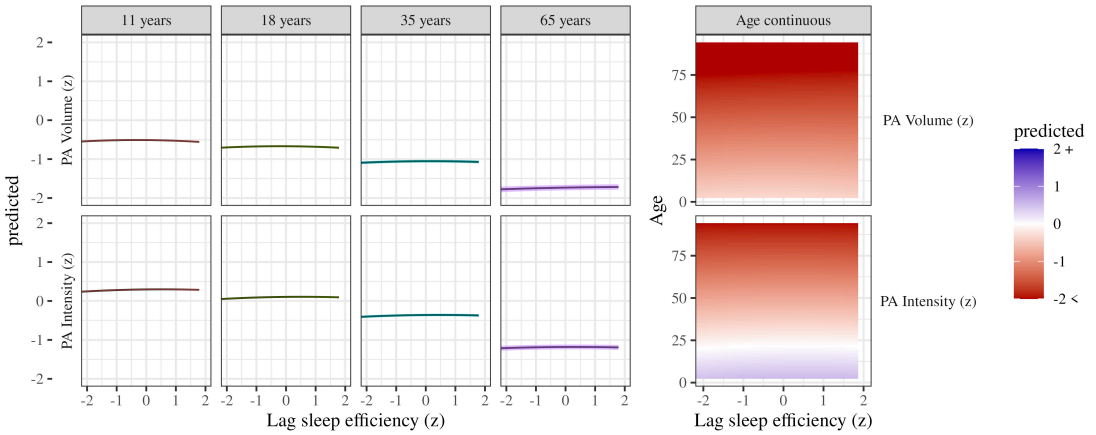

Figure 4: Physical activity by sleep efficiency

The association between sleep onset and next-day physical activity

We estimated the association between sleep onset and next-day physical activity by age. Results, controlling for SES, sex, BMI z-score, and the fixed effects of study ID are presented in Table 5 and Figure 5.

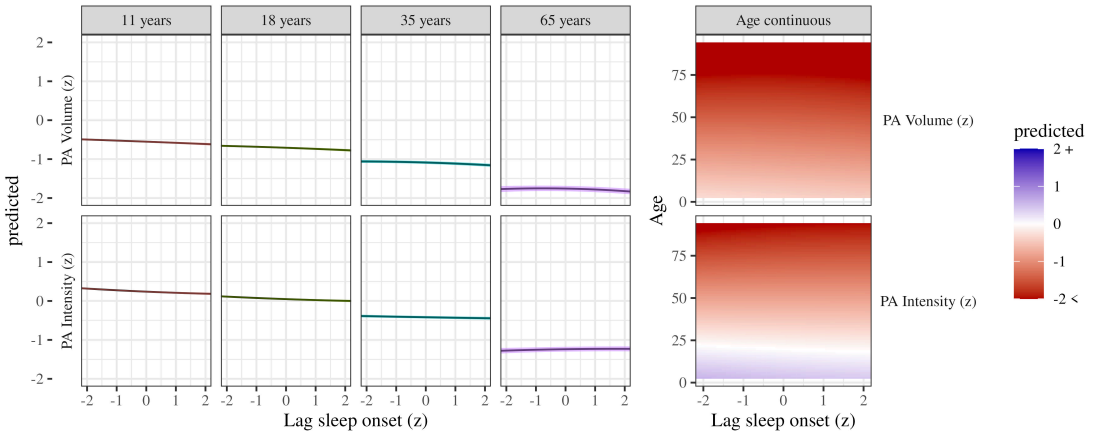

Figure 5: Physical activity by sleep onset

The association between sleep regularity and next-day physical activity

We estimated the association between sleep regularity and next-day physical activity by age. Results, controlling for SES, sex, BMI z-score, and the fixed effects of study ID are presented in Table 5 and Figure 6.

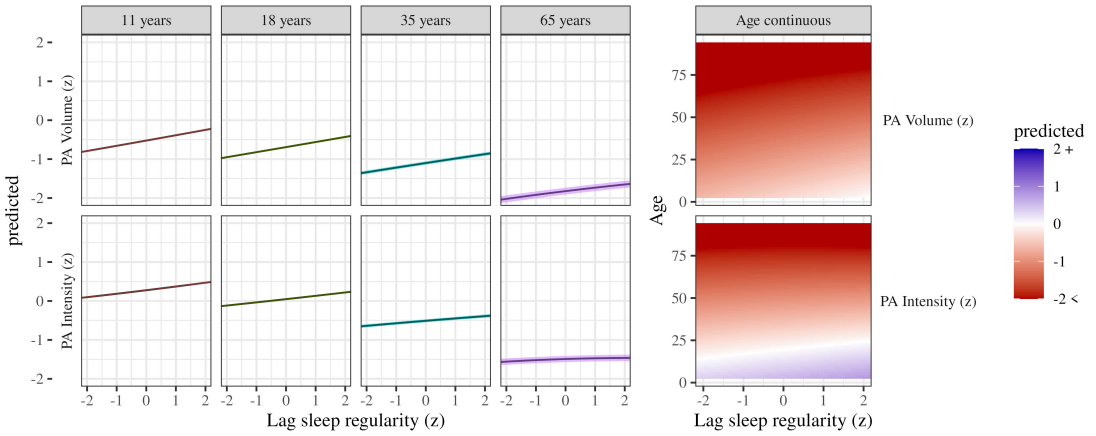

Figure 6: Physical activity by sleep regularity

Log Transforming PA Volume

Based on preliminary modelling on a subset of the data, we chose to leave physical activity volume in its natural form to enhance interpretability. The log scale meant that the variable became uninterpretable towards the edge of its range. Below, we present the results had we not made this decision. These results are all moderated by age, as this was our primary analysis. The associated figures are back-transformed to the scale range.

The association between physical activity volume and sleep

We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by age, and the results are presented in Table 6 and Figure 7.

Table 6: Physical activity predicting sleep controlling for SES, sex, and BMI z-score.

Term	PA volume (ln) β [95% CI]	PA volume (ln) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.17 [−0.08, 0.42]	.185	0.00 [−0.15, 0.14]	.995
Log pa volume	−0.27 [−0.38, −0.16]	< .001	0.05 [0.05, 0.06]	< .001
Age	−0.06 [−0.13, 0.01]	.081	−0.02 [−0.03, −0.01]	.003
Log pa volume²	0.06 [0.04, 0.08]	< .001	0.00 [0.00, 0.01]	.612
Log pa volume × Age	0.05 [0.01, 0.09]	.016	−0.01 [−0.01, −0.01]	< .001
Age × Log pa volume²	−0.01 [−0.02, 0.00]	< .001	−0.01 [−0.01, 0.00]	< .001
Sleep efficiency (z)				
(Intercept)	−1.12 [−1.37, −0.87]	< .001	0.15 [−0.02, 0.32]	.086
Log pa volume	0.67 [0.56, 0.78]	< .001	0.02 [0.02, 0.03]	< .001
Age	0.22 [0.15, 0.29]	< .001	−0.01 [−0.03, 0.00]	.067
Log pa volume²	−0.09 [−0.10, −0.07]	< .001	−0.02 [−0.02, −0.01]	< .001
Log pa volume × Age	−0.09 [−0.13, −0.06]	< .001	−0.01 [−0.02, −0.01]	< .001
Age × Log pa volume²	0.01 [0.00, 0.01]	.003	0.01 [0.00, 0.01]	< .001
Sleep onset (z)				
(Intercept)	−0.37 [−0.68, −0.06]	.018	0.12 [−0.14, 0.37]	.379
Log pa volume	0.47 [0.39, 0.56]	< .001	−0.06 [−0.06, −0.05]	< .001
Age	0.31 [0.26, 0.37]	< .001	0.00 [−0.02, 0.01]	.524
Log pa volume²	−0.09 [−0.10, −0.08]	< .001	−0.01 [−0.02, −0.01]	< .001
Log pa volume × Age	−0.17 [−0.20, −0.14]	< .001	0.01 [0.00, 0.01]	< .001

Term	PA volume (ln) β [95% CI]	PA volume (ln) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Age × Log pa volume²	0.02 [0.02, 0.03]	< .001	0.01 [0.01, 0.01]	< .001
Sleep regularity (z)				
(Intercept)	−4.20 [−4.48, −3.93]	< .001	0.06 [−0.14, 0.25]	.569
Log pa volume	1.98 [1.84, 2.12]	< .001	0.14 [0.13, 0.15]	< .001
Age	−0.39 [−0.47, −0.30]	< .001	−0.08 [−0.10, −0.06]	< .001
Log pa volume²	−0.22 [−0.24, −0.20]	< .001	−0.09 [−0.09, −0.08]	< .001
Log pa volume × Age	0.22 [0.17, 0.27]	< .001	−0.05 [−0.05, −0.05]	< .001
Age × Log pa volume²	−0.03 [−0.04, −0.02]	< .001	0.01 [0.01, 0.01]	< .001

Adjusted for SES, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised and (ln) natural-log transformed; ² denotes a quadratic term and × an interaction. Age was centred at 35.6 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 35.6 and age interactions are the change per 10 years.

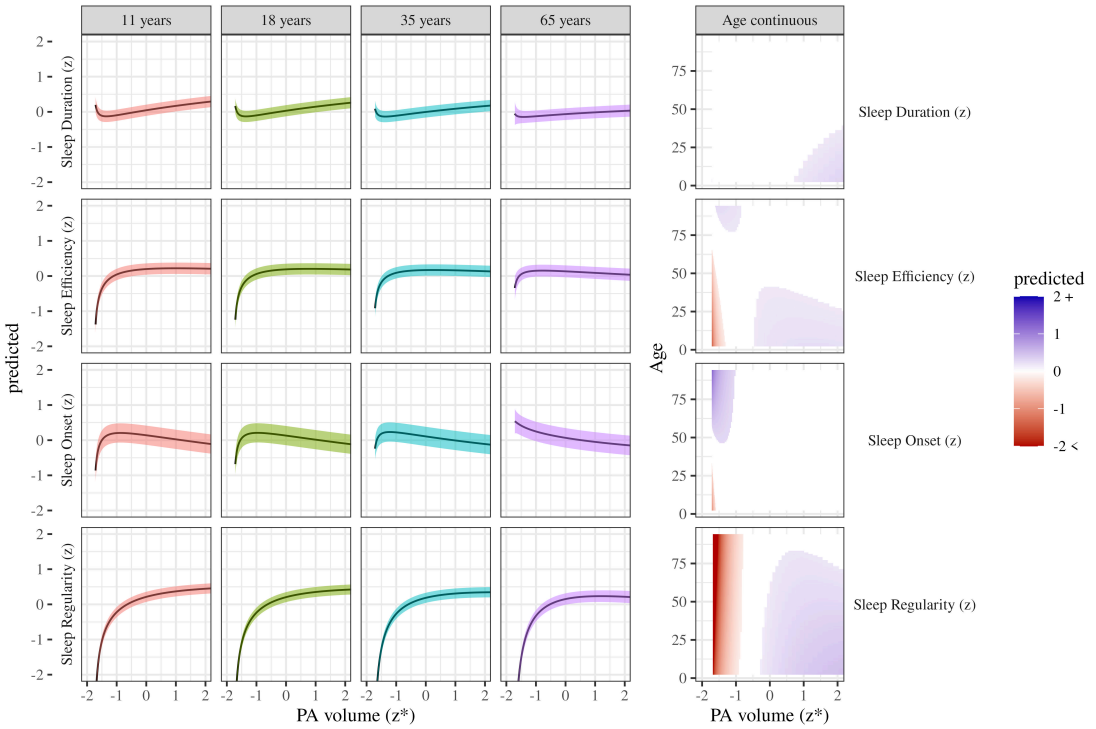

Figure 7: Sleep metrics on Physical activity volume

The association between sleep duration and next-day physical activity

We estimated the association between sleep duration and next-day physical activity by age. Results, controlling for SES, sex, and BMI z-score are presented in Table 7 and Figure 8.

Table 7: Sleep predicting physical activity controlling for SES, sex, and BMI z-score

Term	PA volume (ln) β [95% CI]	PA volume (ln) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	3.66 [3.51, 3.80]	< .001	−0.07 [−0.19, 0.05]	.254
Sleep duration	0.01 [0.00, 0.01]	< .001	0.03 [0.02, 0.03]	< .001
Age	−0.13 [−0.14, −0.12]	< .001	−0.27 [−0.28, −0.26]	< .001
Sleep duration²	−0.01 [−0.01, −0.01]	< .001	−0.01 [−0.01, −0.01]	< .001
Sleep duration × Age	−0.01 [−0.01, −0.01]	< .001	0.00 [0.00, 0.00]	.051
Age × Sleep duration²	0.00 [0.00, 0.00]	.992	0.00 [0.00, 0.00]	.326
Sleep efficiency (z)				
(Intercept)	3.65 [3.51, 3.79]	< .001	−0.07 [−0.20, 0.05]	.242
Sleep efficiency	0.01 [0.00, 0.01]	< .001	0.01 [0.00, 0.01]	.009
Age	−0.13 [−0.14, −0.12]	< .001	−0.27 [−0.28, −0.26]	< .001
Sleep efficiency²	−0.01 [−0.01, 0.00]	< .001	−0.01 [−0.01, 0.00]	< .001
Sleep efficiency × Age	0.00 [0.00, 0.00]	.010	0.00 [0.00, 0.00]	.200
Age × Sleep efficiency²	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	.227
Sleep onset (z)				
(Intercept)	3.66 [3.51, 3.80]	< .001	−0.08 [−0.21, 0.04]	.202
Sleep onset	−0.01 [−0.01, −0.01]	< .001	−0.01 [−0.02, −0.01]	< .001
Age	−0.13 [−0.14, −0.12]	< .001	−0.27 [−0.28, −0.26]	< .001
Sleep onset²	−0.01 [−0.01, 0.00]	< .001	0.00 [0.00, 0.00]	.811
Sleep onset × Age	0.00 [0.00, 0.00]	.001	0.01 [0.01, 0.01]	< .001

Term	PA volume (ln) β [95% CI]	PA volume (ln) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Age × Sleep onset²	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	.020
Sleep regularity (z)				
(Intercept)	3.61 [3.47, 3.75]	< .001	−0.18 [−0.30, −0.05]	.006
Sleep regularity	0.08 [0.08, 0.09]	< .001	0.06 [0.06, 0.07]	< .001
Age	−0.13 [−0.14, −0.13]	< .001	−0.32 [−0.34, −0.31]	< .001
Sleep regularity²	−0.01 [−0.01, −0.01]	< .001	0.00 [0.00, 0.00]	.414
Sleep regularity × Age	0.00 [0.00, 0.00]	< .001	−0.01 [−0.01, −0.01]	< .001
Age × Sleep regularity²	0.00 [0.00, 0.00]	.481	0.00 [0.00, 0.00]	.023

Adjusted for SES, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised and (ln) natural-log transformed; ² denotes a quadratic term and × an interaction. Age was centred at 35.6 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 35.6 and age interactions are the change per 10 years.

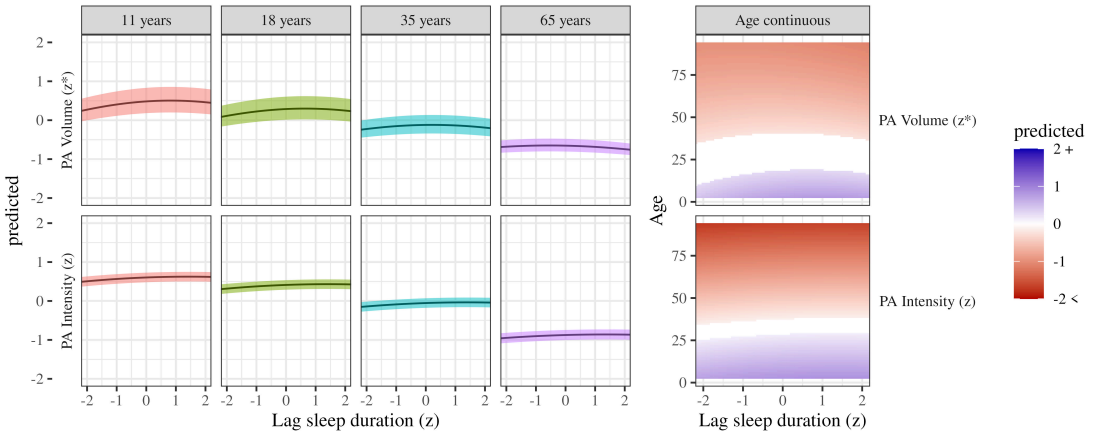

Figure 8: Physical activity by sleep duration

The association between sleep efficiency and next-day physical activity

We estimated the association between sleep efficiency and next-day physical activity by age. Results, controlling for SES, sex, and BMI z-score are presented in Table 7 and Figure 9.

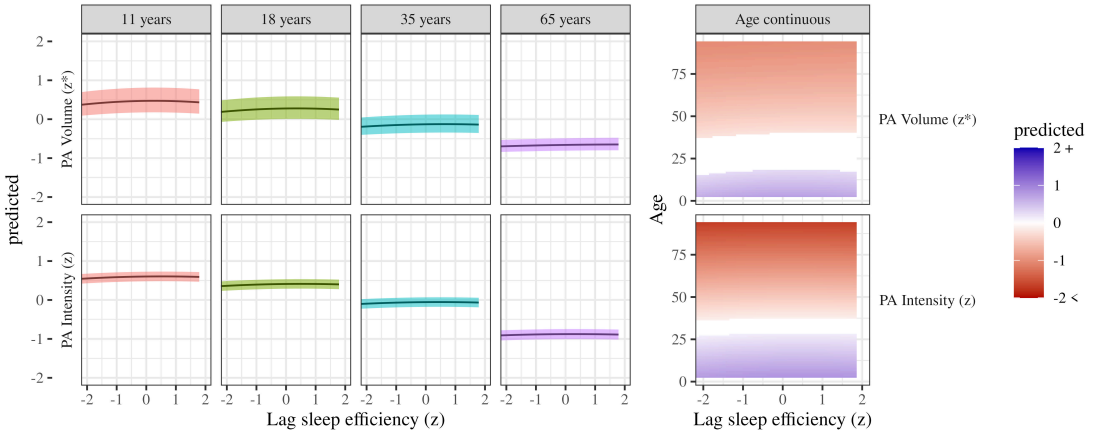

Figure 9: Physical activity by sleep efficiency

The association between sleep onset and next-day physical activity

We estimated the association between sleep onset and next-day physical activity by age. Results, controlling for SES, sex, and BMI z-score are presented in Table 7 and Figure 10.

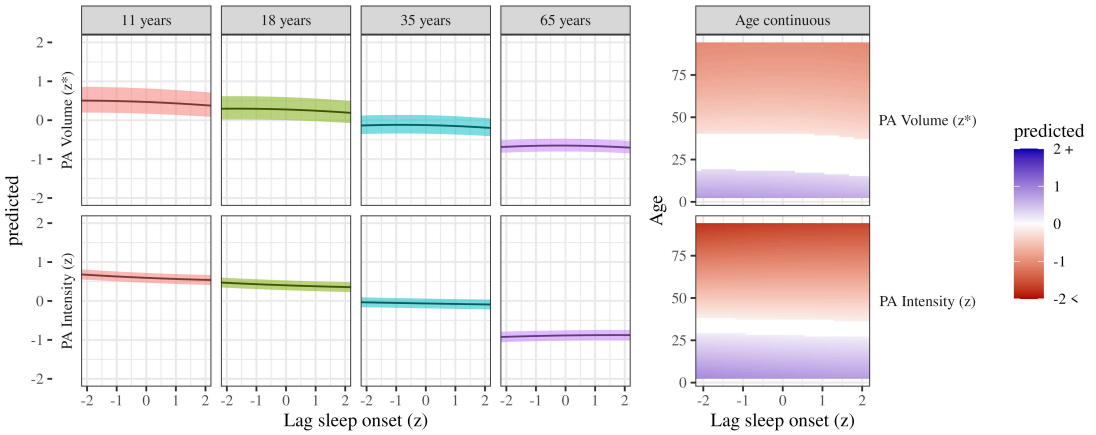

Figure 10: Physical activity by sleep onset

The association between sleep regularity and next-day physical activity

We estimated the association between sleep regularity and next-day physical activity by age. Results, controlling for SES, sex, and BMI z-score are presented in Table 7 and Figure 11.

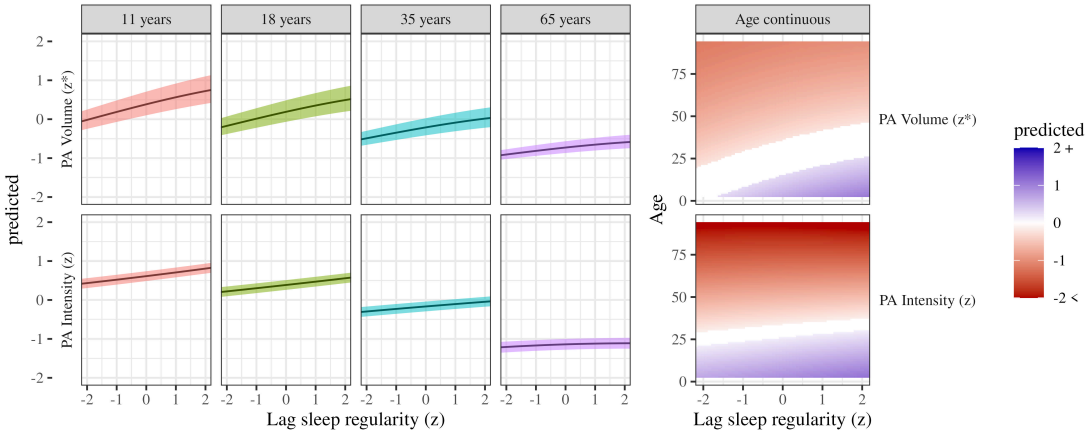

Figure 11: Physical activity by sleep regularity

All Eligible Participants Regardless of Valid Wear Days

The primary analysis required at least 4 days of valid wear time. Here we refit the primary models to every eligible participant, regardless of how many valid days they contributed. Both analyses share the same imputations and standardisation constants. Results, controlling for SES, sex, and BMI z-score, are presented in Tables 8 and 9 and Figures 12 to 17.

Table 8: Physical activity predicting sleep controlling for SES, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.00 [−0.16, 0.15]	.960	−0.01 [−0.15, 0.14]	.944
Physical activity	0.10 [0.09, 0.11]	< .001	0.05 [0.04, 0.06]	< .001
Age	−0.02 [−0.03, −0.01]	.004	−0.02 [−0.03, −0.01]	.002
Physical activity²	−0.02 [−0.02, −0.01]	< .001	0.00 [0.00, 0.01]	.691
Physical activity × Age	−0.02 [−0.02, −0.01]	< .001	−0.01 [−0.01, −0.01]	< .001
Age × Physical activity²	0.00 [−0.01, 0.00]	.002	−0.01 [−0.01, 0.00]	< .001
Sleep efficiency (z)				
(Intercept)	0.15 [−0.01, 0.31]	.067	0.15 [−0.02, 0.31]	.090
Physical activity	0.04 [0.03, 0.04]	< .001	0.02 [0.02, 0.03]	< .001
Age	−0.01 [−0.02, 0.01]	.466	−0.01 [−0.03, 0.00]	.100
Physical activity²	−0.02 [−0.03, −0.01]	< .001	−0.02 [−0.03, −0.01]	< .001
Physical activity × Age	−0.02 [−0.02, −0.02]	< .001	−0.01 [−0.02, −0.01]	< .001
Age × Physical activity²	0.00 [0.00, 0.00]	.149	0.01 [0.00, 0.01]	< .001
Sleep onset (z)				
(Intercept)	0.09 [−0.18, 0.36]	.515	0.11 [−0.14, 0.37]	.386
Physical activity	−0.12 [−0.12, −0.11]	< .001	−0.06 [−0.06, −0.05]	< .001
Age	−0.01 [−0.02, 0.00]	.073	0.00 [−0.01, 0.01]	.744
Physical activity²	0.02 [0.02, 0.03]	< .001	−0.01 [−0.02, −0.01]	< .001
Physical activity × Age	−0.01 [−0.01, −0.01]	< .001	0.01 [0.00, 0.01]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Age × Physical activity²	0.01 [0.01, 0.01]	< .001	0.01 [0.01, 0.01]	< .001
Sleep regularity (z)				
(Intercept)	0.19 [0.04, 0.33]	.012	0.06 [−0.13, 0.25]	.548
Physical activity	0.29 [0.27, 0.30]	< .001	0.14 [0.13, 0.15]	< .001
Age	0.01 [−0.01, 0.02]	.571	−0.08 [−0.10, −0.06]	< .001
Physical activity²	−0.15 [−0.16, −0.14]	< .001	−0.09 [−0.09, −0.08]	< .001
Physical activity × Age	0.00 [−0.01, 0.00]	.063	−0.05 [−0.05, −0.05]	< .001
Age × Physical activity²	−0.03 [−0.04, −0.03]	< .001	0.01 [0.01, 0.01]	< .001

Adjusted for SES, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction. Age was centred at 35.6 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 35.6 and age interactions are the change per 10 years.

Table 9: Sleep predicting physical activity controlling for SES, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.13 [−0.17, 0.44]	.382	−0.07 [−0.19, 0.05]	.248
Sleep duration	0.02 [0.01, 0.02]	< .001	0.03 [0.02, 0.03]	< .001
Age	−0.23 [−0.24, −0.22]	< .001	−0.27 [−0.28, −0.26]	< .001
Sleep duration²	−0.02 [−0.03, −0.02]	< .001	−0.01 [−0.01, −0.01]	< .001
Sleep duration × Age	−0.01 [−0.01, −0.01]	< .001	0.00 [0.00, 0.00]	.045
Age × Sleep duration²	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	.374
Sleep efficiency (z)				
(Intercept)	0.12 [−0.18, 0.42]	.416	−0.07 [−0.20, 0.05]	.236
Sleep efficiency	0.00 [0.00, 0.01]	.292	0.01 [0.00, 0.01]	.010
Age	−0.23 [−0.24, −0.21]	< .001	−0.27 [−0.29, −0.26]	< .001
Sleep efficiency²	−0.01 [−0.01, −0.01]	< .001	−0.01 [−0.01, 0.00]	< .001
Sleep efficiency × Age	0.00 [0.00, 0.01]	< .001	0.00 [0.00, 0.00]	.209
Age × Sleep efficiency²	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	.224
Sleep onset (z)				
(Intercept)	0.13 [−0.18, 0.43]	.414	−0.08 [−0.21, 0.04]	.197
Sleep onset	−0.02 [−0.03, −0.02]	< .001	−0.01 [−0.02, −0.01]	< .001
Age	−0.22 [−0.24, −0.21]	< .001	−0.27 [−0.29, −0.26]	< .001
Sleep onset²	0.00 [−0.01, 0.00]	< .001	0.00 [0.00, 0.00]	.892
Sleep onset × Age	0.00 [0.00, 0.00]	.011	0.01 [0.01, 0.01]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Age × Sleep onset²	0.00 [0.00, 0.00]	.002	0.00 [0.00, 0.00]	.016
Sleep regularity (z)				
(Intercept)	0.04 [−0.27, 0.34]	.819	−0.18 [−0.30, −0.05]	.006
Sleep regularity	0.12 [0.11, 0.12]	< .001	0.06 [0.06, 0.07]	< .001
Age	−0.24 [−0.26, −0.22]	< .001	−0.32 [−0.34, −0.31]	< .001
Sleep regularity²	0.00 [0.00, 0.00]	.386	0.00 [0.00, 0.00]	.364
Sleep regularity × Age	−0.01 [−0.01, −0.01]	< .001	−0.01 [−0.01, −0.01]	< .001
Age × Sleep regularity²	0.00 [0.00, 0.00]	.151	0.00 [0.00, 0.00]	.015

Adjusted for SES, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction. Age was centred at 35.6 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 35.6 and age interactions are the change per 10 years.

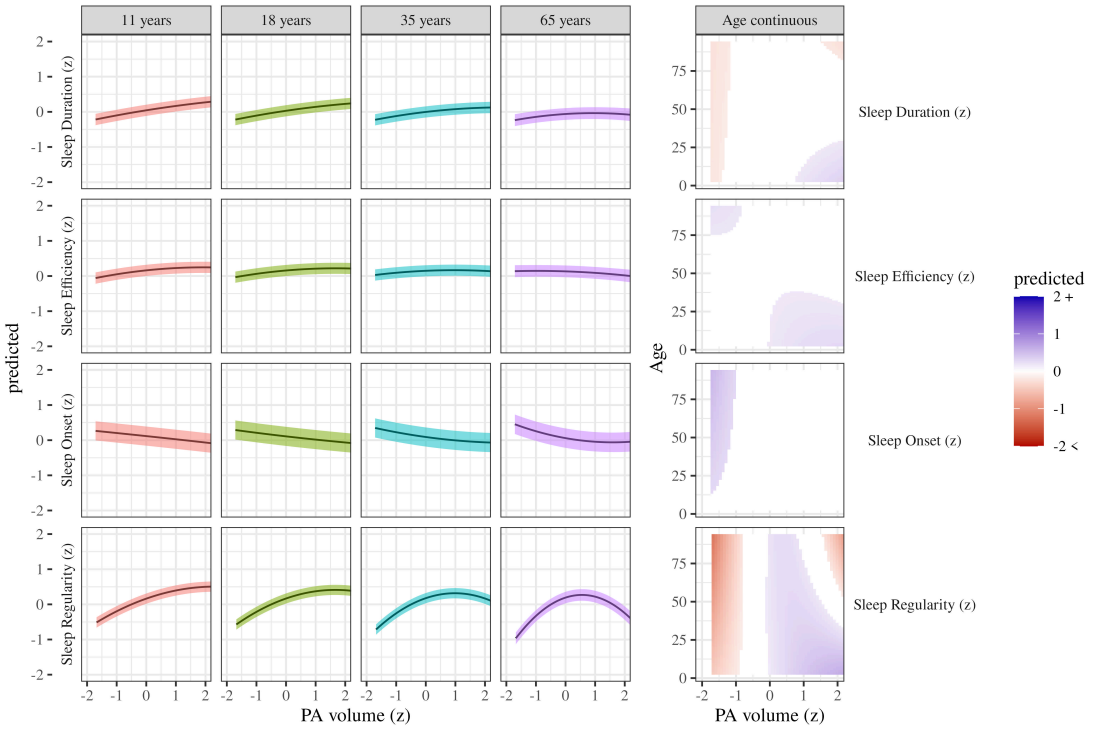

Figure 12: Sleep metrics on physical activity volume (all wear days)

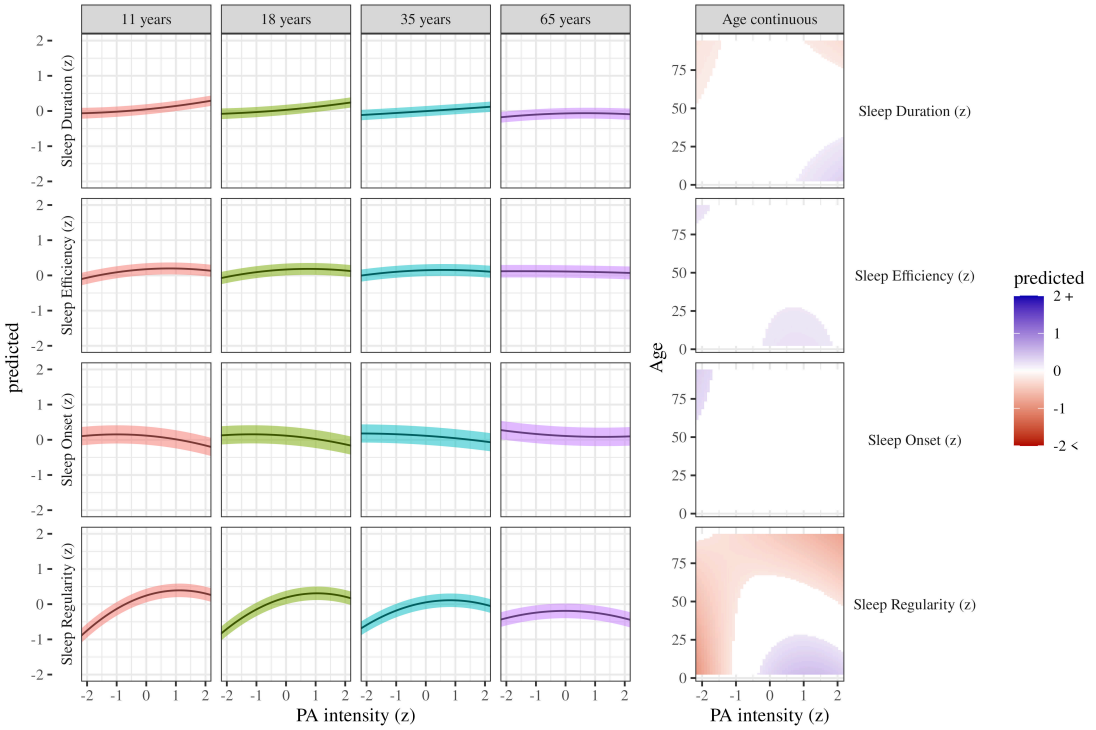

Figure 13: Sleep metrics on physical activity intensity (all wear days)

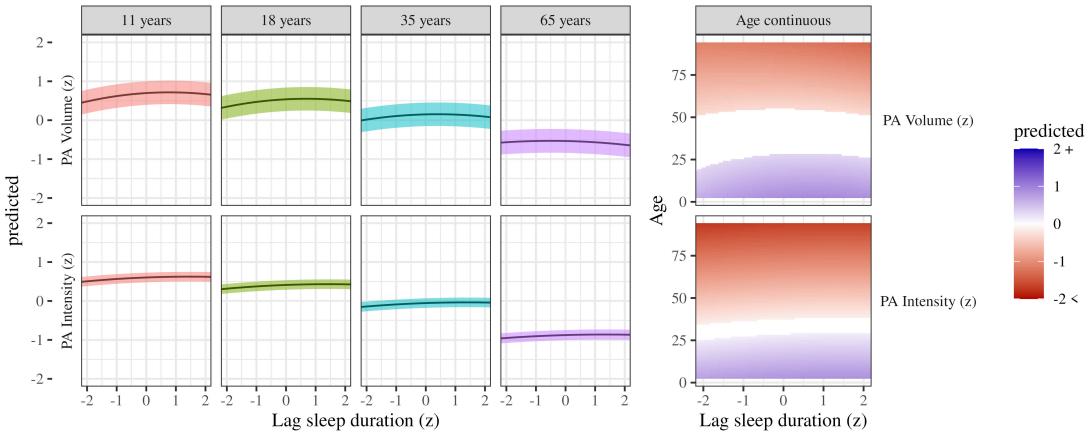

Figure 14: Physical activity by previous night sleep duration (all wear days)

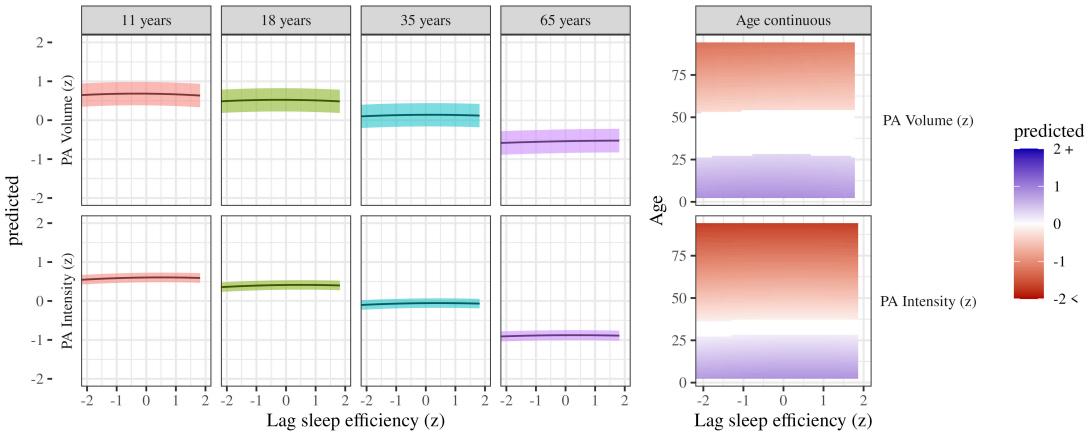

Figure 15: Physical activity by previous night sleep efficiency (all wear days)

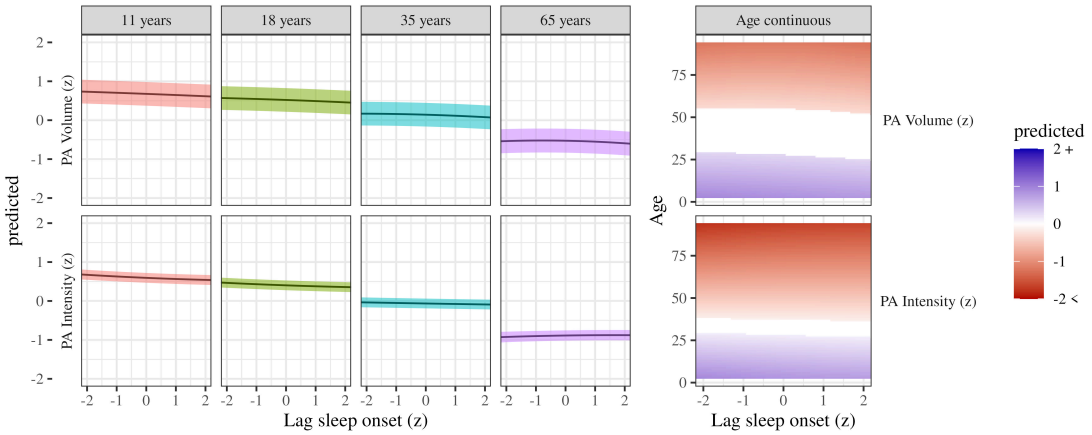

Figure 16: Physical activity by previous night sleep onset (all wear days)

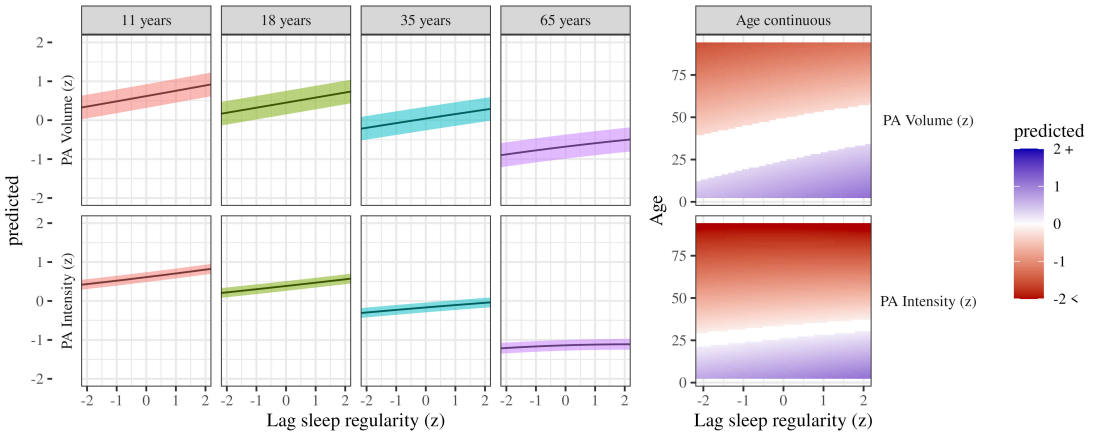

Figure 17: Physical activity by previous night sleep regularity (all wear days)

Participants With at Least Seven Valid Days

Some recommend at least seven nights to estimate sleep regularity. Here we refit the primary models to the 14,981 participants with at least seven days of valid wear time. Results, controlling for SES, sex, and BMI z-score, are presented in Tables 10 and 11 and Figures 18 to 23.

Table 10: Physical activity predicting sleep controlling for SES, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	−0.03 [−0.28, 0.23]	.834	−0.04 [−0.28, 0.20]	.757
Physical activity	0.09 [0.08, 0.10]	< .001	0.05 [0.04, 0.06]	< .001
Age	0.03 [0.01, 0.04]	< .001	0.03 [0.01, 0.04]	< .001
Physical activity²	−0.02 [−0.02, −0.01]	< .001	0.00 [−0.01, 0.00]	.403
Physical activity × Age	−0.01 [−0.01, 0.00]	.004	−0.01 [−0.01, 0.00]	< .001
Age × Physical activity²	0.00 [0.00, 0.00]	.303	−0.01 [−0.01, 0.00]	< .001
Sleep efficiency (z)				
(Intercept)	0.12 [−0.18, 0.43]	.425	0.11 [−0.21, 0.44]	.499
Physical activity	0.06 [0.05, 0.07]	< .001	0.04 [0.03, 0.05]	< .001
Age	−0.05 [−0.06, −0.03]	< .001	−0.05 [−0.07, −0.03]	< .001
Physical activity²	−0.02 [−0.03, −0.02]	< .001	−0.02 [−0.03, −0.02]	< .001
Physical activity × Age	−0.03 [−0.03, −0.03]	< .001	−0.02 [−0.02, −0.01]	< .001
Age × Physical activity²	0.00 [0.00, 0.01]	.021	0.01 [0.00, 0.01]	< .001
Sleep onset (z)				
(Intercept)	0.00 [−0.40, 0.39]	.984	0.02 [−0.36, 0.39]	.937
Physical activity	−0.09 [−0.10, −0.08]	< .001	−0.05 [−0.05, −0.04]	< .001
Age	−0.07 [−0.08, −0.06]	< .001	−0.06 [−0.07, −0.04]	< .001
Physical activity²	0.01 [0.00, 0.01]	< .001	−0.01 [−0.01, 0.00]	.001
Physical activity × Age	−0.03 [−0.03, −0.02]	< .001	0.00 [−0.01, 0.00]	.034

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Age × Physical activity²	0.01 [0.00, 0.01]	< .001	0.01 [0.01, 0.01]	< .001
Sleep regularity (z)				
(Intercept)	0.21 [0.06, 0.36]	.006	0.05 [−0.23, 0.33]	.723
Physical activity	0.28 [0.27, 0.30]	< .001	0.16 [0.14, 0.17]	< .001
Age	0.01 [−0.02, 0.04]	.381	−0.12 [−0.15, −0.09]	< .001
Physical activity²	−0.17 [−0.18, −0.15]	< .001	−0.11 [−0.12, −0.10]	< .001
Physical activity × Age	−0.01 [−0.02, −0.01]	< .001	−0.05 [−0.06, −0.05]	< .001
Age × Physical activity²	−0.04 [−0.05, −0.04]	< .001	0.02 [0.02, 0.02]	< .001

Adjusted for SES, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction. Age was centred at 35.6 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 35.6 and age interactions are the change per 10 years.

Table 11: Sleep predicting physical activity controlling for SES, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	−0.01 [−0.37, 0.34]	.939	−0.10 [−0.25, 0.04]	.167
Sleep duration	0.02 [0.02, 0.03]	< .001	0.04 [0.04, 0.05]	< .001
Age	−0.19 [−0.20, −0.18]	< .001	−0.24 [−0.25, −0.22]	< .001
Sleep duration²	−0.02 [−0.03, −0.02]	< .001	−0.01 [−0.02, −0.01]	< .001
Sleep duration × Age	−0.01 [−0.02, −0.01]	< .001	−0.01 [−0.01, 0.00]	< .001
Age × Sleep duration²	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	< .001
Sleep efficiency (z)				
(Intercept)	−0.02 [−0.37, 0.33]	.903	−0.11 [−0.25, 0.04]	.160
Sleep efficiency	0.01 [0.01, 0.02]	< .001	0.01 [0.00, 0.02]	.001
Age	−0.18 [−0.20, −0.17]	< .001	−0.23 [−0.24, −0.22]	< .001
Sleep efficiency²	−0.01 [−0.01, 0.00]	< .001	−0.01 [−0.01, 0.00]	< .001
Sleep efficiency × Age	0.00 [0.00, 0.00]	.852	0.00 [0.00, 0.00]	.138
Age × Sleep efficiency²	0.00 [0.00, 0.00]	.037	0.00 [0.00, 0.00]	.016
Sleep onset (z)				
(Intercept)	−0.05 [−0.42, 0.33]	.799	−0.10 [−0.25, 0.06]	.217
Sleep onset	−0.05 [−0.06, −0.04]	< .001	−0.02 [−0.03, −0.01]	< .001
Age	−0.19 [−0.21, −0.18]	< .001	−0.23 [−0.25, −0.22]	< .001
Sleep onset²	−0.02 [−0.02, −0.01]	< .001	0.00 [−0.01, 0.00]	.065
Sleep onset × Age	0.01 [0.01, 0.01]	< .001	0.01 [0.01, 0.01]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Age × Sleep onset²	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	.116
Sleep regularity (z)				
(Intercept)	−0.12 [−0.47, 0.24]	.524	−0.22 [−0.38, −0.05]	.009
Sleep regularity	0.11 [0.10, 0.11]	< .001	0.07 [0.06, 0.08]	< .001
Age	−0.18 [−0.20, −0.16]	< .001	−0.30 [−0.32, −0.28]	< .001
Sleep regularity²	0.00 [0.00, 0.00]	.232	0.00 [−0.01, 0.00]	.443
Sleep regularity × Age	−0.01 [−0.01, 0.00]	< .001	−0.02 [−0.02, −0.01]	< .001
Age × Sleep regularity²	0.00 [0.00, 0.00]	.269	0.00 [0.00, 0.00]	.915

Adjusted for SES, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction. Age was centred at 35.6 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 35.6 and age interactions are the change per 10 years.

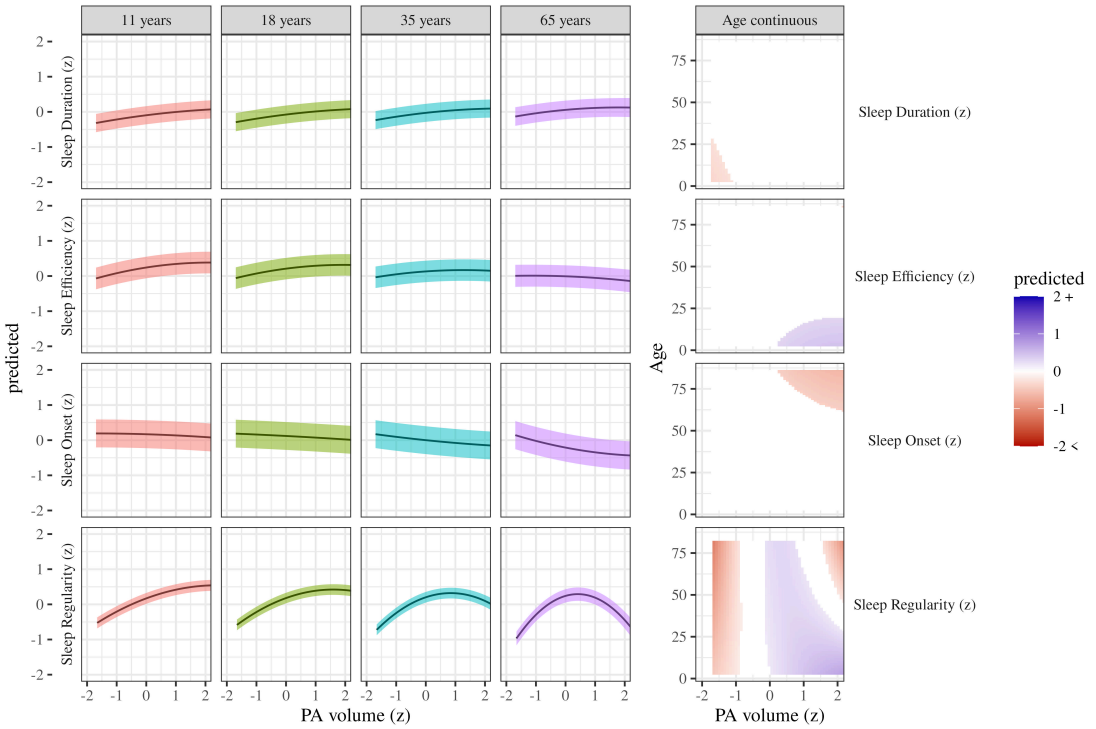

Figure 18: Sleep metrics on physical activity volume (at least seven valid days)

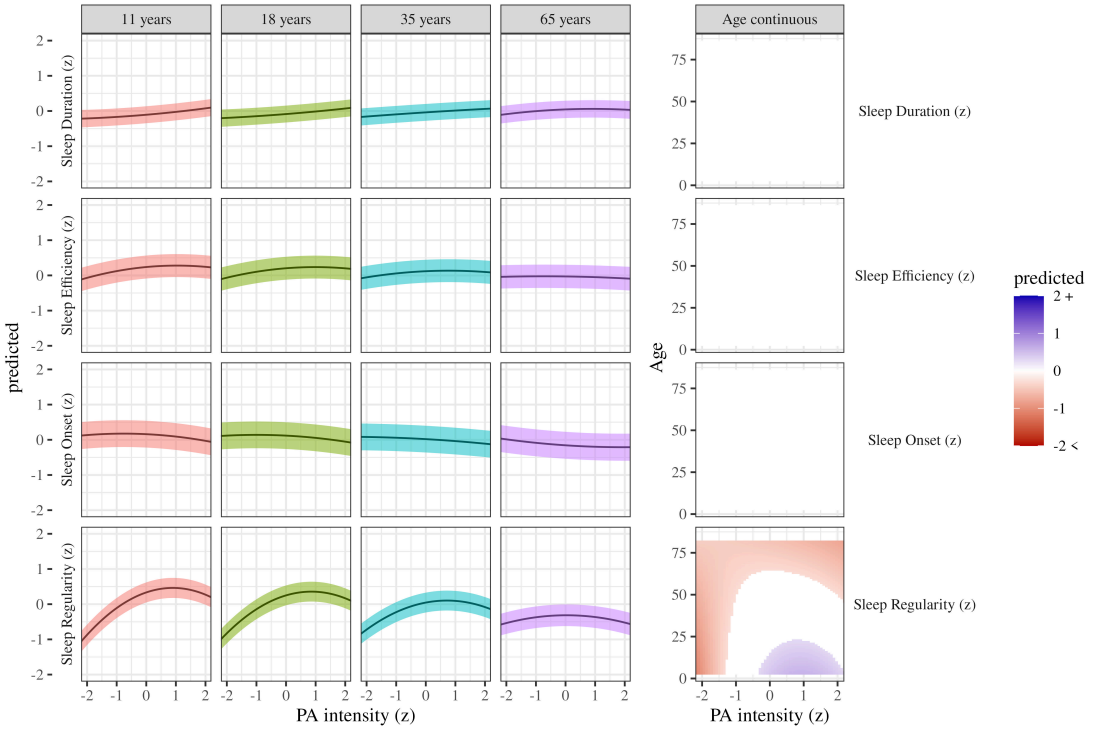

Figure 19: Sleep metrics on physical activity intensity (at least seven valid days)

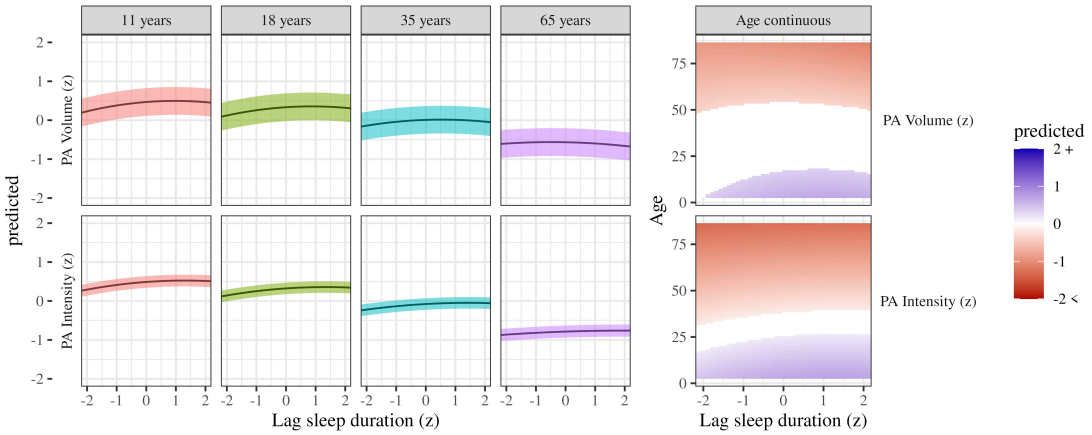

Figure 20: Physical activity by previous night sleep duration (at least seven valid days)

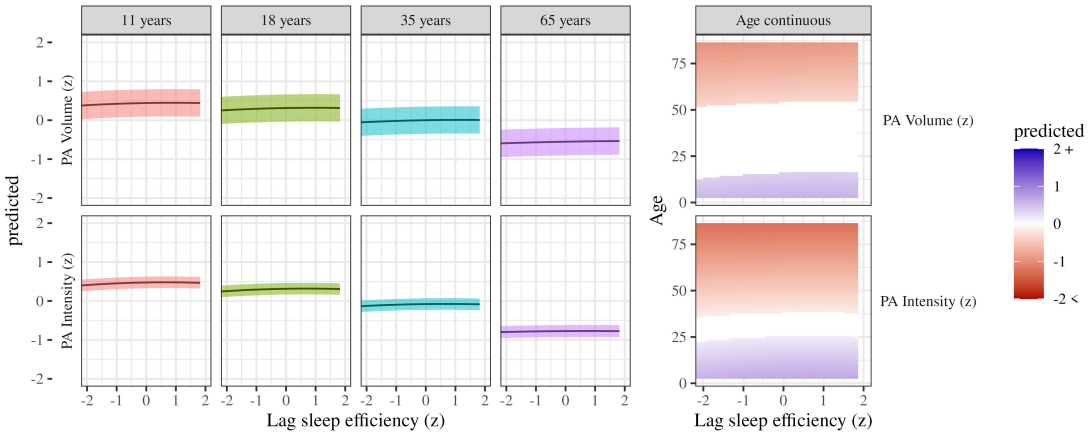

Figure 21: Physical activity by previous night sleep efficiency (at least seven valid days)

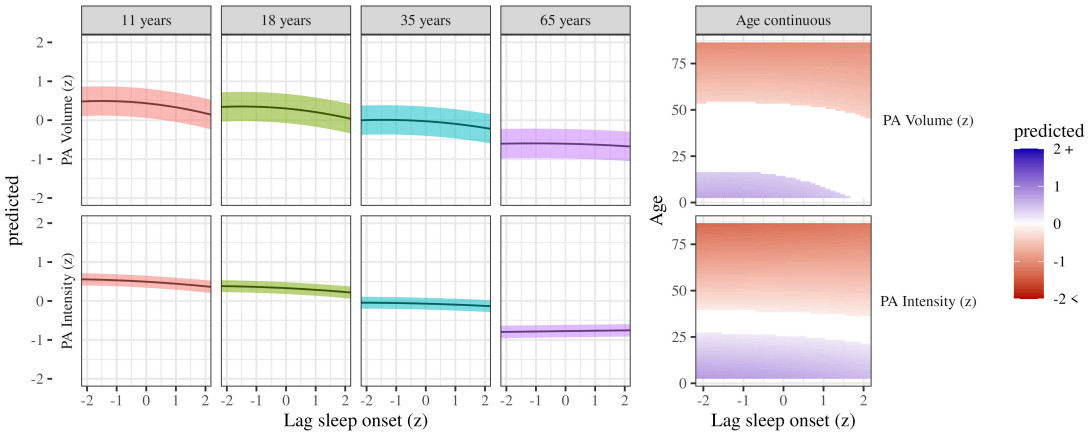

Figure 22: Physical activity by previous night sleep onset (at least seven valid days)

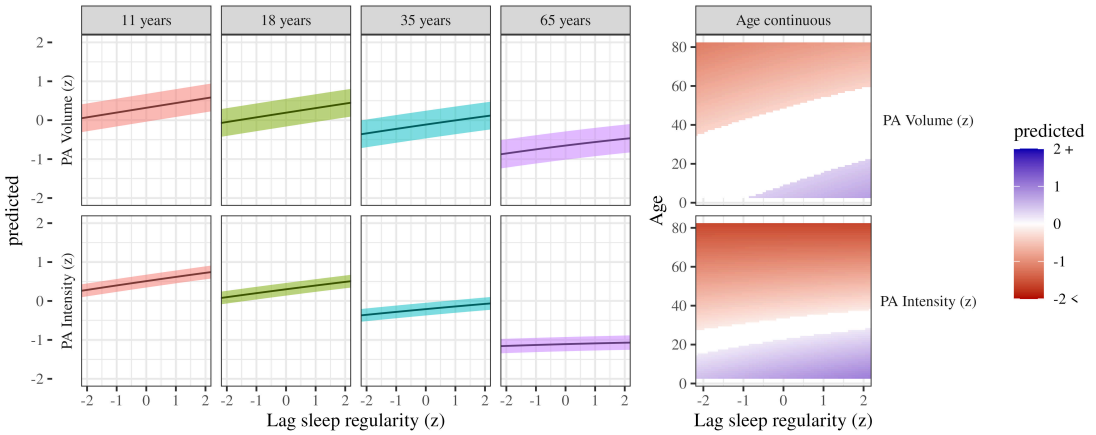

Figure 23: Physical activity by previous night sleep regularity (at least seven valid days)

Moderation

In the following section we present models for a range of moderators which were pre-defined in our protocol.

BMI z-score

The association between physical activity volume and sleep by BMI z-score

We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by BMI z-score, and the results are presented in Table 12 and Figure 24

Table 12: Physical activity predicting sleep controlling for SES, age, and sex.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	−0.04 [−0.19, 0.11]	.603	−0.05 [−0.18, 0.09]	.524
Physical activity	0.11 [0.10, 0.12]	< .001	0.06 [0.05, 0.07]	< .001
BMI z-score	0.00 [−0.01, 0.00]	.249	−0.02 [−0.02, −0.01]	< .001
Physical activity²	0.00 [−0.01, 0.00]	.037	0.01 [0.01, 0.02]	< .001
Physical activity × BMI z-score	0.01 [0.00, 0.02]	.003	0.01 [0.01, 0.02]	< .001
BMI z-score × Physical activity²	0.00 [0.00, 0.00]	.154	0.00 [0.00, 0.01]	.016
Sleep efficiency (z)				
(Intercept)	0.21 [0.07, 0.36]	.005	0.20 [0.05, 0.36]	.011
Physical activity	0.06 [0.05, 0.07]	< .001	0.01 [0.00, 0.01]	.072
BMI z-score	−0.01 [−0.02, 0.00]	.012	−0.02 [−0.03, −0.01]	< .001
Physical activity²	−0.02 [−0.02, −0.01]	< .001	0.00 [0.00, 0.00]	.947
Physical activity × BMI z-score	0.02 [0.02, 0.03]	< .001	0.01 [0.01, 0.02]	< .001
BMI z-score × Physical activity²	0.00 [−0.01, 0.00]	.001	0.00 [−0.01, 0.00]	.576
Sleep onset (z)				
(Intercept)	0.15 [−0.13, 0.43]	.286	0.15 [−0.11, 0.41]	.260
Physical activity	−0.10 [−0.11, −0.09]	< .001	−0.08 [−0.08, −0.07]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
BMI z-score	0.00 [−0.01, 0.00]	.218	0.00 [0.00, 0.01]	.547
Physical activity²	0.00 [0.00, 0.00]	.594	−0.01 [−0.01, −0.01]	< .001
Physical activity × BMI z-score	−0.01 [−0.01, 0.00]	.006	−0.01 [−0.02, −0.01]	< .001
BMI z-score × Physical activity²	0.00 [0.00, 0.00]	.920	0.00 [−0.01, 0.00]	.010
Sleep regularity (z)				
(Intercept)	0.14 [0.00, 0.29]	.058	0.14 [−0.04, 0.31]	.119
Physical activity	0.31 [0.30, 0.32]	< .001	0.10 [0.09, 0.11]	< .001
BMI z-score	−0.05 [−0.06, −0.05]	< .001	−0.06 [−0.07, −0.05]	< .001
Physical activity²	−0.06 [−0.06, −0.05]	< .001	−0.02 [−0.02, −0.02]	< .001
Physical activity × BMI z-score	0.04 [0.03, 0.05]	< .001	0.03 [0.02, 0.03]	< .001
BMI z-score × Physical activity²	−0.01 [−0.01, 0.00]	< .001	−0.01 [−0.01, 0.00]	.004

Adjusted for SES, age, and sex. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction.

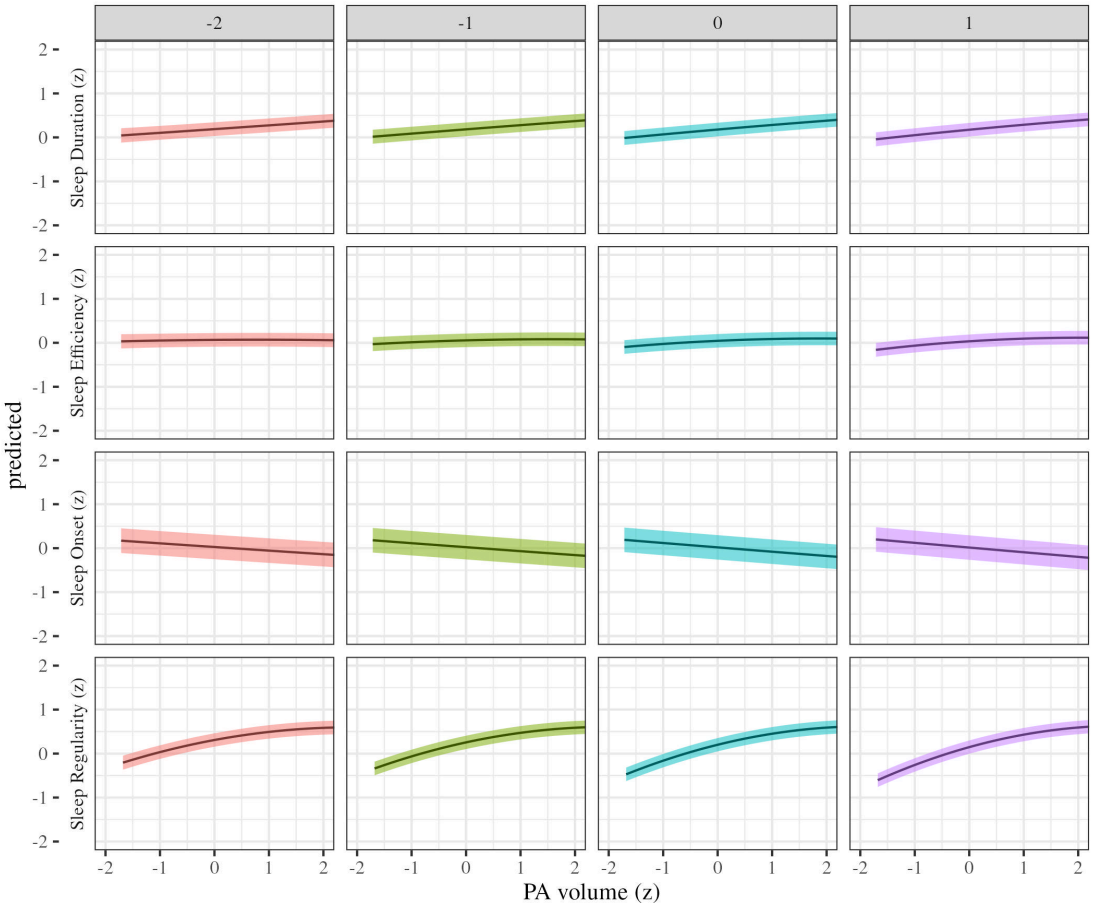

Figure 24: Sleep metrics on Physical activity volume by BMI z-score

The association between physical activity intensity and sleep by BMI z-score

We estimated how physical activity intensity is associated with sleep across BMI z-score. We present the results in Table 12 and Figure 25.

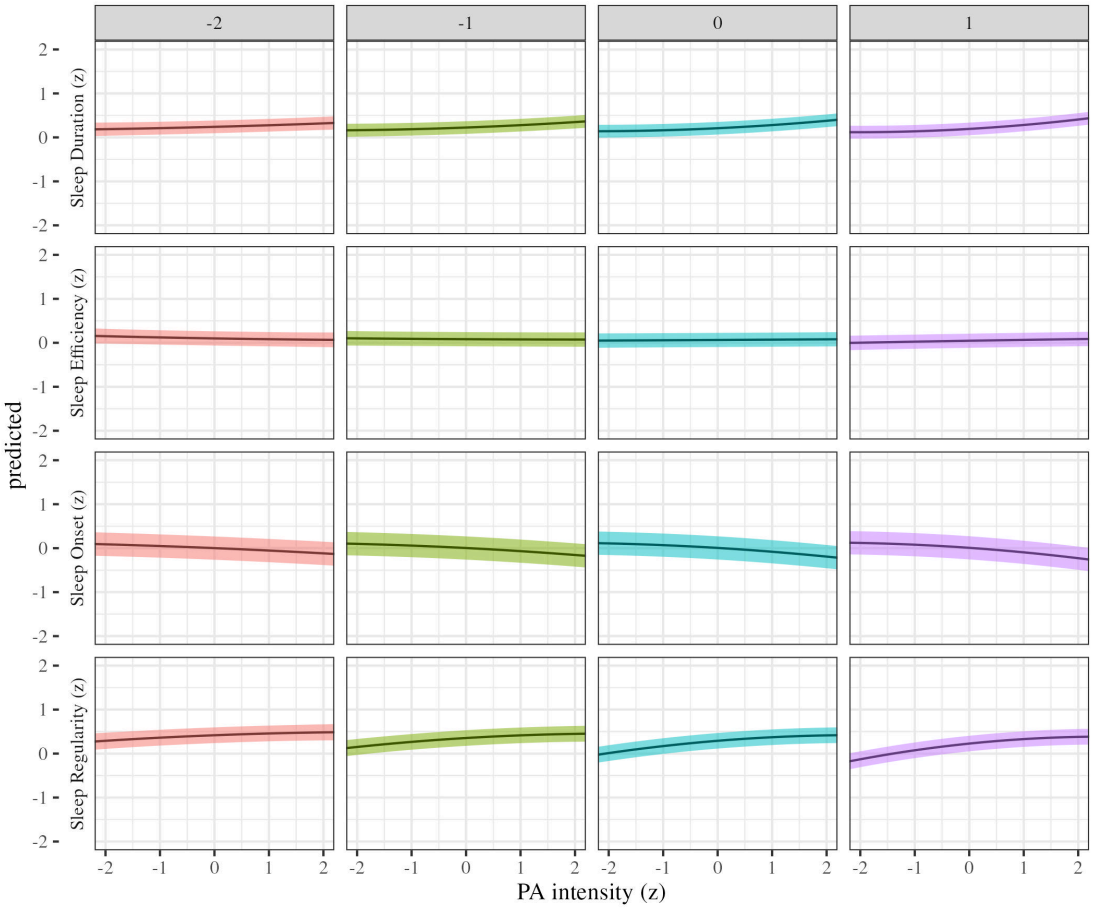

Figure 25: Sleep metrics on Physical activity intensity moderated by BMI z-score

The association between sleep duration and next-day physical activity

We estimated the association between sleep duration and next-day physical activity by BMI z-score. Results, controlling for SES, age and sex are presented in Table 13 and Figure 26.

Table 13: Sleep predicting physical activity controlling for SES, age, and sex

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.07 [−0.25, 0.39]	.661	−0.16 [−0.29, −0.02]	.025
Sleep duration	0.02 [0.02, 0.03]	< .001	0.02 [0.02, 0.03]	< .001
BMI z-score	−0.05 [−0.06, −0.04]	< .001	−0.06 [−0.07, −0.05]	< .001
Sleep duration²	−0.02 [−0.02, −0.02]	< .001	−0.01 [−0.01, −0.01]	< .001
Sleep duration × BMI z-score	0.00 [0.00, 0.01]	.026	0.00 [0.00, 0.01]	.041
BMI z-score × Sleep duration²	0.00 [0.00, 0.00]	.783	0.00 [0.00, 0.01]	.019
Sleep efficiency (z)				
(Intercept)	0.06 [−0.26, 0.38]	.723	−0.16 [−0.30, −0.03]	.020
Sleep efficiency	0.00 [0.00, 0.01]	.552	0.01 [0.00, 0.01]	.002
BMI z-score	−0.05 [−0.06, −0.04]	< .001	−0.06 [−0.07, −0.05]	< .001
Sleep efficiency²	−0.01 [−0.01, −0.01]	< .001	−0.01 [−0.01, −0.01]	< .001
Sleep efficiency × BMI z-score	−0.01 [−0.01, 0.00]	.017	0.00 [−0.01, 0.00]	.245
BMI z-score × Sleep efficiency²	0.00 [0.00, 0.00]	.783	0.00 [0.00, 0.00]	.854
Sleep onset (z)				
(Intercept)	0.06 [−0.27, 0.38]	.730	−0.17 [−0.30, −0.03]	.019
Sleep onset	−0.02 [−0.03, −0.01]	< .001	−0.02 [−0.03, −0.01]	< .001
BMI z-score	−0.05 [−0.06, −0.05]	< .001	−0.06 [−0.07, −0.05]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep onset²	0.00 [−0.01, 0.00]	.020	0.00 [0.00, 0.01]	.212
Sleep onset × BMI z-score	0.00 [−0.01, 0.00]	.191	0.00 [0.00, 0.01]	.682
BMI z-score × Sleep onset²	0.00 [0.00, 0.01]	.026	0.00 [0.00, 0.00]	.838
Sleep regularity (z)				
(Intercept)	0.04 [−0.26, 0.35]	.784	−0.17 [−0.30, −0.04]	.009
Sleep regularity	0.12 [0.12, 0.13]	< .001	0.07 [0.07, 0.08]	< .001
BMI z-score	−0.04 [−0.05, −0.04]	< .001	−0.05 [−0.06, −0.05]	< .001
Sleep regularity²	0.00 [−0.01, 0.00]	.083	0.00 [−0.01, 0.00]	.045
Sleep regularity × BMI z-score	0.00 [−0.01, 0.00]	.272	0.00 [0.00, 0.01]	.324
BMI z-score × Sleep regularity²	0.00 [0.00, 0.00]	.685	0.00 [0.00, 0.00]	.479

Adjusted for SES, age, and sex. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction.

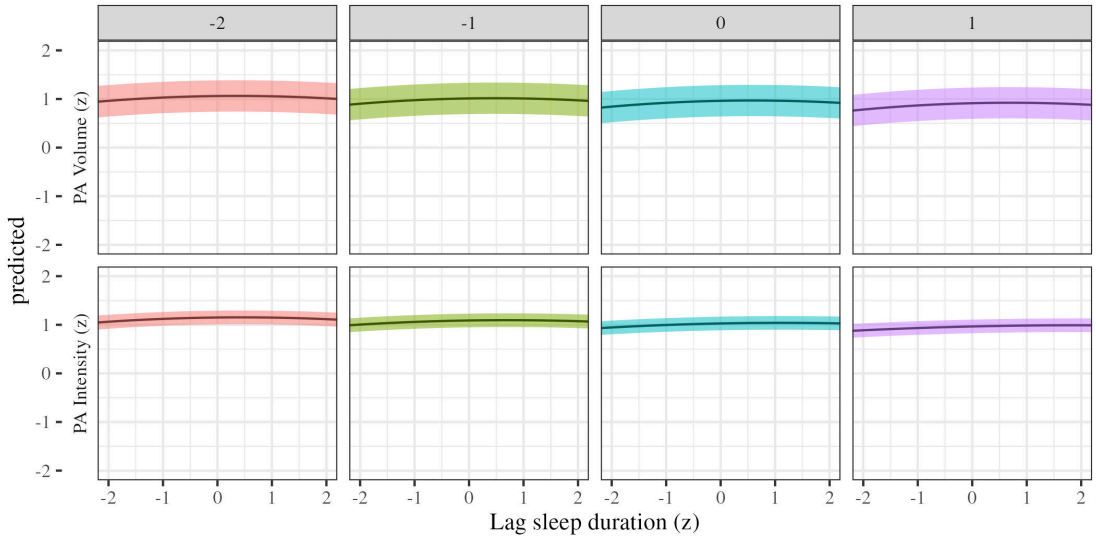

Figure 26: Physical activity by sleep duration moderated by BMI z-score

The association between sleep efficiency and next-day physical activity

We estimated the association between sleep efficiency and next-day physical activity by BMI z-score. Results, controlling for SES, age and sex are presented in Table 13 and Figure 27.

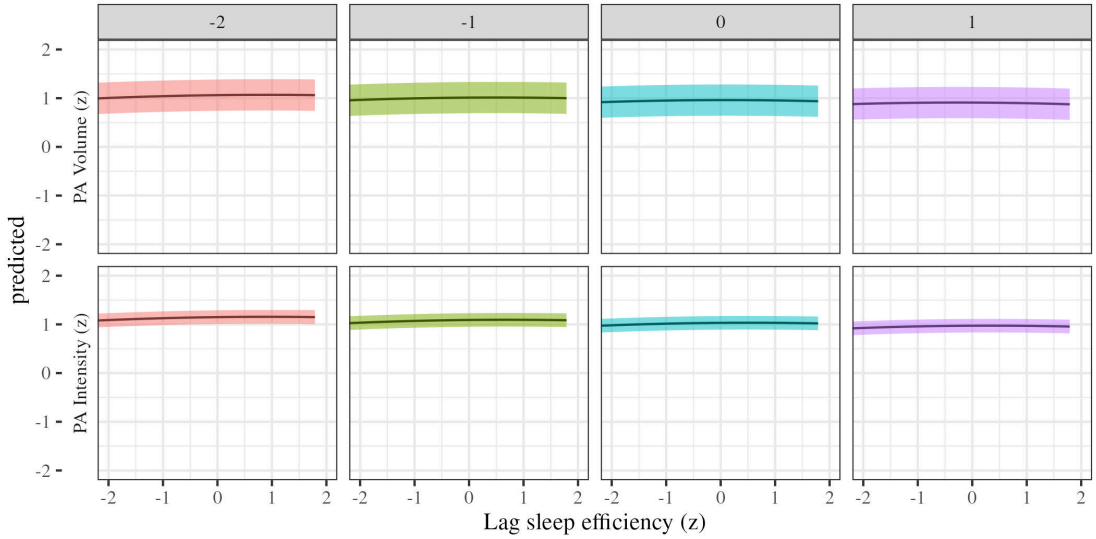

Figure 27: Physical activity by sleep efficiency moderated by BMI z-score

The association between sleep onset and next-day physical activity

We estimated the association between sleep onset and next-day physical activity by BMI z-score. Results, controlling for SES, age and sex are presented in Table 13 and Figure 28.

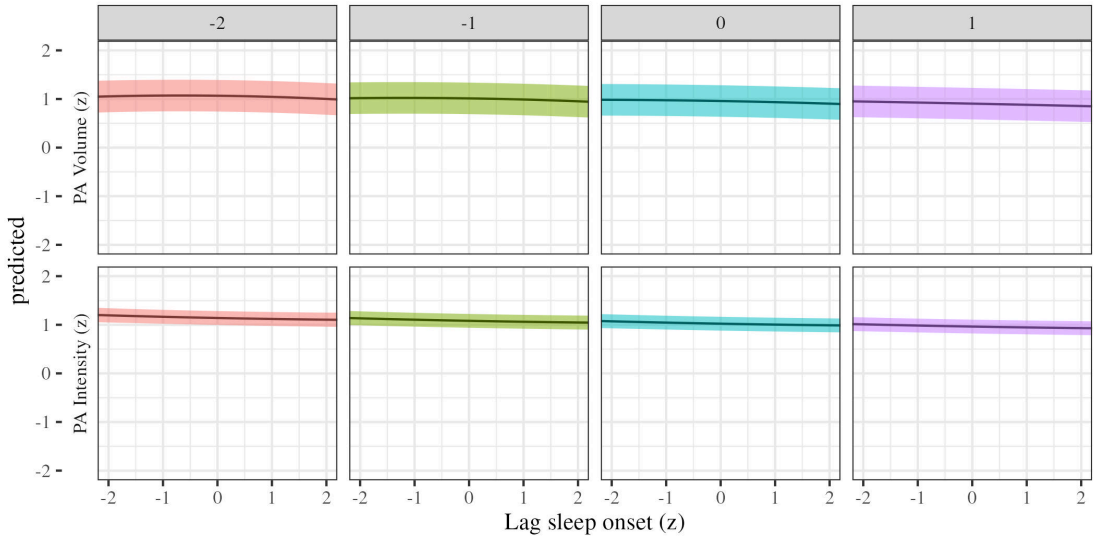

Figure 28: Physical activity by sleep onset moderated by BMI z-score

The association between sleep regularity and next-day physical activity

We estimated the association between sleep regularity and next-day physical activity by BMI z-score. Results are presented in Table 13 and Figure 29.

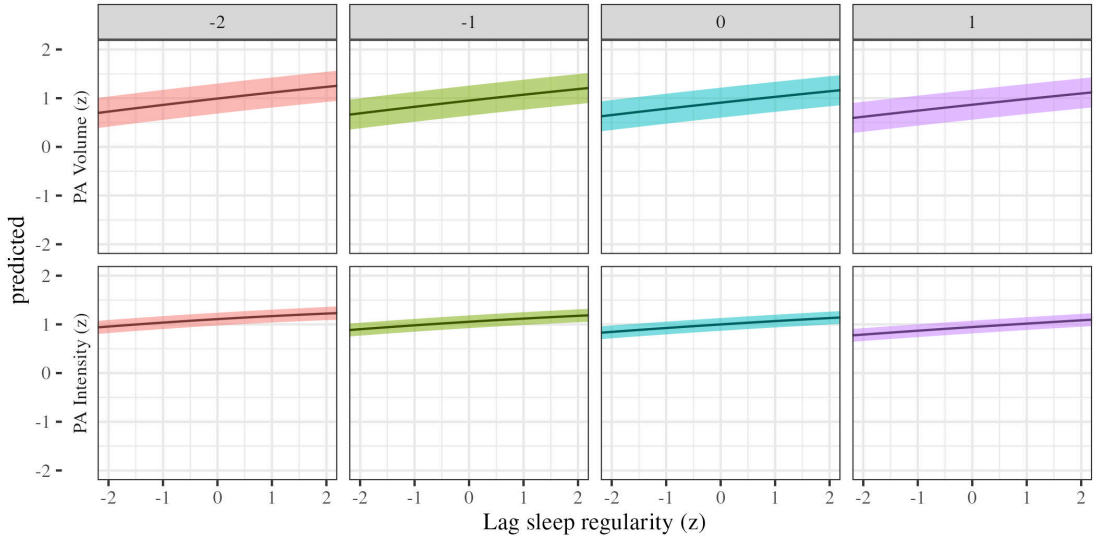

Figure 29: Physical activity by sleep regularity moderated by BMI z-score

SES

The association between physical activity volume and sleep by SES

We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by SES, and the results are presented in Table 14 and Figure 30

Table 14: Physical activity predicting sleep controlling for age, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.01 [−0.19, 0.20]	.946	0.01 [−0.17, 0.20]	.884
Physical activity	0.09 [0.08, 0.11]	< .001	0.06 [0.05, 0.07]	< .001
Medium	0.01 [−0.01, 0.03]	.324	0.01 [−0.01, 0.04]	.240
High	0.03 [0.00, 0.05]	.022	0.01 [−0.01, 0.03]	.412
Physical activity²	0.00 [−0.01, 0.00]	.522	0.01 [0.00, 0.02]	.035
Physical activity × Medium	0.01 [−0.01, 0.03]	.300	0.00 [−0.01, 0.02]	.631
Physical activity × High	0.02 [0.00, 0.04]	.094	0.00 [−0.02, 0.01]	.761
Medium × Physical activity²	0.00 [0.00, 0.01]	.439	−0.01 [−0.02, 0.00]	.239
High × Physical activity²	−0.01 [−0.01, 0.00]	.151	0.00 [−0.01, 0.01]	.719
Sleep efficiency (z)				
(Intercept)	0.15 [−0.04, 0.33]	.129	0.13 [−0.06, 0.33]	.191
Physical activity	0.03 [0.01, 0.04]	< .001	−0.01 [−0.02, 0.01]	.253
Medium	0.03 [0.00, 0.05]	.021	0.04 [0.01, 0.06]	.002
High	0.04 [0.01, 0.06]	.002	0.03 [0.01, 0.06]	.014
Physical activity²	−0.01 [−0.02, −0.01]	< .001	0.01 [0.00, 0.01]	.193
Physical activity × Medium	0.01 [−0.01, 0.02]	.620	0.01 [0.00, 0.03]	.096
Physical activity × High	0.04 [0.02, 0.06]	< .001	0.03 [0.01, 0.04]	.001
Medium × Physical activity²	0.00 [0.00, 0.01]	.252	−0.01 [−0.02, 0.00]	.128
High × Physical activity²	−0.01 [−0.01, 0.00]	.051	−0.01 [−0.02, 0.00]	.299

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep onset (z)				
(Intercept)	0.06 [−0.26, 0.39]	.701	0.05 [−0.25, 0.36]	.746
Physical activity	−0.14 [−0.16, −0.13]	< .001	−0.09 [−0.10, −0.08]	< .001
Medium	0.04 [0.02, 0.05]	< .001	0.04 [0.02, 0.06]	< .001
High	0.02 [0.00, 0.04]	.045	0.03 [0.01, 0.05]	.006
Physical activity²	0.01 [0.00, 0.01]	.010	0.00 [−0.01, 0.00]	.221
Physical activity × Medium	0.03 [0.01, 0.04]	.001	0.01 [0.00, 0.02]	.093
Physical activity × High	0.06 [0.04, 0.08]	< .001	0.03 [0.02, 0.04]	< .001
Medium × Physical activity²	0.00 [−0.01, 0.00]	.431	0.00 [−0.01, 0.01]	.826
High × Physical activity²	0.00 [−0.01, 0.00]	.084	0.00 [−0.01, 0.00]	.243
Sleep regularity (z)				
(Intercept)	0.15 [−0.01, 0.32]	.068	0.15 [−0.03, 0.34]	.104
Physical activity	0.34 [0.32, 0.36]	< .001	0.08 [0.07, 0.10]	< .001
Medium	0.01 [−0.02, 0.04]	.385	0.00 [−0.03, 0.02]	.743
High	0.04 [0.01, 0.07]	.003	0.03 [0.00, 0.06]	.027
Physical activity²	−0.06 [−0.07, −0.06]	< .001	−0.03 [−0.04, −0.02]	< .001
Physical activity × Medium	−0.03 [−0.06, −0.01]	.008	0.00 [−0.02, 0.01]	.677
Physical activity × High	−0.03 [−0.06, −0.01]	.009	0.03 [0.01, 0.05]	.001
Medium × Physical activity²	0.01 [0.00, 0.01]	.097	0.01 [0.00, 0.02]	.114
High × Physical activity²	0.01 [0.00, 0.02]	.056	0.00 [−0.02, 0.01]	.487

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p

Adjusted for age, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction.

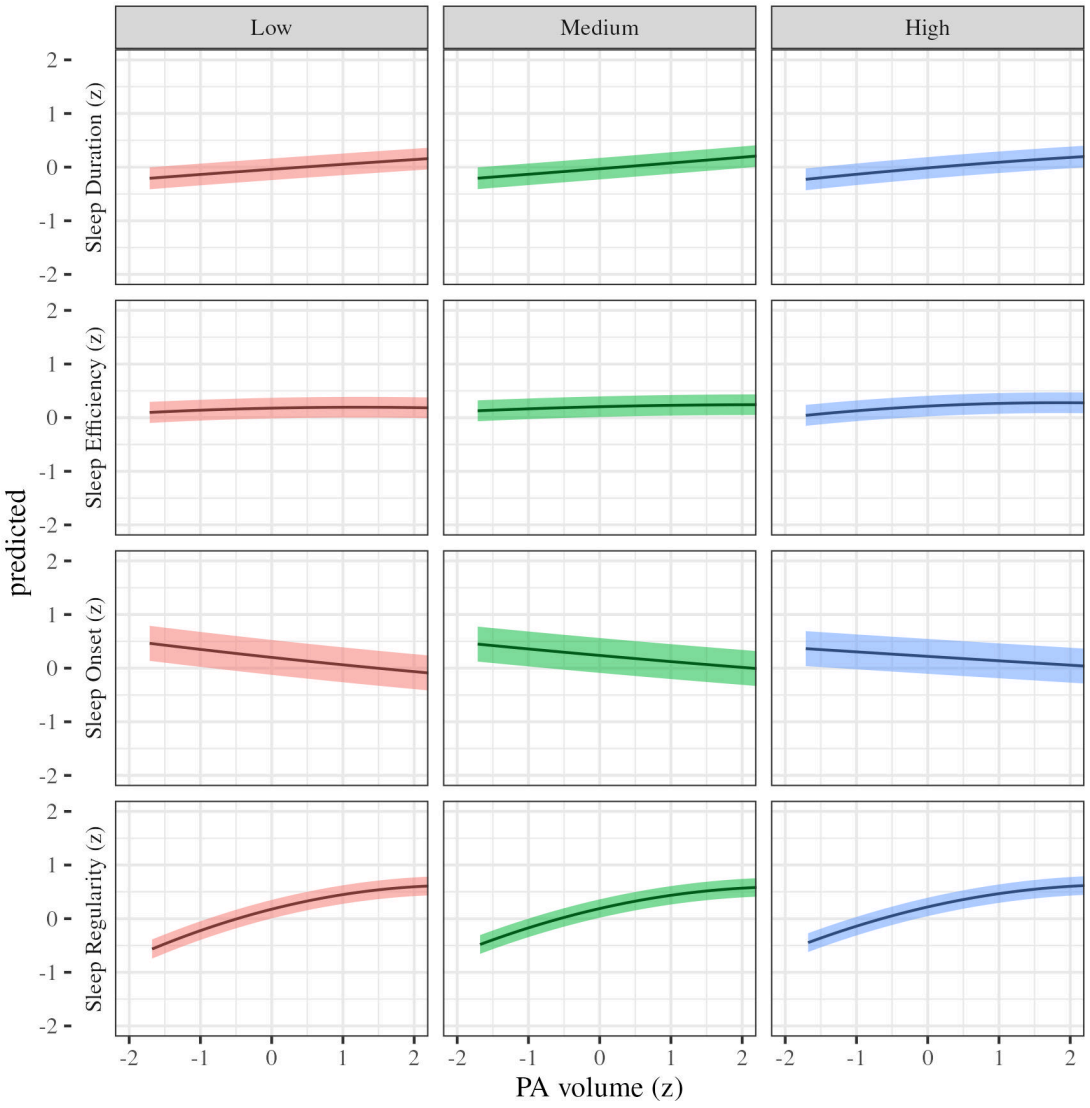

Figure 30: Sleep metrics on Physical activity volume by SES

The association between physical activity intensity and sleep by SES

We estimated how physical activity intensity is associated with sleep across SES. We present the results in Table 14 and Figure 31.

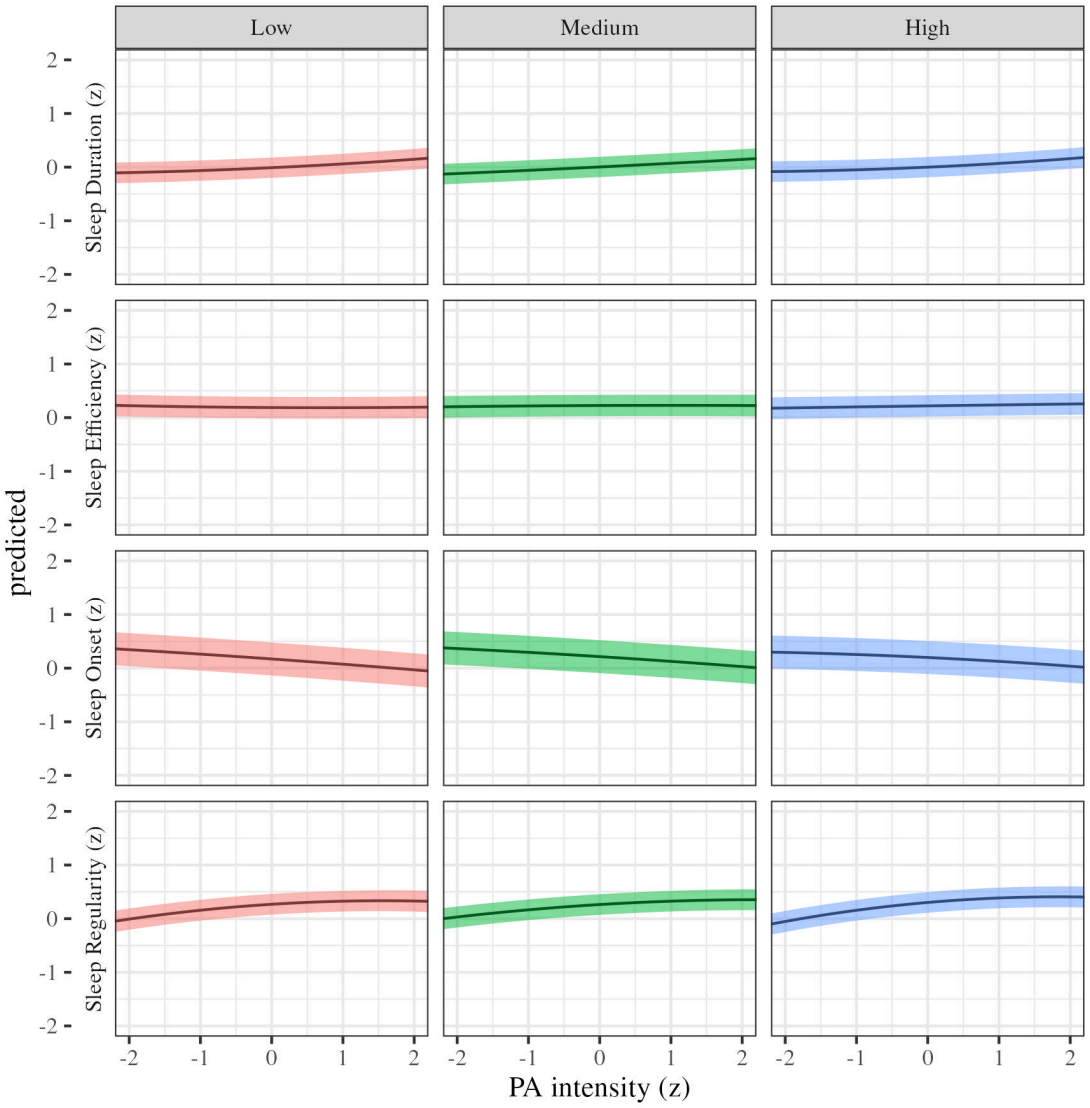

Figure 31: Sleep metrics on Physical activity intensity moderated by SES

The association between sleep duration and next-day physical activity

We estimated the association between sleep duration and next-day physical activity by SES. Results, controlling for age, sex and BMI z-score are presented in Table 15 and Figure 32.

Table 15: Sleep predicting physical activity controlling for age, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.16 [−0.14, 0.47]	.295	−0.06 [−0.19, 0.06]	.312
Sleep duration	−0.01 [−0.01, 0.00]	.087	0.02 [0.01, 0.03]	< .001
Medium	−0.06 [−0.08, −0.04]	< .001	0.01 [−0.01, 0.02]	.453
High	−0.08 [−0.10, −0.06]	< .001	0.02 [0.00, 0.03]	.088
Sleep duration²	−0.02 [−0.02, −0.02]	< .001	−0.01 [−0.01, −0.01]	< .001
Sleep duration × Medium	0.02 [0.01, 0.03]	< .001	0.00 [−0.01, 0.01]	.666
Sleep duration × High	0.03 [0.02, 0.04]	< .001	0.01 [0.00, 0.02]	.029
Medium × Sleep duration²	0.00 [0.00, 0.01]	.527	0.00 [0.00, 0.01]	.360
High × Sleep duration²	0.00 [0.00, 0.01]	.417	0.00 [0.00, 0.01]	.658
Sleep efficiency (z)				
(Intercept)	0.15 [−0.16, 0.45]	.347	−0.07 [−0.20, 0.05]	.265
Sleep efficiency	0.01 [0.00, 0.02]	.029	0.00 [−0.01, 0.01]	.463
Medium	−0.06 [−0.07, −0.04]	< .001	0.01 [−0.01, 0.03]	.177
High	−0.07 [−0.09, −0.05]	< .001	0.03 [0.01, 0.05]	.003
Sleep efficiency²	0.00 [−0.01, 0.00]	.096	0.00 [−0.01, 0.00]	.053
Sleep efficiency × Medium	−0.01 [−0.02, 0.01]	.290	0.01 [0.00, 0.02]	.229
Sleep efficiency × High	0.00 [−0.01, 0.01]	.712	0.00 [−0.01, 0.02]	.578
Medium × Sleep efficiency²	0.00 [−0.01, 0.00]	.304	0.00 [−0.01, 0.00]	.580
High × Sleep efficiency²	−0.01 [−0.01, 0.00]	.025	−0.01 [−0.01, 0.00]	.003

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep onset (z)				
(Intercept)	0.16 [−0.15, 0.47]	.312	−0.07 [−0.20, 0.05]	.263
Sleep onset	−0.02 [−0.03, −0.01]	< .001	0.00 [−0.01, 0.01]	.836
Medium	−0.07 [−0.09, −0.05]	< .001	0.01 [−0.01, 0.03]	.302
High	−0.09 [−0.11, −0.07]	< .001	0.01 [−0.01, 0.03]	.154
Sleep onset²	−0.01 [−0.02, −0.01]	< .001	0.00 [−0.01, 0.01]	.986
Sleep onset × Medium	0.01 [0.00, 0.02]	.051	−0.01 [−0.02, 0.01]	.394
Sleep onset × High	0.00 [−0.01, 0.02]	.672	−0.02 [−0.03, −0.01]	.003
Medium × Sleep onset²	0.01 [0.00, 0.02]	.002	0.00 [−0.01, 0.01]	.815
High × Sleep onset²	0.01 [0.00, 0.02]	.022	0.00 [0.00, 0.01]	.226
Sleep regularity (z)				
(Intercept)	0.09 [−0.23, 0.40]	.598	−0.15 [−0.27, −0.04]	.010
Sleep regularity	0.13 [0.12, 0.14]	< .001	0.08 [0.07, 0.09]	< .001
Medium	−0.04 [−0.06, −0.02]	< .001	0.02 [0.00, 0.04]	.022
High	−0.06 [−0.08, −0.04]	< .001	0.03 [0.01, 0.05]	.001
Sleep regularity²	0.00 [0.00, 0.01]	.714	0.00 [0.00, 0.01]	.513
Sleep regularity × Medium	−0.01 [−0.03, 0.00]	.050	−0.02 [−0.04, −0.01]	< .001
Sleep regularity × High	−0.01 [−0.02, 0.00]	.104	0.00 [−0.01, 0.02]	.469
Medium × Sleep regularity²	−0.01 [−0.01, 0.00]	.124	0.00 [−0.01, 0.00]	.236
High × Sleep regularity²	−0.01 [−0.01, 0.00]	.143	−0.01 [−0.02, 0.00]	.041

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p

Adjusted for age, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction.

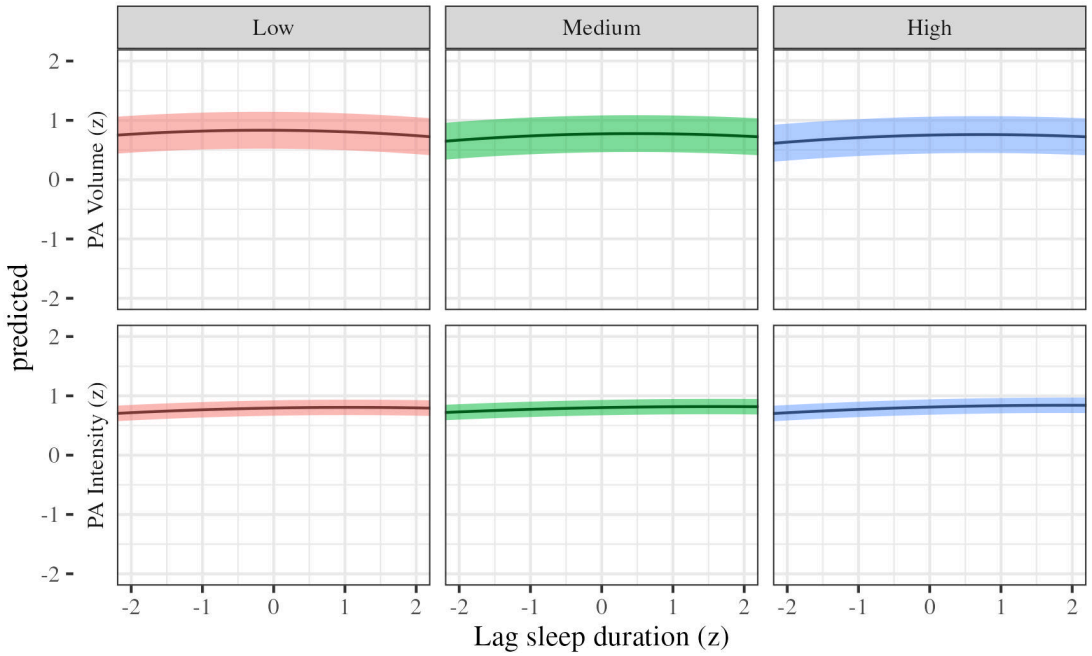

Figure 32: Physical activity by sleep duration moderated by SES

The association between sleep efficiency and next-day physical activity

We estimated the association between sleep efficiency and next-day physical activity by SES. Results, controlling for age, sex and BMI z-score are presented in Table 15 and Figure 33.

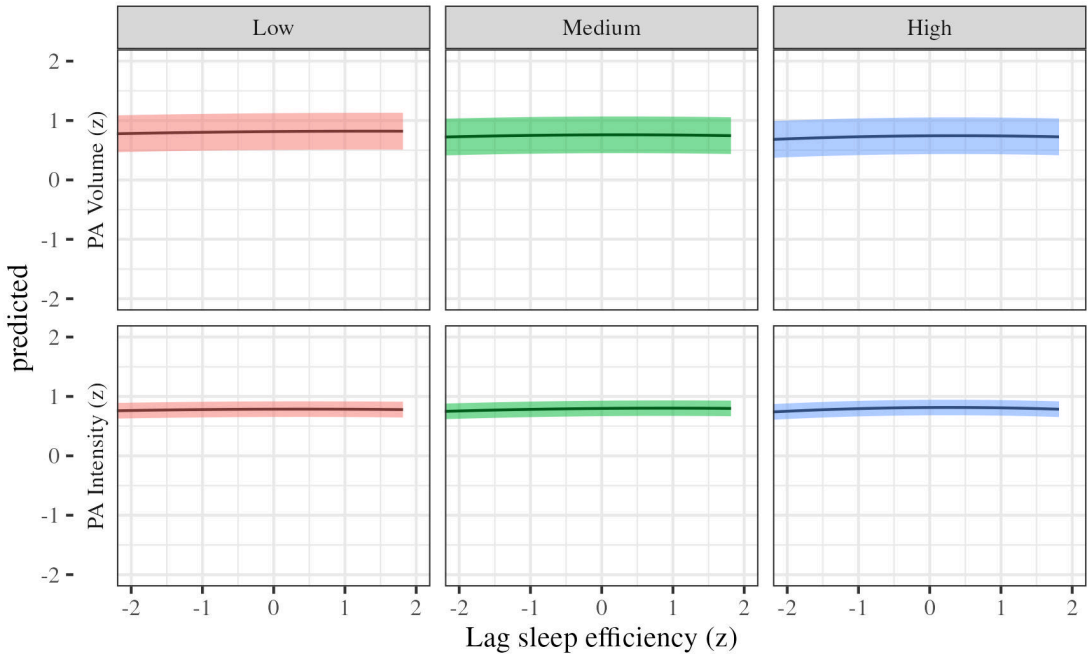

Figure 33: Physical activity by sleep efficiency moderated by SES

The association between sleep onset and next-day physical activity

We estimated the association between sleep onset and next-day physical activity by SES. Results, controlling for age, sex and BMI z-score are presented in Table 15 and Figure 34.

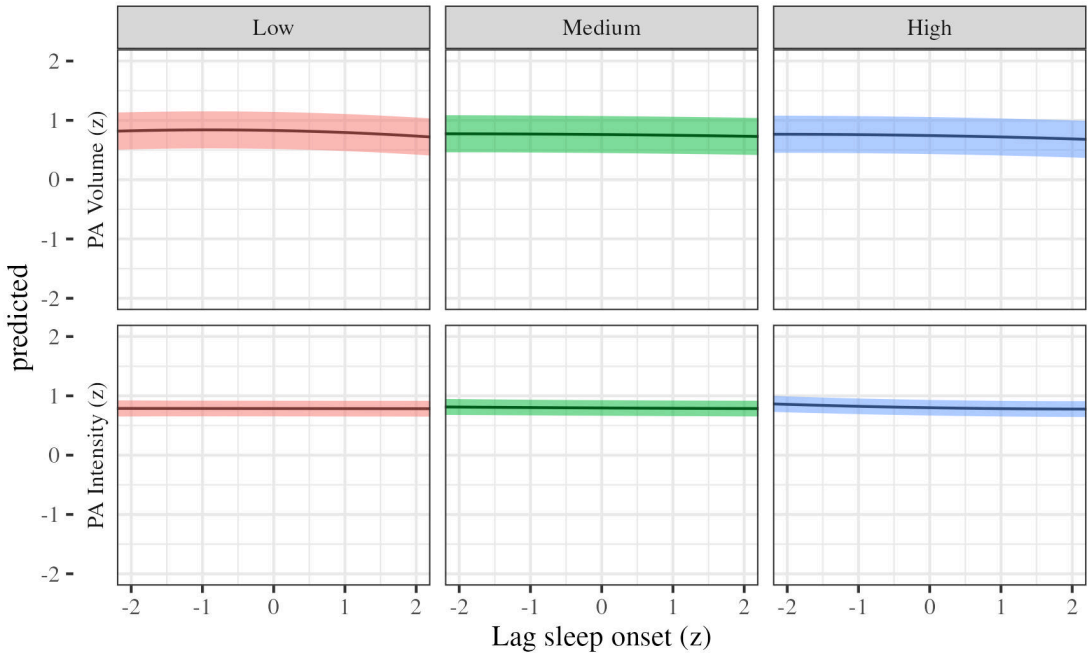

Figure 34: Physical activity by sleep onset moderated by SES

The association between sleep regularity and next-day physical activity

We estimated the association between sleep regularity and next-day physical activity by SES. Results are presented in Table 15 and Figure 35.

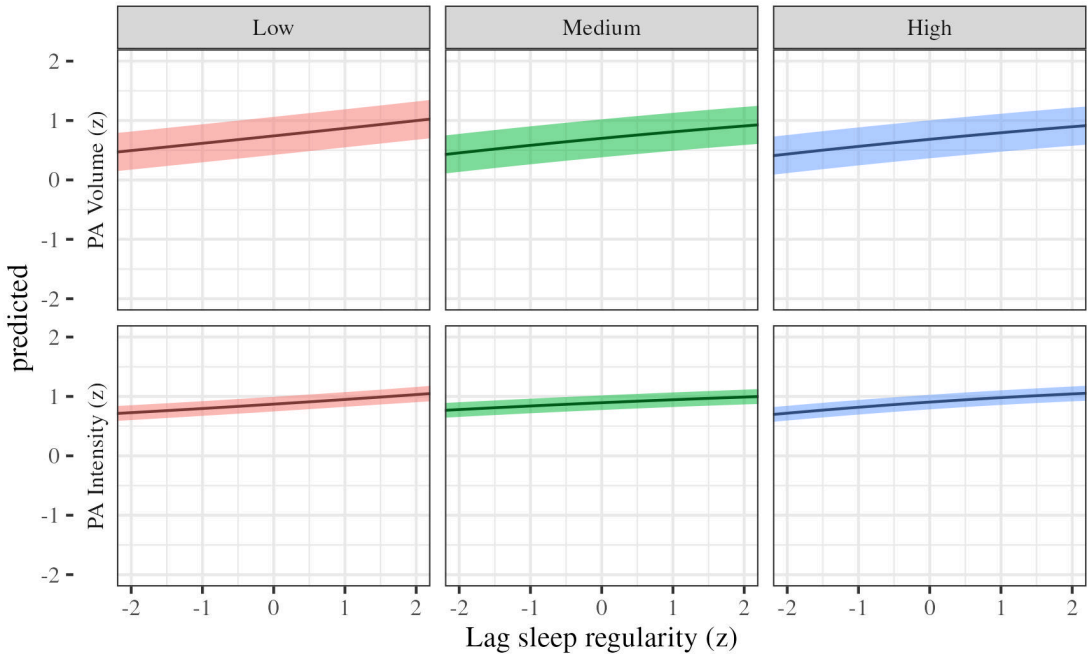

Figure 35: Physical activity by sleep regularity moderated by SES

sex

The association between physical activity volume and sleep by sex

We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by sex, and the results are presented in Table 16 and Figure 36

Table 16: Physical activity predicting sleep controlling for SES, age, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.01 [−0.15, 0.17]	.865	0.01 [−0.13, 0.16]	.857
Physical activity	0.12 [0.11, 0.13]	< .001	0.07 [0.06, 0.08]	< .001
Male	−0.13 [−0.15, −0.12]	< .001	−0.13 [−0.15, −0.11]	< .001
Physical activity²	0.00 [−0.01, 0.00]	.089	0.01 [0.00, 0.02]	< .001
Physical activity × Male	−0.02 [−0.04, −0.01]	.004	0.00 [−0.01, 0.01]	.996
Male × Physical activity²	0.00 [0.00, 0.01]	.444	0.00 [−0.01, 0.01]	.486
Sleep efficiency (z)				
(Intercept)	0.19 [0.03, 0.34]	.020	0.17 [0.00, 0.33]	.044
Physical activity	0.05 [0.04, 0.06]	< .001	−0.01 [−0.02, 0.00]	.058
Male	−0.18 [−0.19, −0.16]	< .001	−0.16 [−0.18, −0.15]	< .001
Physical activity²	−0.02 [−0.02, −0.02]	< .001	0.00 [−0.01, 0.01]	.991
Physical activity × Male	−0.01 [−0.02, 0.01]	.392	0.03 [0.02, 0.04]	< .001
Male × Physical activity²	0.01 [0.00, 0.02]	< .001	0.00 [−0.01, 0.01]	.517
Sleep onset (z)				
(Intercept)	0.12 [−0.16, 0.39]	.397	0.11 [−0.15, 0.36]	.415
Physical activity	−0.12 [−0.13, −0.11]	< .001	−0.09 [−0.10, −0.08]	< .001
Male	0.01 [0.00, 0.02]	.125	0.02 [0.01, 0.03]	.004
Physical activity²	0.00 [0.00, 0.00]	.637	0.00 [−0.01, 0.00]	.054
Physical activity × Male	0.01 [0.00, 0.02]	.026	0.02 [0.01, 0.03]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Male × Physical activity²	0.00 [0.00, 0.01]	.476	−0.01 [−0.01, 0.00]	.013
Sleep regularity (z)				
(Intercept)	0.16 [0.01, 0.30]	.034	0.14 [−0.03, 0.31]	.108
Physical activity	0.33 [0.32, 0.35]	< .001	0.09 [0.08, 0.10]	< .001
Male	−0.18 [−0.19, −0.16]	< .001	−0.15 [−0.17, −0.13]	< .001
Physical activity²	−0.07 [−0.08, −0.07]	< .001	−0.03 [−0.03, −0.02]	< .001
Physical activity × Male	−0.03 [−0.04, −0.01]	.003	0.03 [0.02, 0.04]	< .001
Male × Physical activity²	0.02 [0.01, 0.03]	< .001	0.01 [0.00, 0.02]	.012

Adjusted for SES, age, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction.

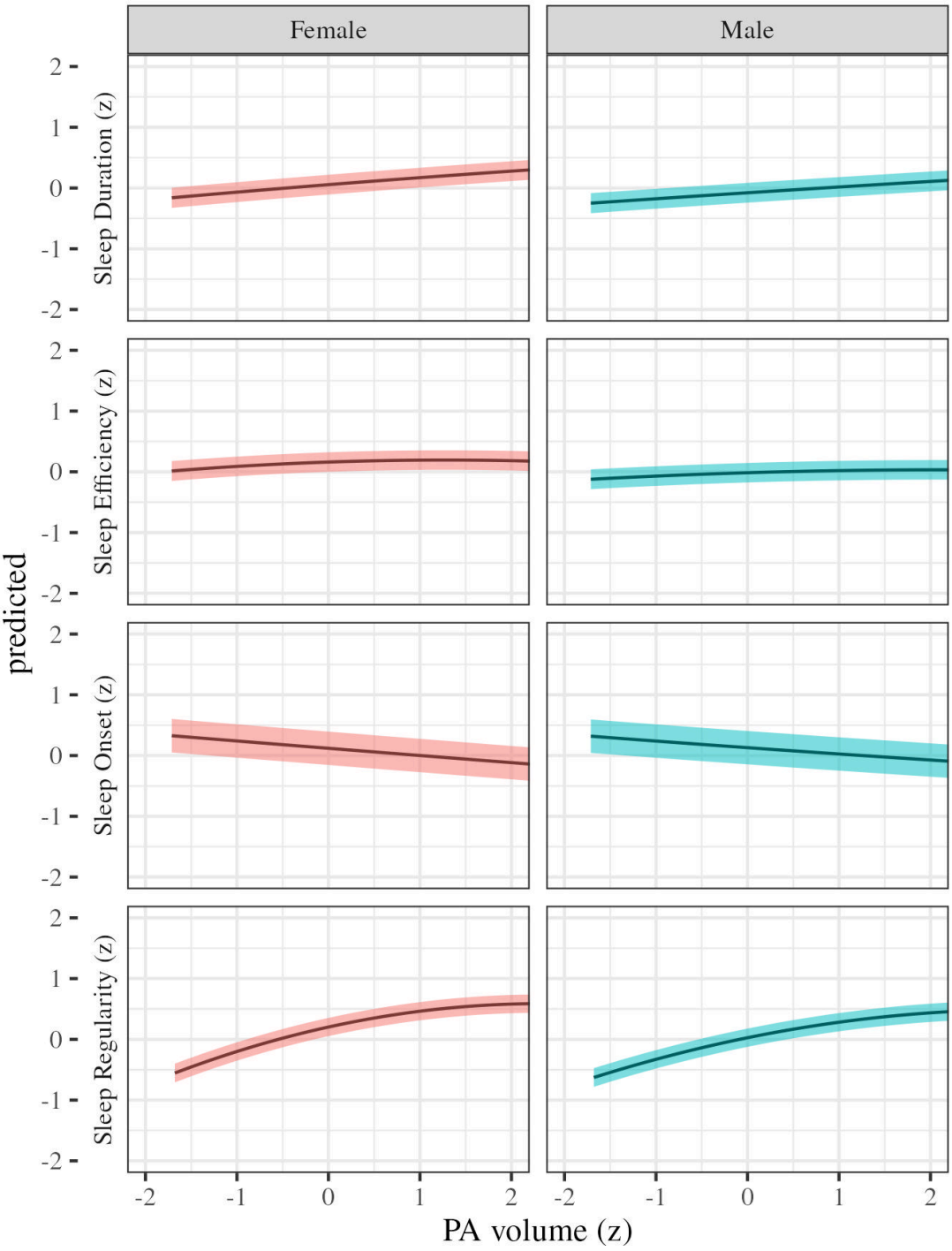

Figure 36: Sleep metrics on Physical activity volume by sex

The association between physical activity intensity and sleep by sex

We estimated how physical activity intensity is associated with sleep across sex. We present the results in Table 16 and Figure 37.

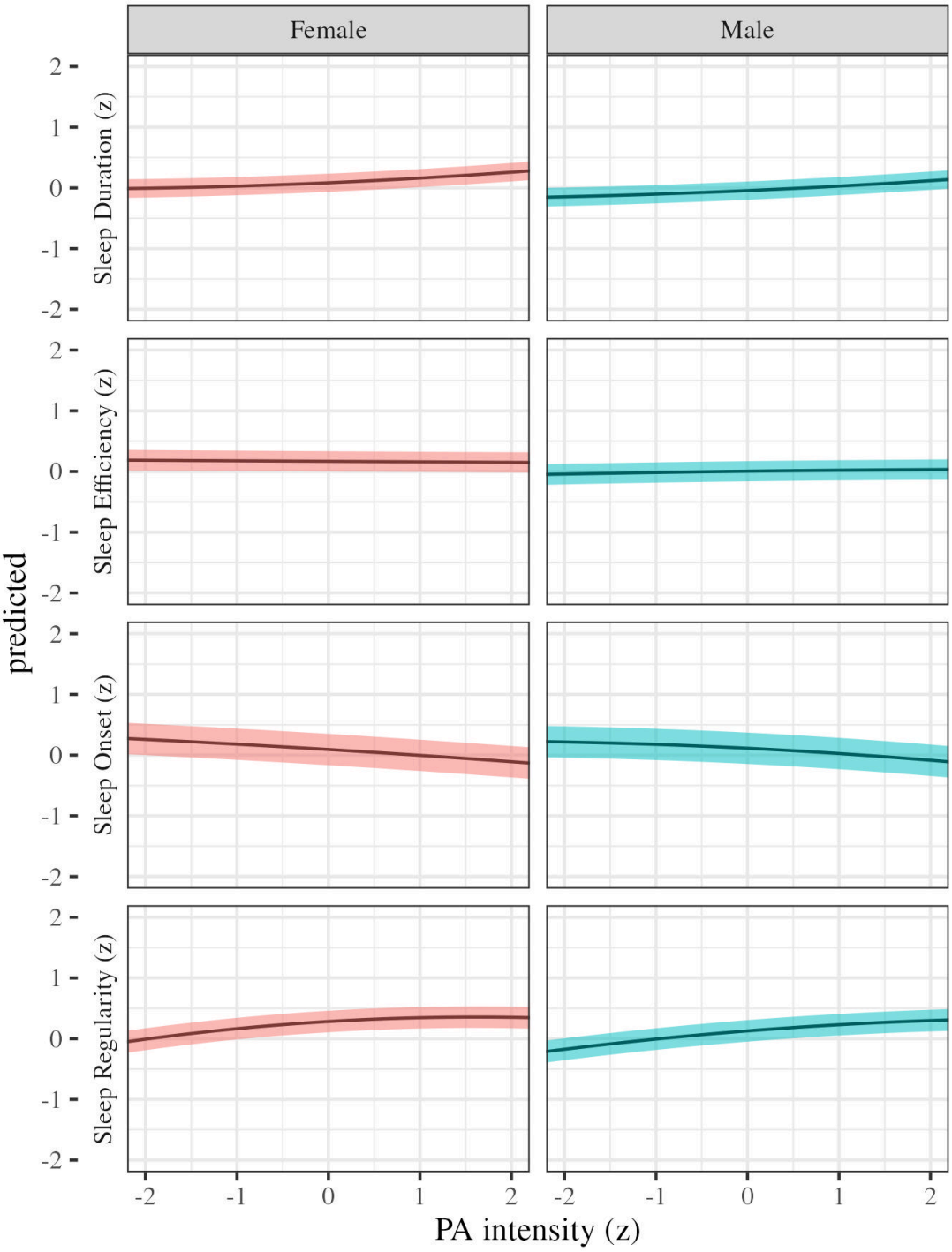

Figure 37: Sleep metrics on Physical activity intensity moderated by sex

The association between sleep duration and next-day physical activity

We estimated the association between sleep duration and next-day physical activity by sex. Results, controlling for SES, age and BMI z-score are presented in Table 17 and Figure 38.

Table 17: Sleep predicting physical activity controlling for SES, age, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.14 [−0.16, 0.44]	.366	−0.07 [−0.19, 0.05]	.253
Sleep duration	0.01 [0.00, 0.01]	.027	0.02 [0.02, 0.03]	< .001
Male	0.16 [0.15, 0.17]	< .001	0.22 [0.21, 0.24]	< .001
Sleep duration²	−0.02 [−0.02, −0.01]	< .001	−0.01 [−0.01, 0.00]	< .001
Sleep duration × Male	0.02 [0.01, 0.02]	< .001	0.00 [0.00, 0.01]	.459
Male × Sleep duration²	0.00 [−0.01, 0.00]	.832	0.00 [−0.01, 0.00]	.782
Sleep efficiency (z)				
(Intercept)	0.12 [−0.18, 0.42]	.419	−0.07 [−0.20, 0.05]	.243
Sleep efficiency	0.02 [0.01, 0.02]	< .001	0.01 [0.00, 0.02]	< .001
Male	0.16 [0.15, 0.17]	< .001	0.22 [0.21, 0.23]	< .001
Sleep efficiency²	−0.01 [−0.01, 0.00]	< .001	−0.01 [−0.01, −0.01]	< .001
Sleep efficiency × Male	−0.02 [−0.03, −0.01]	< .001	−0.01 [−0.02, 0.00]	.068
Male × Sleep efficiency²	0.00 [−0.01, 0.00]	.100	0.00 [0.00, 0.01]	.612
Sleep onset (z)				
(Intercept)	0.13 [−0.18, 0.43]	.410	−0.08 [−0.20, 0.05]	.215
Sleep onset	0.00 [−0.01, 0.01]	.550	−0.01 [−0.02, 0.00]	.006
Male	0.15 [0.14, 0.17]	< .001	0.22 [0.21, 0.23]	< .001
Sleep onset²	0.00 [−0.01, 0.00]	.011	0.00 [0.00, 0.01]	.256
Sleep onset × Male	−0.05 [−0.06, −0.04]	< .001	−0.01 [−0.02, 0.00]	.073

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Male × Sleep onset²	0.00 [0.00, 0.01]	.336	0.00 [0.00, 0.01]	.575
Sleep regularity (z)				
(Intercept)	0.04 [−0.27, 0.34]	.814	−0.18 [−0.30, −0.05]	.007
Sleep regularity	0.11 [0.10, 0.11]	< .001	0.07 [0.06, 0.08]	< .001
Male	0.20 [0.19, 0.22]	< .001	0.24 [0.23, 0.26]	< .001
Sleep regularity²	0.00 [−0.01, 0.00]	.022	0.00 [−0.01, 0.00]	.137
Sleep regularity × Male	0.04 [0.03, 0.05]	< .001	0.00 [−0.01, 0.01]	.501
Male × Sleep regularity²	0.01 [0.00, 0.01]	.015	0.00 [0.00, 0.01]	.584

Adjusted for SES, age, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction.

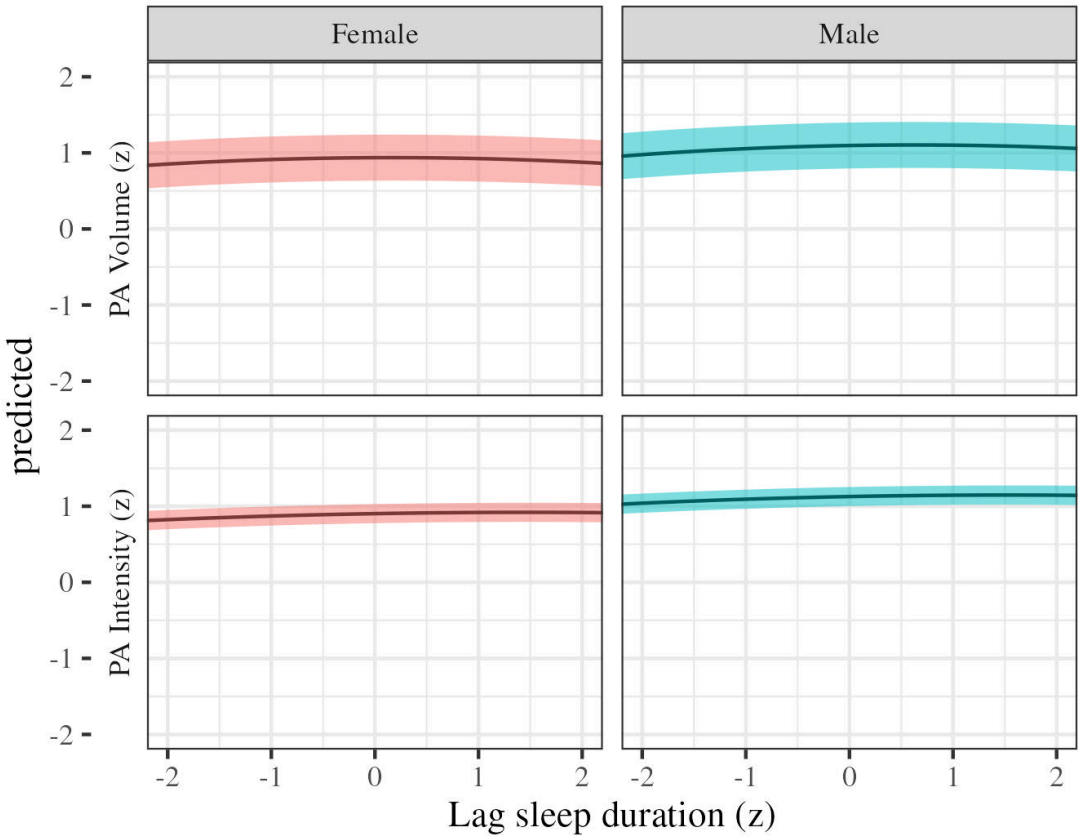

Figure 38: Physical activity by sleep duration moderated by sex

The association between sleep efficiency and next-day physical activity

We estimated the association between sleep efficiency and next-day physical activity by sex. Results, controlling for SES, age and BMI z-score are presented in Table 17 and Figure 39.

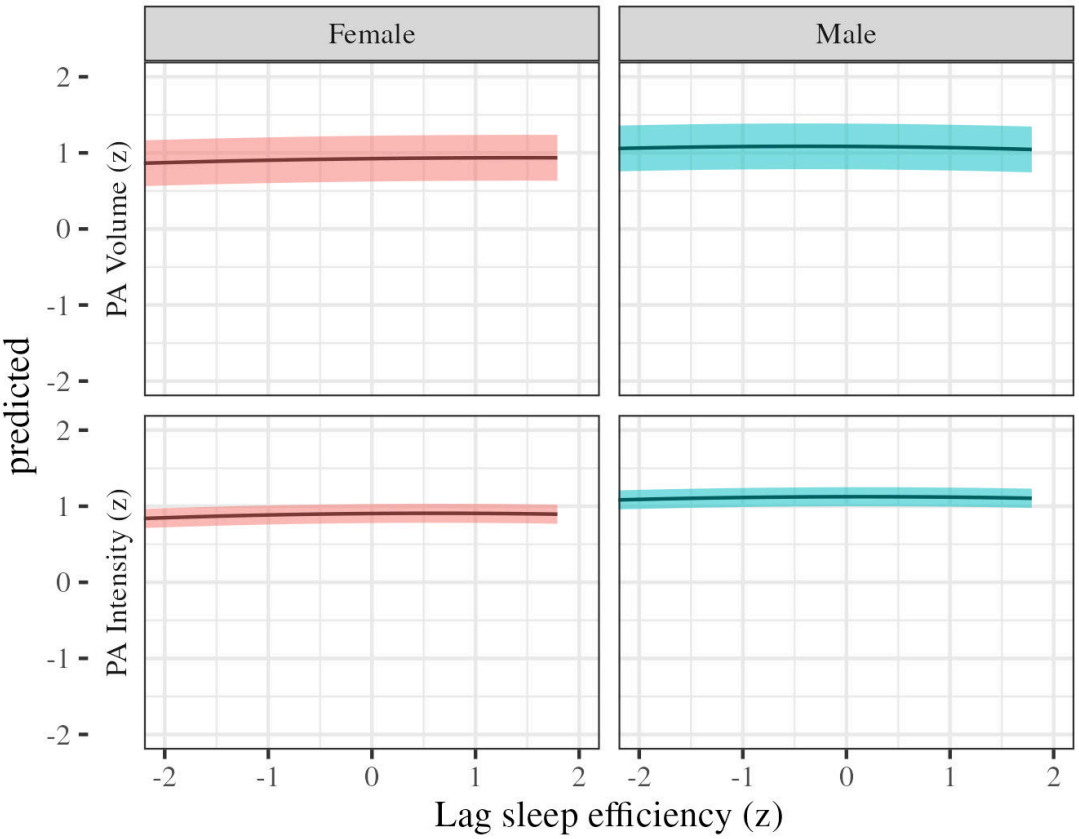

Figure 39: Physical activity by sleep efficiency moderated by sex

The association between sleep onset and next-day physical activity

We estimated the association between sleep onset and next-day physical activity by sex. Results, controlling for SES, age and BMI z-score are presented in Table 17 and Figure 40.

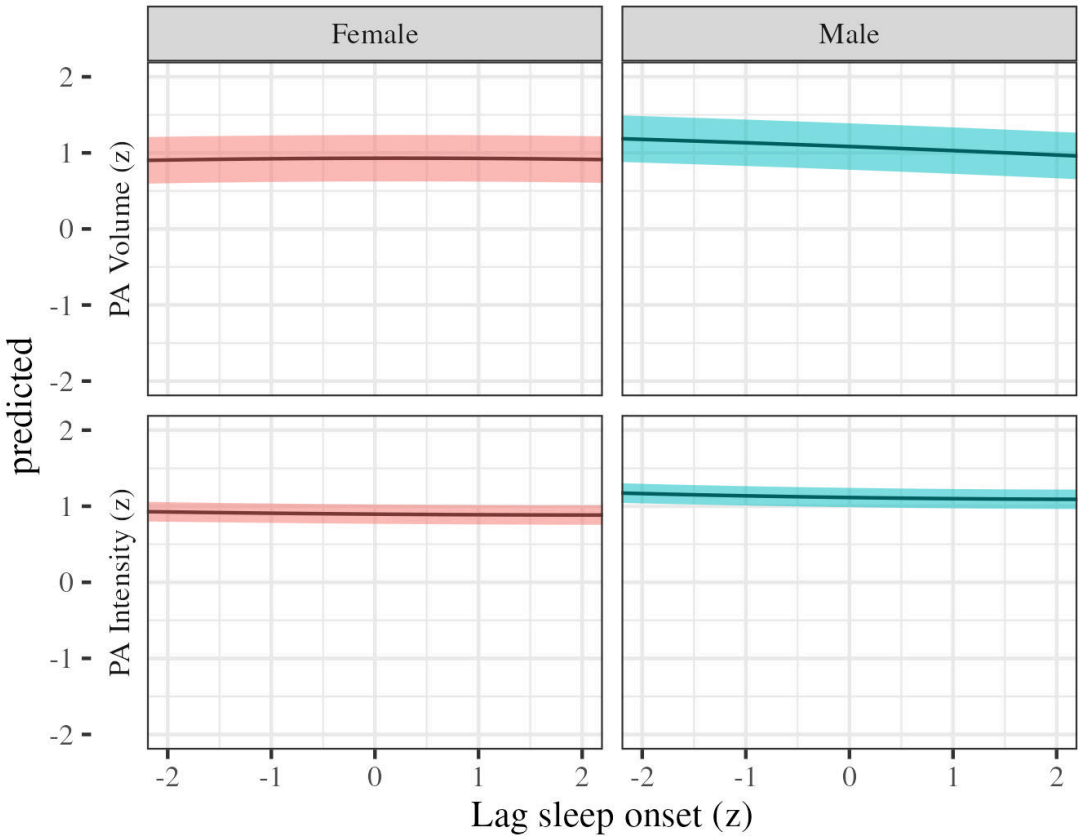

Figure 40: Physical activity by sleep onset moderated by sex

The association between sleep regularity and next-day physical activity

We estimated the association between sleep regularity and next-day physical activity by sex. Results are presented in Table 17 and Figure 41.

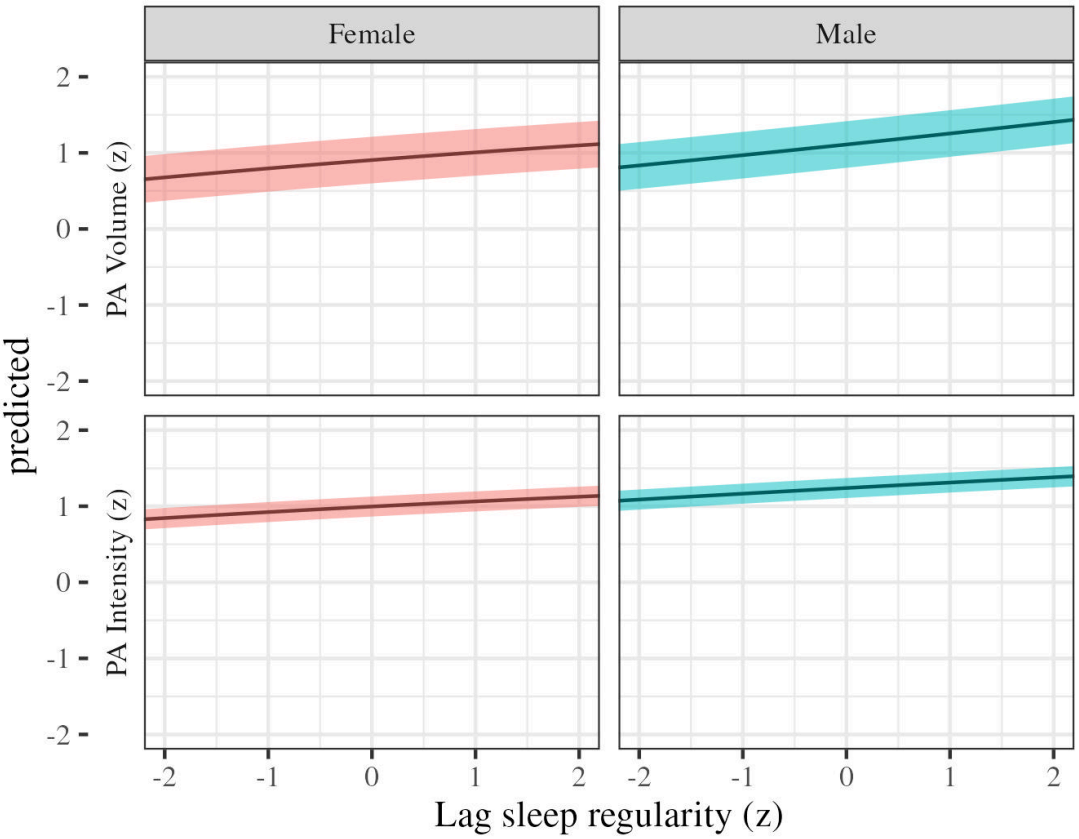

Figure 41: Physical activity by sleep regularity moderated by sex

weekday

The association between physical activity volume and sleep by weekday

We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by weekday, and the results are presented in Table 18 and Figure 42

Table 18: Physical activity predicting sleep controlling for SES, age, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.10 [−0.06, 0.26]	.215	0.11 [−0.03, 0.26]	.133
Physical activity	0.10 [0.09, 0.12]	< .001	0.03 [0.02, 0.04]	< .001
Monday	−0.11 [−0.13, −0.10]	< .001	−0.12 [−0.14, −0.10]	< .001
Saturday	−0.01 [−0.03, 0.00]	.160	−0.03 [−0.05, −0.01]	< .001
Sunday	−0.14 [−0.15, −0.12]	< .001	−0.14 [−0.16, −0.12]	< .001
Thursday	−0.12 [−0.14, −0.11]	< .001	−0.13 [−0.15, −0.11]	< .001
Tuesday	−0.13 [−0.15, −0.12]	< .001	−0.15 [−0.17, −0.13]	< .001
Wednesday	−0.14 [−0.15, −0.12]	< .001	−0.15 [−0.17, −0.13]	< .001
Physical activity²	−0.02 [−0.02, −0.01]	< .001	−0.01 [−0.02, 0.00]	.038
Physical activity × Monday	0.01 [−0.01, 0.03]	.504	0.07 [0.05, 0.08]	< .001
Physical activity × Saturday	0.03 [0.02, 0.05]	< .001	0.00 [−0.01, 0.02]	.753
Physical activity × Sunday	−0.01 [−0.03, 0.01]	.500	0.04 [0.03, 0.05]	< .001
Physical activity × Thursday	−0.01 [−0.03, 0.01]	.144	0.06 [0.05, 0.07]	< .001
Physical activity × Tuesday	0.00 [−0.02, 0.02]	.698	0.08 [0.06, 0.09]	< .001
Physical activity × Wednesday	−0.02 [−0.04, 0.00]	.119	0.06 [0.04, 0.07]	< .001
Monday × Physical activity²	0.02 [0.01, 0.02]	< .001	0.02 [0.00, 0.03]	.007
Saturday × Physical activity²	−0.01 [−0.02, 0.00]	.033	0.01 [−0.01, 0.02]	.337
Sunday × Physical activity²	0.03 [0.02, 0.03]	< .001	0.02 [0.01, 0.03]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Thursday × Physical activity²	0.02 [0.01, 0.03]	< .001	0.02 [0.01, 0.04]	< .001
Tuesday × Physical activity²	0.02 [0.01, 0.03]	< .001	0.03 [0.02, 0.04]	< .001
Wednesday × Physical activity²	0.02 [0.02, 0.03]	< .001	0.03 [0.02, 0.04]	< .001
Sleep efficiency (z)				
(Intercept)	0.15 [0.00, 0.31]	.057	0.15 [−0.02, 0.31]	.078
Physical activity	0.04 [0.03, 0.06]	< .001	−0.02 [−0.03, −0.01]	.002
Monday	0.06 [0.04, 0.07]	< .001	0.07 [0.05, 0.09]	< .001
Saturday	−0.03 [−0.04, −0.01]	< .001	−0.06 [−0.08, −0.05]	< .001
Sunday	0.00 [−0.02, 0.01]	.938	0.00 [−0.02, 0.01]	.733
Thursday	0.05 [0.04, 0.07]	< .001	0.06 [0.05, 0.08]	< .001
Tuesday	0.05 [0.03, 0.06]	< .001	0.06 [0.04, 0.08]	< .001
Wednesday	0.06 [0.04, 0.07]	< .001	0.07 [0.05, 0.09]	< .001
Physical activity²	−0.01 [−0.01, 0.00]	.015	0.00 [0.00, 0.01]	.349
Physical activity × Monday	−0.03 [−0.05, −0.01]	.009	0.03 [0.01, 0.04]	< .001
Physical activity × Saturday	0.06 [0.04, 0.08]	< .001	0.01 [0.00, 0.03]	.028
Physical activity × Sunday	−0.01 [−0.03, 0.01]	.440	0.01 [−0.01, 0.02]	.411
Physical activity × Thursday	−0.03 [−0.05, −0.01]	.002	0.02 [0.00, 0.03]	.018
Physical activity × Tuesday	−0.02 [−0.04, 0.00]	.036	0.02 [0.01, 0.04]	< .001
Physical activity × Wednesday	−0.01 [−0.03, 0.01]	.206	0.02 [0.01, 0.04]	.001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Monday × Physical activity²	0.00 [−0.01, 0.00]	.345	−0.02 [−0.03, −0.01]	.003
Saturday × Physical activity²	−0.02 [−0.03, −0.01]	< .001	0.01 [0.00, 0.02]	.024
Sunday × Physical activity²	0.00 [−0.01, 0.01]	.794	−0.01 [−0.02, 0.01]	.322
Thursday × Physical activity²	0.00 [−0.01, 0.01]	.790	−0.01 [−0.02, 0.00]	.035
Tuesday × Physical activity²	0.00 [−0.01, 0.01]	.542	−0.01 [−0.03, 0.00]	.014
Wednesday × Physical activity²	0.00 [−0.01, 0.01]	.486	−0.02 [−0.03, 0.00]	.008
Sleep onset (z)				
(Intercept)	0.20 [−0.07, 0.47]	.147	0.19 [−0.07, 0.44]	.153
Physical activity	−0.07 [−0.08, −0.06]	< .001	−0.06 [−0.07, −0.05]	< .001
Monday	−0.16 [−0.17, −0.15]	< .001	−0.14 [−0.15, −0.13]	< .001
Saturday	0.09 [0.07, 0.10]	< .001	0.07 [0.05, 0.08]	< .001
Sunday	−0.15 [−0.16, −0.14]	< .001	−0.14 [−0.15, −0.13]	< .001
Thursday	−0.13 [−0.14, −0.12]	< .001	−0.11 [−0.13, −0.10]	< .001
Tuesday	−0.14 [−0.16, −0.13]	< .001	−0.12 [−0.14, −0.11]	< .001
Wednesday	−0.14 [−0.15, −0.13]	< .001	−0.12 [−0.14, −0.11]	< .001
Physical activity²	0.00 [0.00, 0.00]	.890	0.00 [−0.01, 0.00]	.572
Physical activity × Monday	−0.09 [−0.11, −0.08]	< .001	−0.02 [−0.03, −0.01]	< .001
Physical activity × Saturday	0.05 [0.03, 0.06]	< .001	0.01 [0.00, 0.02]	.021
Physical activity × Sunday	−0.06 [−0.07, −0.05]	< .001	−0.02 [−0.03, −0.01]	< .001
Physical activity × Thursday	−0.07 [−0.09, −0.06]	< .001	−0.03 [−0.04, −0.02]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Physical activity × Tuesday	−0.08 [−0.10, −0.07]	< .001	−0.02 [−0.03, −0.01]	< .001
Physical activity × Wednesday	−0.08 [−0.10, −0.07]	< .001	−0.02 [−0.03, −0.01]	< .001
Monday × Physical activity²	0.01 [0.00, 0.02]	< .001	0.00 [−0.01, 0.00]	.272
Saturday × Physical activity²	−0.02 [−0.02, −0.01]	< .001	0.00 [−0.01, 0.01]	.811
Sunday × Physical activity²	0.00 [0.00, 0.01]	.163	0.00 [0.00, 0.01]	.430
Thursday × Physical activity²	0.01 [0.00, 0.01]	.021	−0.01 [−0.02, 0.00]	.046
Tuesday × Physical activity²	0.01 [0.00, 0.01]	.003	−0.01 [−0.02, 0.00]	.009
Wednesday × Physical activity²	0.01 [0.00, 0.01]	.003	−0.01 [−0.02, 0.00]	.062
Sleep regularity (z)				
(Intercept)	−0.07 [−0.22, 0.08]	.341	−0.07 [−0.25, 0.11]	.453
Physical activity	0.28 [0.26, 0.29]	< .001	0.00 [−0.02, 0.01]	.484
Monday	0.43 [0.41, 0.45]	< .001	0.46 [0.44, 0.48]	< .001
Saturday	−0.03 [−0.05, −0.01]	< .001	−0.06 [−0.08, −0.04]	< .001
Sunday	−0.01 [−0.03, 0.00]	.152	−0.07 [−0.09, −0.05]	< .001
Thursday	0.34 [0.32, 0.35]	< .001	0.36 [0.35, 0.38]	< .001
Tuesday	0.42 [0.41, 0.44]	< .001	0.45 [0.43, 0.47]	< .001
Wednesday	0.41 [0.39, 0.42]	< .001	0.45 [0.43, 0.47]	< .001
Physical activity²	−0.04 [−0.05, −0.03]	< .001	0.00 [−0.01, 0.01]	.770
Physical activity × Monday	−0.03 [−0.05, 0.00]	.018	0.11 [0.10, 0.13]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Physical activity × Saturday	−0.01 [−0.03, 0.01]	.463	0.02 [0.01, 0.04]	.001
Physical activity × Sunday	0.03 [0.01, 0.05]	.004	0.06 [0.05, 0.08]	< .001
Physical activity × Thursday	−0.01 [−0.03, 0.01]	.213	0.08 [0.06, 0.09]	< .001
Physical activity × Tuesday	−0.02 [−0.04, 0.00]	.092	0.11 [0.10, 0.13]	< .001
Physical activity × Wednesday	−0.03 [−0.05, −0.01]	.014	0.08 [0.07, 0.10]	< .001
Monday × Physical activity²	−0.02 [−0.03, −0.01]	< .001	−0.07 [−0.08, −0.05]	< .001
Saturday × Physical activity²	0.00 [−0.01, 0.01]	.820	0.00 [−0.01, 0.01]	.922
Sunday × Physical activity²	−0.01 [−0.02, 0.00]	.055	0.01 [−0.01, 0.02]	.286
Thursday × Physical activity²	−0.01 [−0.02, 0.00]	.040	−0.05 [−0.06, −0.03]	< .001
Tuesday × Physical activity²	−0.01 [−0.02, 0.00]	.086	−0.05 [−0.06, −0.04]	< .001
Wednesday × Physical activity²	−0.01 [−0.02, 0.00]	.004	−0.07 [−0.08, −0.05]	< .001

Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction.

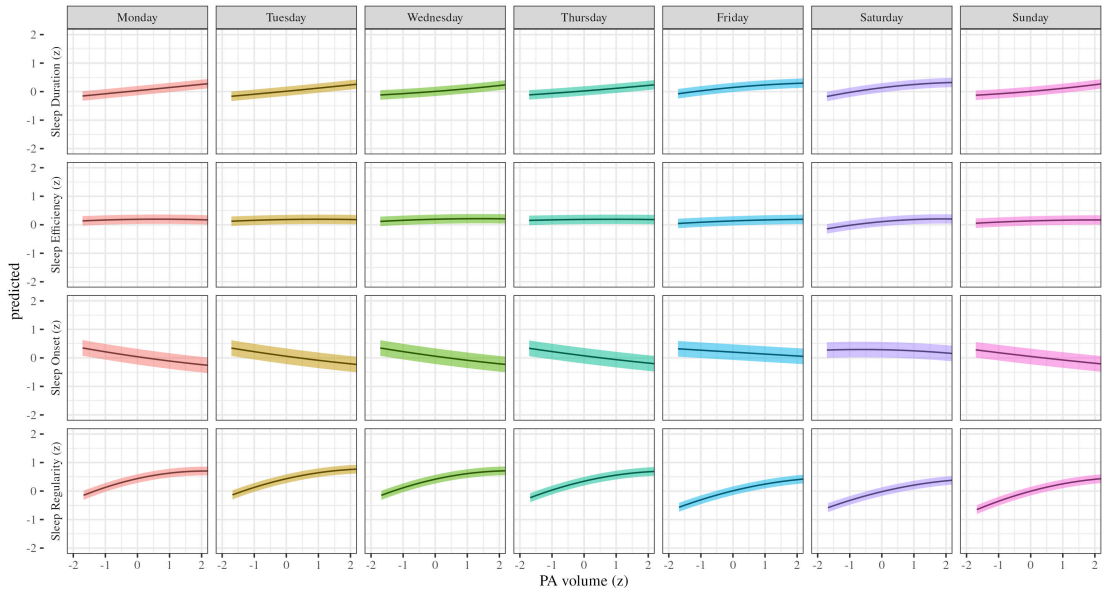

Figure 42: Sleep metrics on Physical activity volume by weekday

The association between physical activity intensity and sleep by weekday

We estimated how physical activity intensity is associated with sleep across weekday. We present the results in Table 18 and Figure 43.

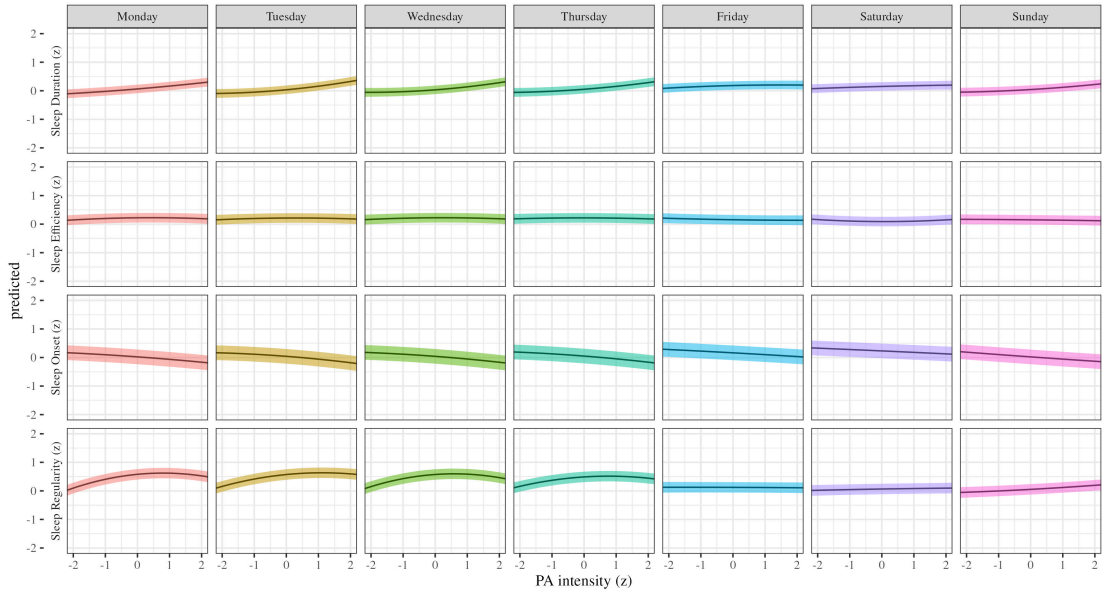

Figure 43: Sleep metrics on Physical activity intensity moderated by weekday

The association between sleep duration and next-day physical activity

We estimated the association between sleep duration and next-day physical activity by weekday. Results, controlling for SES, age, sex and BMI z-score are presented in Table 19 and Figure 44.

Table 19: Sleep predicting physical activity controlling for SES, age, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.21 [−0.10, 0.51]	.182	−0.04 [−0.16, 0.09]	.545
Sleep duration	0.03 [0.02, 0.04]	< .001	0.06 [0.05, 0.07]	< .001
Monday	−0.07 [−0.08, −0.06]	< .001	−0.01 [−0.03, 0.00]	.032
Saturday	−0.12 [−0.13, −0.11]	< .001	−0.12 [−0.13, −0.11]	< .001
Sunday	−0.18 [−0.20, −0.17]	< .001	−0.13 [−0.14, −0.11]	< .001
Thursday	0.00 [−0.01, 0.01]	.883	0.03 [0.01, 0.04]	< .001
Tuesday	−0.02 [−0.03, −0.01]	.003	0.02 [0.01, 0.03]	.004
Wednesday	−0.04 [−0.05, −0.03]	< .001	0.02 [0.01, 0.04]	< .001
Sleep duration²	−0.02 [−0.03, −0.02]	< .001	0.00 [−0.01, 0.00]	.667
Sleep duration × Monday	−0.01 [−0.02, 0.00]	.023	−0.02 [−0.03, −0.01]	< .001
Sleep duration × Saturday	−0.05 [−0.06, −0.04]	< .001	−0.08 [−0.09, −0.06]	< .001
Sleep duration × Sunday	−0.03 [−0.04, −0.02]	< .001	−0.07 [−0.08, −0.06]	< .001
Sleep duration × Thursday	0.01 [0.00, 0.02]	.033	0.01 [0.00, 0.02]	.170
Sleep duration × Tuesday	0.01 [−0.01, 0.02]	.291	−0.01 [−0.02, 0.00]	.102
Sleep duration × Wednesday	0.01 [−0.01, 0.02]	.383	0.01 [0.00, 0.02]	.143
Monday × Sleep duration²	0.01 [0.00, 0.02]	.003	0.00 [−0.01, 0.01]	.855
Saturday × Sleep duration²	0.00 [0.00, 0.01]	.360	−0.01 [−0.01, 0.00]	.043
Sunday × Sleep duration²	0.01 [0.00, 0.01]	.032	0.00 [−0.01, 0.00]	.369
Thursday × Sleep duration²	0.00 [−0.01, 0.01]	.940	−0.01 [−0.02, 0.00]	.028

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Tuesday × Sleep duration²	0.00 [−0.01, 0.01]	.969	−0.01 [−0.02, 0.00]	.032
Wednesday × Sleep duration²	0.01 [0.00, 0.02]	.070	0.00 [−0.01, 0.00]	.354
Sleep efficiency (z)				
(Intercept)	0.19 [−0.11, 0.49]	.223	−0.03 [−0.16, 0.09]	.585
Sleep efficiency	−0.01 [−0.02, 0.00]	.070	−0.02 [−0.03, −0.01]	< .001
Monday	−0.07 [−0.08, −0.05]	< .001	−0.02 [−0.03, 0.00]	.015
Saturday	−0.12 [−0.13, −0.11]	< .001	−0.14 [−0.15, −0.12]	< .001
Sunday	−0.17 [−0.18, −0.16]	< .001	−0.13 [−0.14, −0.12]	< .001
Thursday	0.01 [−0.01, 0.02]	.439	0.02 [0.00, 0.03]	.014
Tuesday	−0.01 [−0.02, 0.00]	.126	0.02 [0.00, 0.03]	.021
Wednesday	−0.03 [−0.04, −0.02]	< .001	0.02 [0.01, 0.03]	.004
Sleep efficiency²	0.00 [−0.01, 0.00]	.130	−0.01 [−0.01, 0.00]	.003
Sleep efficiency × Monday	0.02 [0.01, 0.03]	.001	0.02 [0.01, 0.03]	< .001
Sleep efficiency × Saturday	0.02 [0.01, 0.04]	< .001	0.06 [0.05, 0.07]	< .001
Sleep efficiency × Sunday	0.00 [−0.01, 0.02]	.555	0.06 [0.05, 0.08]	< .001
Sleep efficiency × Thursday	0.00 [−0.01, 0.02]	.657	0.00 [−0.01, 0.01]	.963
Sleep efficiency × Tuesday	0.00 [−0.01, 0.02]	.556	0.01 [0.00, 0.03]	.032
Sleep efficiency × Wednesday	0.01 [0.00, 0.02]	.185	0.01 [0.00, 0.02]	.165
Monday × Sleep efficiency²	0.00 [0.00, 0.01]	.244	0.00 [−0.01, 0.01]	.762
Saturday × Sleep efficiency²	0.00 [0.00, 0.01]	.203	0.01 [0.00, 0.01]	.025
Sunday × Sleep efficiency²	−0.01 [−0.01, 0.00]	.017	0.00 [0.00, 0.01]	.570

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Thursday × Sleep efficiency²	0.00 [−0.01, 0.00]	.201	0.00 [−0.01, 0.01]	.840
Tuesday × Sleep efficiency²	−0.01 [−0.02, 0.00]	.013	0.00 [−0.01, 0.00]	.407
Wednesday × Sleep efficiency²	0.00 [−0.01, 0.01]	.948	0.00 [−0.01, 0.01]	.567
Sleep onset (z)				
(Intercept)	0.18 [−0.12, 0.49]	.232	−0.06 [−0.18, 0.07]	.380
Sleep onset	−0.04 [−0.05, −0.03]	< .001	−0.05 [−0.06, −0.04]	< .001
Monday	−0.06 [−0.08, −0.05]	< .001	−0.01 [−0.03, 0.00]	.049
Saturday	−0.11 [−0.12, −0.10]	< .001	−0.10 [−0.11, −0.09]	< .001
Sunday	−0.17 [−0.18, −0.16]	< .001	−0.11 [−0.12, −0.10]	< .001
Thursday	0.00 [−0.02, 0.01]	.734	0.01 [0.00, 0.03]	.045
Tuesday	−0.01 [−0.03, 0.00]	.025	0.02 [0.01, 0.03]	.002
Wednesday	−0.03 [−0.05, −0.02]	< .001	0.02 [0.01, 0.03]	.003
Sleep onset²	0.00 [−0.01, 0.00]	.316	0.01 [0.00, 0.02]	< .001
Sleep onset × Monday	0.04 [0.02, 0.05]	< .001	0.04 [0.03, 0.05]	< .001
Sleep onset × Saturday	0.07 [0.06, 0.08]	< .001	0.12 [0.11, 0.13]	< .001
Sleep onset × Sunday	0.05 [0.04, 0.06]	< .001	0.13 [0.12, 0.14]	< .001
Sleep onset × Thursday	0.00 [−0.02, 0.01]	.453	−0.01 [−0.02, 0.00]	.228
Sleep onset × Tuesday	0.01 [0.00, 0.02]	.106	0.02 [0.01, 0.03]	< .001
Sleep onset × Wednesday	0.01 [0.00, 0.02]	.117	0.00 [−0.01, 0.02]	.417
Monday × Sleep onset²	0.01 [0.00, 0.01]	.163	0.00 [−0.01, 0.01]	.865

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Saturday × Sleep onset²	−0.01 [−0.01, 0.00]	.051	−0.03 [−0.04, −0.02]	< .001
Sunday × Sleep onset²	0.00 [−0.01, 0.00]	.261	−0.03 [−0.03, −0.02]	< .001
Thursday × Sleep onset²	0.00 [−0.01, 0.01]	.639	0.00 [−0.01, 0.01]	.674
Tuesday × Sleep onset²	−0.01 [−0.01, 0.00]	.180	−0.01 [−0.02, 0.00]	.014
Wednesday × Sleep onset²	0.00 [−0.01, 0.01]	.701	0.00 [−0.01, 0.01]	.640
Sleep regularity (z)				
(Intercept)	0.08 [−0.23, 0.39]	.603	−0.13 [−0.26, 0.00]	.049
Sleep regularity	0.13 [0.12, 0.14]	< .001	0.04 [0.03, 0.05]	< .001
Monday	−0.03 [−0.05, −0.02]	< .001	0.00 [−0.01, 0.01]	.986
Saturday	−0.08 [−0.09, −0.06]	< .001	−0.13 [−0.15, −0.12]	< .001
Sunday	−0.11 [−0.13, −0.10]	< .001	−0.14 [−0.16, −0.13]	< .001
Thursday	−0.01 [−0.02, 0.01]	.352	0.03 [0.01, 0.04]	< .001
Tuesday	−0.03 [−0.05, −0.02]	< .001	0.00 [−0.01, 0.02]	.607
Wednesday	−0.04 [−0.05, −0.02]	< .001	0.02 [0.00, 0.04]	.014
Sleep regularity²	0.00 [−0.01, 0.01]	.685	−0.01 [−0.01, 0.00]	.099
Sleep regularity × Monday	−0.04 [−0.05, −0.03]	< .001	−0.02 [−0.03, 0.00]	.013
Sleep regularity × Saturday	−0.01 [−0.03, 0.00]	.039	0.02 [0.01, 0.03]	.005
Sleep regularity × Sunday	−0.03 [−0.04, −0.01]	< .001	0.01 [0.00, 0.03]	.049
Sleep regularity × Thursday	−0.02 [−0.03, 0.00]	.014	0.00 [−0.02, 0.01]	.681
Sleep regularity × Tuesday	−0.01 [−0.02, 0.01]	.248	0.01 [0.00, 0.03]	.051
Sleep regularity × Wednesday	−0.02 [−0.04, −0.01]	.001	0.01 [0.00, 0.02]	.209

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Monday × Sleep regularity²	0.01 [0.00, 0.01]	.257	0.00 [−0.01, 0.01]	.972
Saturday × Sleep regularity²	−0.01 [−0.02, 0.00]	.003	0.00 [−0.01, 0.01]	.697
Sunday × Sleep regularity²	−0.02 [−0.03, −0.01]	< .001	0.00 [−0.01, 0.01]	.978
Thursday × Sleep regularity²	0.00 [−0.01, 0.01]	.610	−0.01 [−0.02, 0.00]	.185
Tuesday × Sleep regularity²	0.01 [0.00, 0.02]	.188	0.01 [0.00, 0.02]	.290
Wednesday × Sleep regularity²	−0.01 [−0.02, 0.00]	.238	0.00 [−0.01, 0.01]	.913

Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction.

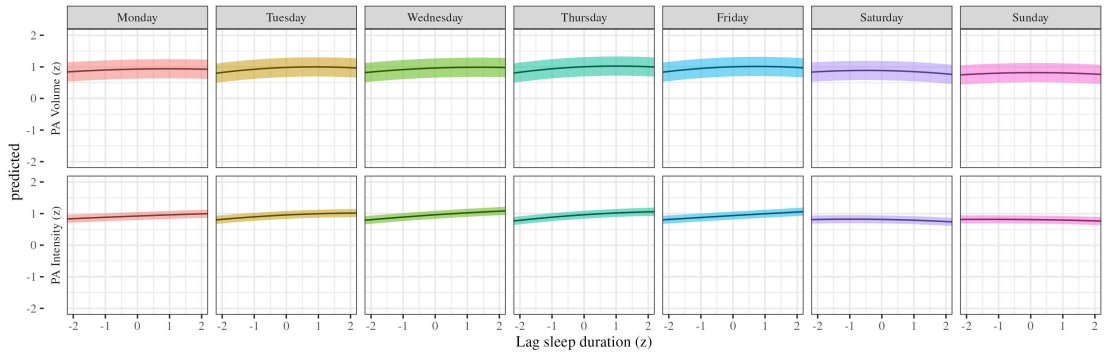

Figure 44: Physical activity by sleep duration moderated by weekday

The association between sleep efficiency and next-day physical activity

We estimated the association between sleep efficiency and next-day physical activity by weekday. Results, controlling for SES, age, sex and BMI z-score are presented in Table 19 and Figure 45.

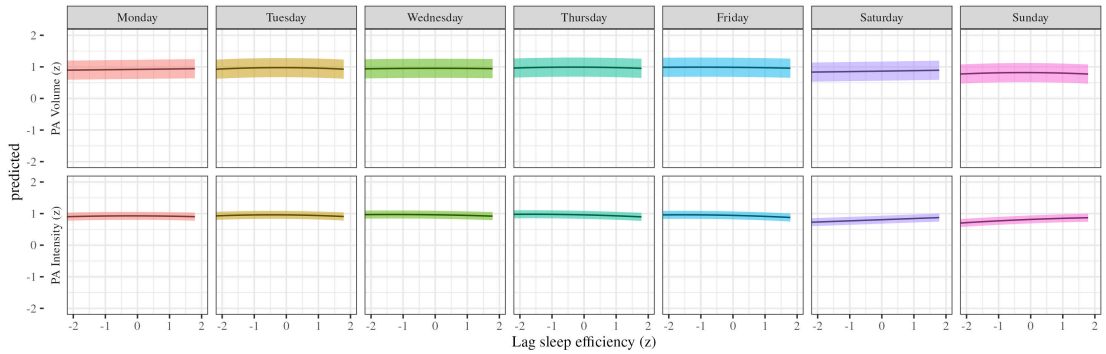

Figure 45: Physical activity by sleep efficiency moderated by weekday

The association between sleep onset and next-day physical activity

We estimated the association between sleep onset and next-day physical activity by weekday. Results, controlling for SES, age, sex and BMI z-score are presented in Table 19 and Figure 46.

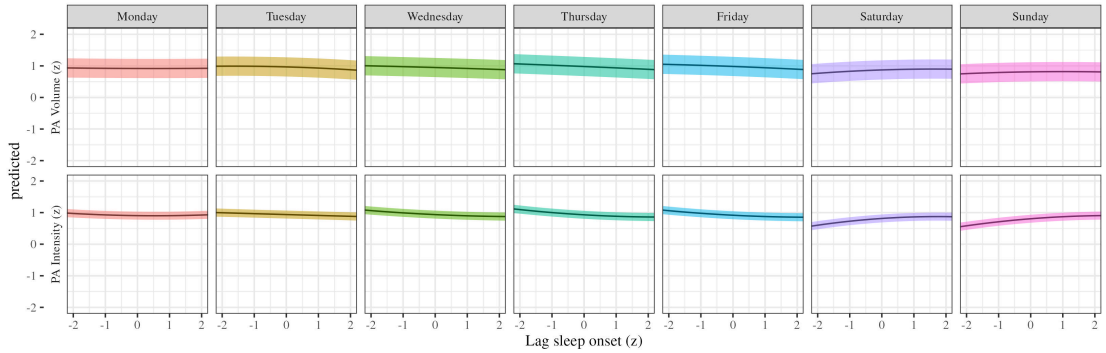

Figure 46: Physical activity by sleep onset moderated by weekday

The association between sleep regularity and next-day physical activity

We estimated the association between sleep regularity and next-day physical activity by weekday. Results are presented in Table 19 and Figure 47.

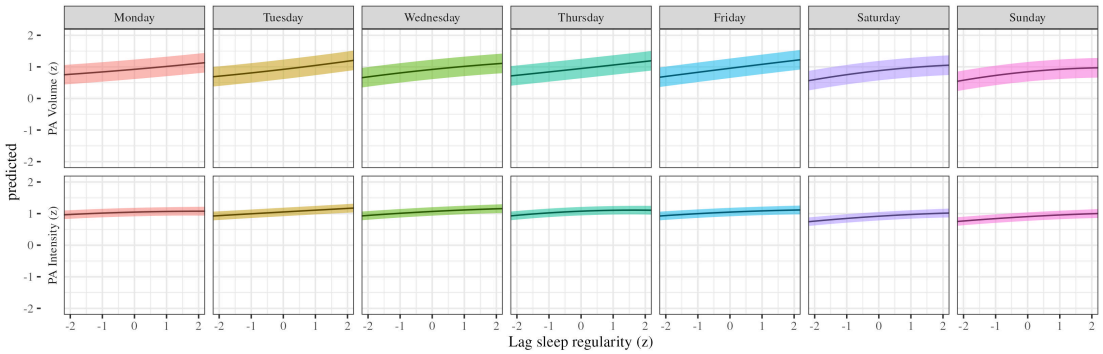

Figure 47: Physical activity by sleep regularity moderated by weekday

season

The association between physical activity volume and sleep by season

We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by season, and the results are presented in Table 20 and Figure 48

Table 20: Physical activity predicting sleep controlling for SES, age, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.02 [−0.14, 0.18]	.768	0.02 [−0.13, 0.17]	.792
Physical activity	0.12 [0.10, 0.13]	< .001	0.06 [0.05, 0.07]	< .001
Spring	−0.02 [−0.05, 0.00]	.026	−0.01 [−0.03, 0.01]	.289
Summer	−0.10 [−0.12, −0.08]	< .001	−0.08 [−0.11, −0.06]	< .001
Winter	0.08 [0.06, 0.10]	< .001	0.07 [0.05, 0.09]	< .001
Physical activity²	0.00 [0.00, 0.01]	.781	0.02 [0.01, 0.02]	< .001
Physical activity × Spring	−0.02 [−0.04, 0.00]	.020	0.00 [−0.02, 0.01]	.638
Physical activity × Summer	−0.02 [−0.04, 0.01]	.132	0.00 [−0.02, 0.02]	.870
Physical activity × Winter	0.00 [−0.02, 0.02]	.754	0.01 [0.00, 0.03]	.065
Spring × Physical activity²	0.00 [−0.01, 0.00]	.284	−0.01 [−0.02, 0.00]	.014
Summer × Physical activity²	0.00 [−0.01, 0.01]	.894	−0.01 [−0.02, 0.00]	.162
Winter × Physical activity²	−0.01 [−0.02, 0.00]	.004	−0.01 [−0.02, 0.00]	.099
Sleep efficiency (z)				
(Intercept)	0.17 [0.02, 0.33]	.029	0.16 [0.00, 0.32]	.056
Physical activity	0.06 [0.04, 0.07]	< .001	0.01 [0.00, 0.02]	.018
Spring	0.00 [−0.02, 0.03]	.712	0.01 [−0.02, 0.03]	.488
Summer	−0.02 [−0.04, 0.00]	.080	−0.01 [−0.03, 0.02]	.555
Winter	0.04 [0.01, 0.06]	.001	0.04 [0.02, 0.07]	< .001
Physical activity²	−0.02 [−0.02, −0.01]	< .001	0.00 [−0.01, 0.01]	.694

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Physical activity × Spring	−0.03 [−0.05, −0.01]	.009	−0.02 [−0.04, −0.01]	.006
Physical activity × Summer	−0.02 [−0.04, 0.01]	.156	−0.01 [−0.03, 0.01]	.211
Physical activity × Winter	0.00 [−0.02, 0.02]	.995	0.00 [−0.02, 0.01]	.907
Spring × Physical activity²	0.01 [0.00, 0.01]	.118	0.00 [−0.01, 0.01]	.837
Summer × Physical activity²	0.01 [0.00, 0.01]	.123	−0.01 [−0.02, 0.01]	.398
Winter × Physical activity²	0.00 [0.00, 0.01]	.569	−0.01 [−0.02, 0.00]	.122
Sleep onset (z)				
(Intercept)	0.10 [−0.17, 0.37]	.473	0.09 [−0.16, 0.35]	.471
Physical activity	−0.12 [−0.13, −0.11]	< .001	−0.08 [−0.09, −0.07]	< .001
Spring	0.02 [0.00, 0.04]	.052	0.02 [0.00, 0.04]	.064
Summer	0.07 [0.05, 0.09]	< .001	0.06 [0.04, 0.08]	< .001
Winter	−0.01 [−0.03, 0.01]	.210	0.00 [−0.01, 0.02]	.768
Physical activity²	−0.01 [−0.01, 0.00]	.001	−0.01 [−0.02, −0.01]	< .001
Physical activity × Spring	0.01 [−0.01, 0.02]	.499	0.00 [−0.01, 0.01]	.997
Physical activity × Summer	0.04 [0.02, 0.06]	< .001	0.02 [0.01, 0.04]	< .001
Physical activity × Winter	0.00 [−0.02, 0.01]	.915	−0.01 [−0.02, 0.00]	.100
Spring × Physical activity²	0.02 [0.01, 0.02]	< .001	0.01 [0.00, 0.02]	.007
Summer × Physical activity²	0.01 [0.00, 0.01]	.012	0.02 [0.01, 0.03]	< .001
Winter × Physical activity²	0.01 [0.01, 0.02]	< .001	0.00 [−0.01, 0.01]	.679
Sleep regularity (z)				
(Intercept)	0.18 [0.03, 0.32]	.016	0.19 [0.01, 0.37]	.034

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Physical activity	0.28 [0.27, 0.30]	< .001	0.12 [0.11, 0.13]	< .001
Spring	−0.08 [−0.10, −0.05]	< .001	−0.12 [−0.15, −0.09]	< .001
Summer	0.03 [0.00, 0.06]	.024	0.01 [−0.02, 0.04]	.586
Winter	−0.03 [−0.05, 0.00]	.029	−0.04 [−0.07, −0.02]	< .001
Physical activity²	−0.05 [−0.05, −0.04]	< .001	−0.04 [−0.04, −0.03]	< .001
Physical activity × Spring	0.05 [0.03, 0.07]	< .001	−0.04 [−0.06, −0.02]	< .001
Physical activity × Summer	0.07 [0.04, 0.09]	< .001	−0.01 [−0.03, 0.01]	.359
Physical activity × Winter	0.04 [0.01, 0.06]	.001	0.00 [−0.02, 0.01]	.614
Spring × Physical activity²	−0.02 [−0.03, −0.01]	< .001	0.03 [0.01, 0.04]	< .001
Summer × Physical activity²	−0.01 [−0.02, 0.00]	.008	0.03 [0.02, 0.04]	< .001
Winter × Physical activity²	−0.01 [−0.02, 0.00]	.030	0.00 [−0.01, 0.01]	.861

Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction.

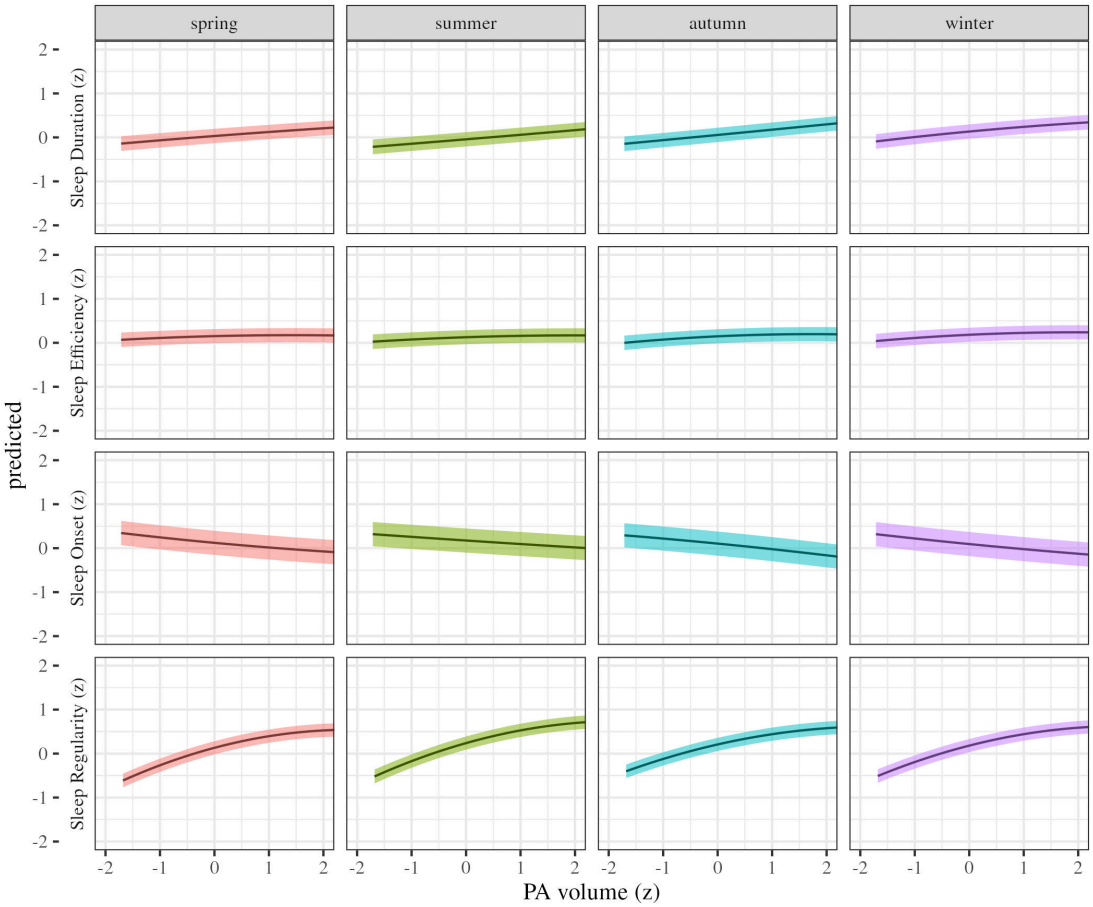

Figure 48: Sleep metrics on Physical activity volume by season

The association between physical activity intensity and sleep by season

We estimated how physical activity intensity is associated with sleep across season. We present the results in Table 20 and Figure 49.

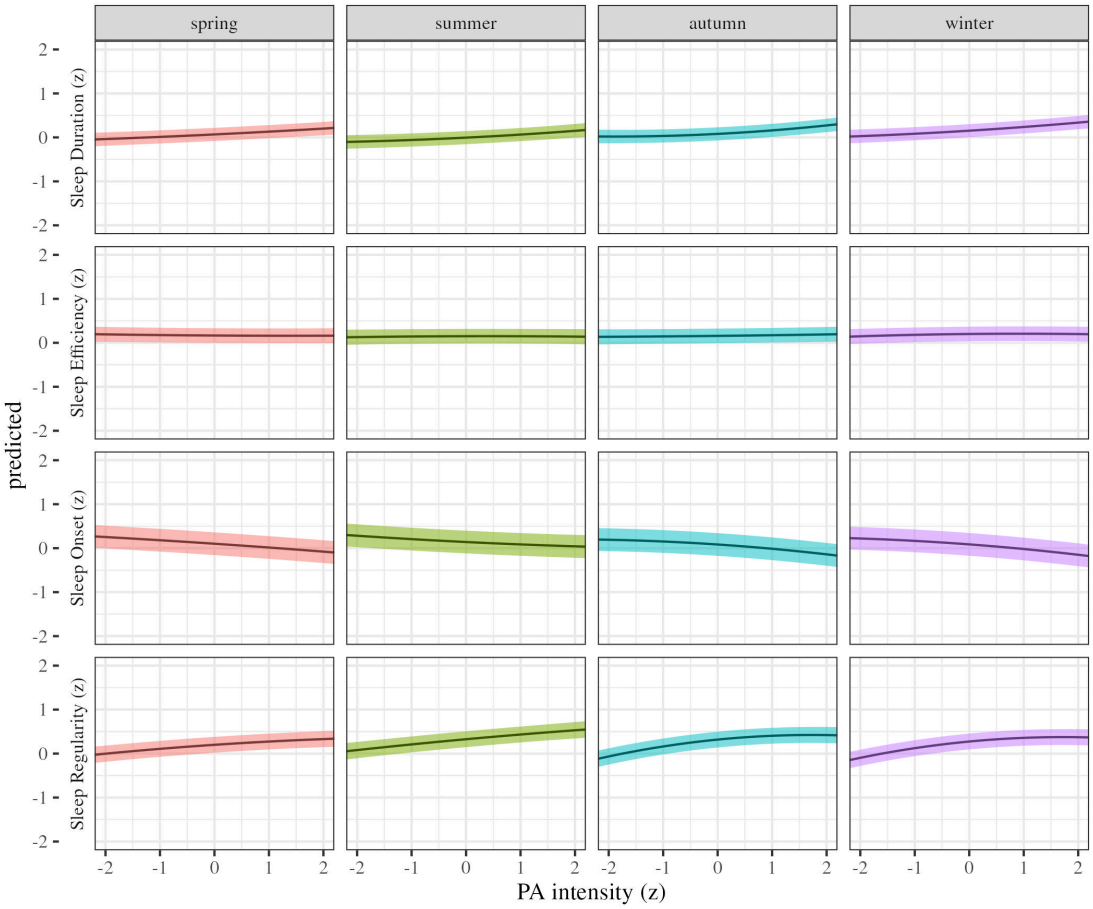

Figure 49: Sleep metrics on Physical activity intensity moderated by season

The association between sleep duration and next-day physical activity

We estimated the association between sleep duration and next-day physical activity by season. Results, controlling for SES, age, sex and BMI z-score are presented in Table 21 and Figure 50.

Table 21: Sleep predicting physical activity controlling for SES, age, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.14 [−0.16, 0.44]	.368	−0.06 [−0.18, 0.07]	.385
Sleep duration	0.04 [0.03, 0.05]	< .001	0.04 [0.03, 0.05]	< .001
Spring	0.00 [−0.01, 0.02]	.627	−0.01 [−0.03, 0.01]	.174
Summer	0.02 [0.00, 0.04]	.035	−0.04 [−0.05, −0.02]	< .001
Winter	−0.04 [−0.06, −0.02]	< .001	−0.03 [−0.05, −0.02]	< .001
Sleep duration²	−0.02 [−0.02, −0.02]	< .001	−0.01 [−0.01, 0.00]	.004
Sleep duration × Spring	−0.05 [−0.06, −0.04]	< .001	−0.03 [−0.04, −0.02]	< .001
Sleep duration × Summer	−0.05 [−0.06, −0.03]	< .001	−0.03 [−0.04, −0.02]	< .001
Sleep duration × Winter	0.00 [−0.01, 0.01]	.408	0.00 [−0.01, 0.01]	.543
Spring × Sleep duration²	0.00 [−0.01, 0.01]	.843	0.00 [−0.01, 0.00]	.165
Summer × Sleep duration²	0.00 [−0.01, 0.01]	.718	−0.01 [−0.02, 0.00]	.021
Winter × Sleep duration²	0.00 [0.00, 0.01]	.456	0.00 [0.00, 0.01]	.583
Sleep efficiency (z)				
(Intercept)	0.13 [−0.17, 0.43]	.391	−0.05 [−0.17, 0.08]	.441
Sleep efficiency	0.01 [0.00, 0.02]	.113	0.01 [0.00, 0.02]	.018
Spring	0.00 [−0.02, 0.02]	.853	−0.02 [−0.04, −0.01]	.008
Summer	0.02 [0.00, 0.04]	.101	−0.05 [−0.07, −0.03]	< .001
Winter	−0.05 [−0.06, −0.03]	< .001	−0.04 [−0.05, −0.02]	< .001
Sleep efficiency²	−0.01 [−0.01, −0.01]	< .001	−0.01 [−0.02, −0.01]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep efficiency × Spring	0.00 [−0.02, 0.01]	.591	−0.01 [−0.02, 0.00]	.245
Sleep efficiency × Summer	−0.01 [−0.02, 0.01]	.445	0.00 [−0.02, 0.01]	.701
Sleep efficiency × Winter	0.00 [−0.01, 0.01]	.709	−0.01 [−0.02, 0.01]	.272
Spring × Sleep efficiency²	0.01 [0.00, 0.01]	.050	0.01 [0.00, 0.01]	.059
Summer × Sleep efficiency²	0.00 [0.00, 0.01]	.172	0.01 [0.00, 0.01]	.067
Winter × Sleep efficiency²	0.01 [0.00, 0.01]	.037	0.00 [0.00, 0.01]	.093
Sleep onset (z)				
(Intercept)	0.13 [−0.17, 0.44]	.396	−0.06 [−0.19, 0.07]	.352
Sleep onset	−0.02 [−0.02, −0.01]	.001	−0.01 [−0.02, −0.01]	.002
Spring	0.01 [−0.01, 0.03]	.366	−0.02 [−0.03, 0.00]	.102
Summer	0.01 [−0.01, 0.03]	.223	−0.05 [−0.06, −0.03]	< .001
Winter	−0.04 [−0.06, −0.02]	< .001	−0.03 [−0.05, −0.01]	< .001
Sleep onset²	0.00 [−0.01, 0.00]	.255	0.00 [0.00, 0.01]	.173
Sleep onset × Spring	−0.02 [−0.03, −0.01]	< .001	0.00 [−0.01, 0.01]	.844
Sleep onset × Summer	0.03 [0.01, 0.04]	< .001	0.02 [0.01, 0.04]	.002
Sleep onset × Winter	−0.03 [−0.05, −0.02]	< .001	−0.01 [−0.02, 0.00]	.104
Spring × Sleep onset²	0.00 [−0.01, 0.00]	.220	0.00 [−0.01, 0.00]	.561
Summer × Sleep onset²	0.00 [−0.01, 0.01]	.485	0.00 [−0.01, 0.01]	.616
Winter × Sleep onset²	0.00 [−0.01, 0.01]	.972	0.00 [−0.01, 0.01]	.896
Sleep regularity (z)				
(Intercept)	0.03 [−0.28, 0.34]	.850	−0.16 [−0.29, −0.03]	.014

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep regularity	0.12 [0.11, 0.13]	< .001	0.08 [0.08, 0.09]	< .001
Spring	0.01 [−0.01, 0.03]	.193	−0.01 [−0.03, 0.01]	.416
Summer	0.03 [0.00, 0.05]	.025	−0.04 [−0.06, −0.02]	< .001
Winter	−0.05 [−0.07, −0.03]	< .001	−0.03 [−0.04, −0.01]	.006
Sleep regularity²	−0.01 [−0.01, 0.00]	.040	0.00 [−0.01, 0.00]	.632
Sleep regularity × Spring	0.01 [−0.01, 0.02]	.356	−0.02 [−0.04, −0.01]	< .001
Sleep regularity × Summer	−0.02 [−0.03, 0.00]	.008	−0.03 [−0.05, −0.02]	< .001
Sleep regularity × Winter	0.02 [0.01, 0.03]	.005	0.00 [−0.01, 0.01]	.703
Spring × Sleep regularity²	0.01 [0.00, 0.01]	.128	0.00 [−0.01, 0.00]	.421
Summer × Sleep regularity²	0.00 [−0.01, 0.01]	.899	−0.01 [−0.02, 0.00]	.026
Winter × Sleep regularity²	0.01 [0.00, 0.02]	.012	0.01 [0.00, 0.01]	.085

Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction.

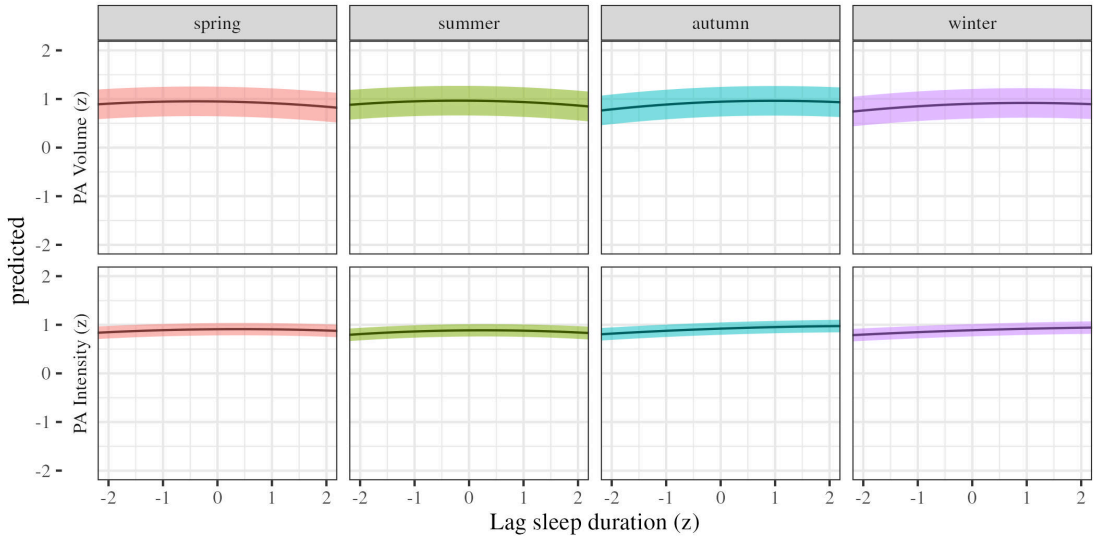

Figure 50: Physical activity by sleep duration moderated by season

The association between sleep efficiency and next-day physical activity

We estimated the association between sleep efficiency and next-day physical activity by season. Results, controlling for SES, age, sex and BMI z-score are presented in Table 21 and Figure 51.

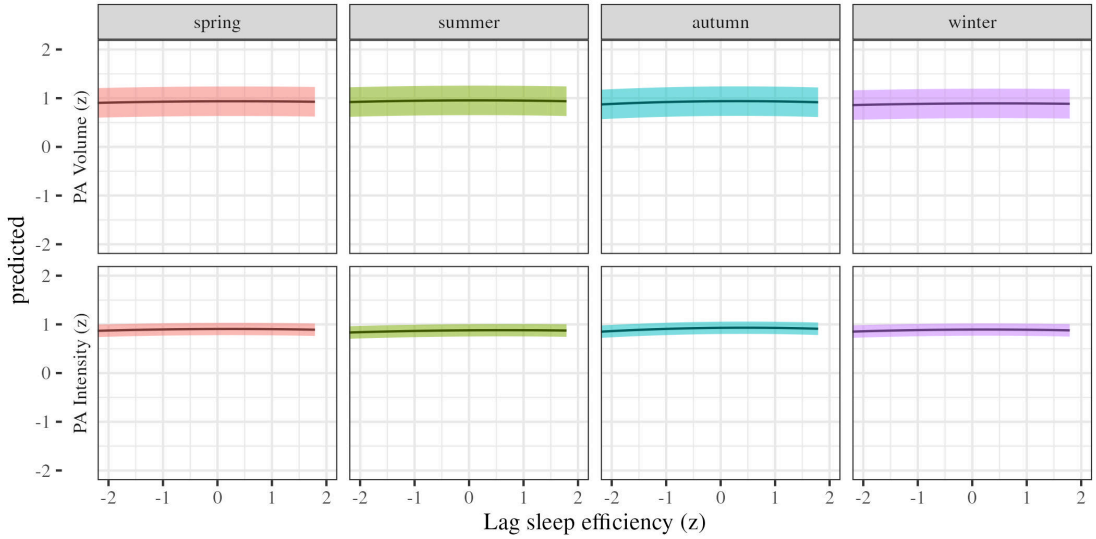

Figure 51: Physical activity by sleep efficiency moderated by season

The association between sleep onset and next-day physical activity

We estimated the association between sleep onset and next-day physical activity by season. Results, controlling for SES, age, sex and BMI z-score are presented in Table 21 and Figure 52.

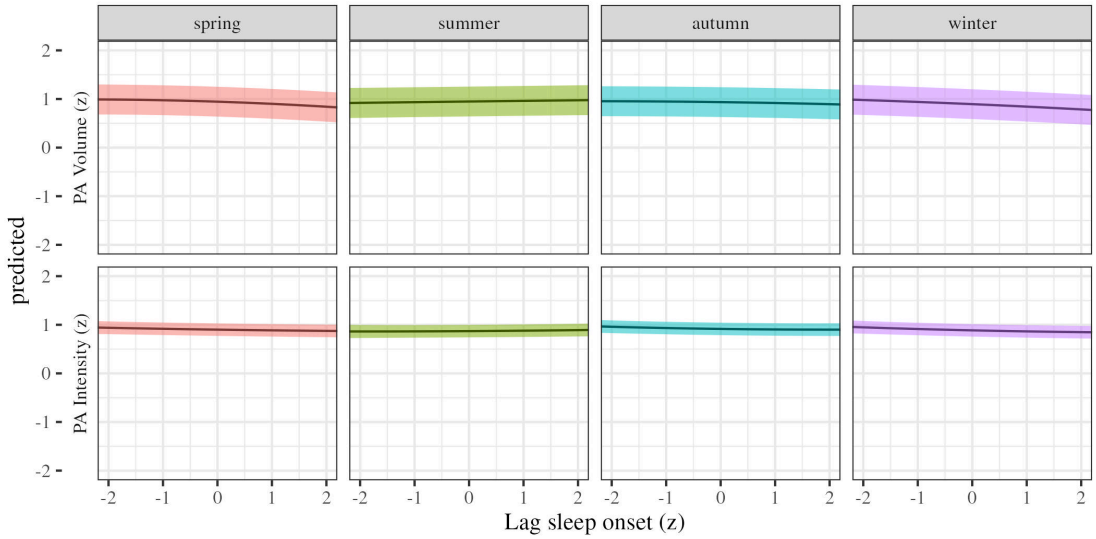

Figure 52: Physical activity by sleep onset moderated by season

The association between sleep regularity and next-day physical activity

We estimated the association between sleep regularity and next-day physical activity by season. Results are presented in Table 21 and Figure 53.

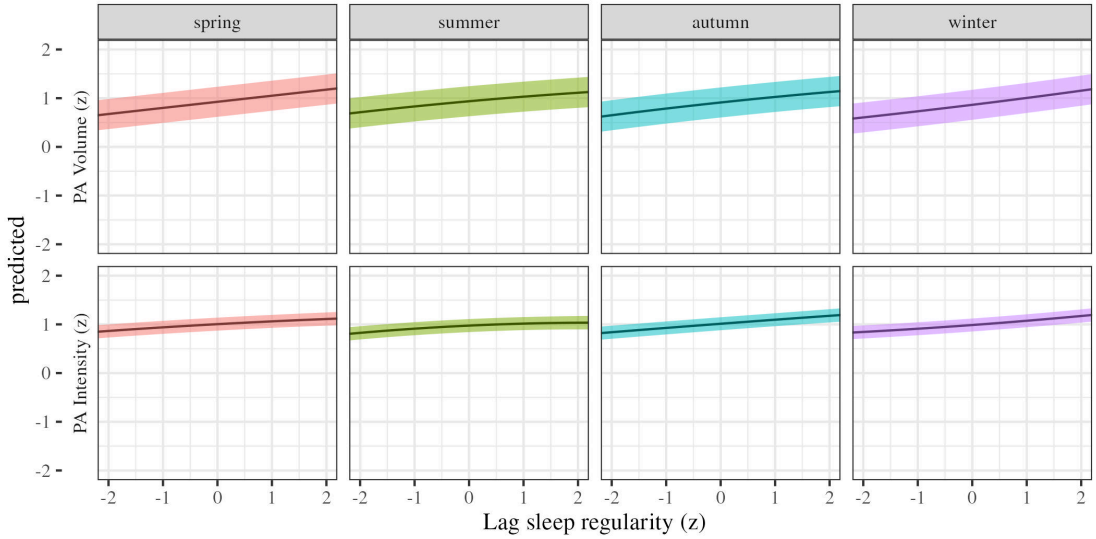

Figure 53: Physical activity by sleep regularity moderated by season

region

The association between physical activity volume and sleep by region

We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by region, and the results are presented in Table 22 and Figure 54

Table 22: Physical activity predicting sleep controlling for SES, age, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.19 [0.02, 0.36]	.031	0.23 [0.06, 0.39]	.006
Physical activity	0.03 [0.00, 0.05]	.037	0.02 [−0.01, 0.04]	.252
Europe	−0.18 [−0.24, −0.12]	< .001	−0.20 [−0.26, −0.14]	< .001
Africa	−0.19 [−0.26, −0.12]	< .001	−0.22 [−0.29, −0.15]	< .001
Asia	−0.44 [−0.52, −0.36]	< .001	−0.50 [−0.57, −0.43]	< .001
North America	−0.07 [−0.14, 0.00]	.047	−0.09 [−0.16, −0.03]	.005
South America	−0.36 [−0.43, −0.30]	< .001	−0.46 [−0.52, −0.40]	< .001
Physical activity²	0.00 [0.00, 0.01]	.105	0.04 [0.02, 0.05]	< .001
Physical activity × Europe	0.06 [0.03, 0.09]	< .001	0.04 [0.01, 0.07]	.008
Physical activity × Africa	0.06 [0.00, 0.11]	.039	0.01 [−0.04, 0.07]	.599
Physical activity × Asia	0.07 [−0.02, 0.16]	.135	0.06 [0.01, 0.11]	.023
Physical activity × North America	0.07 [0.02, 0.12]	.007	0.02 [−0.02, 0.07]	.317
Physical activity × South America	0.16 [0.13, 0.19]	< .001	0.06 [0.02, 0.09]	.001
Europe × Physical activity²	−0.01 [−0.02, 0.00]	.020	−0.04 [−0.05, −0.02]	< .001
Africa × Physical activity²	−0.04 [−0.09, 0.02]	.189	−0.02 [−0.05, 0.02]	.355
Asia × Physical activity²	−0.06 [−0.14, 0.01]	.102	−0.04 [−0.09, 0.00]	.060
North America × Physical activity²	0.00 [−0.02, 0.02]	.856	−0.02 [−0.06, 0.01]	.151

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
South America × Physical activity²	−0.02 [−0.02, −0.01]	< .001	0.00 [−0.02, 0.02]	.797
Sleep efficiency (z)				
(Intercept)	0.29 [0.12, 0.46]	< .001	0.33 [0.16, 0.50]	< .001
Physical activity	−0.01 [−0.04, 0.01]	.336	0.00 [−0.03, 0.03]	.937
Europe	−0.15 [−0.21, −0.08]	< .001	−0.18 [−0.24, −0.11]	< .001
Africa	0.02 [−0.05, 0.09]	.624	−0.03 [−0.11, 0.04]	.337
Asia	−0.20 [−0.29, −0.12]	< .001	−0.34 [−0.41, −0.26]	< .001
North America	0.03 [−0.04, 0.11]	.361	−0.09 [−0.16, −0.02]	.012
South America	−0.21 [−0.27, −0.14]	< .001	−0.30 [−0.37, −0.24]	< .001
Physical activity²	0.00 [−0.01, 0.01]	.981	−0.01 [−0.03, 0.00]	.054
Physical activity × Europe	0.03 [0.00, 0.05]	.060	0.00 [−0.03, 0.03]	.946
Physical activity × Africa	0.05 [−0.01, 0.10]	.097	−0.01 [−0.06, 0.04]	.723
Physical activity × Asia	0.01 [−0.08, 0.10]	.866	0.07 [0.01, 0.12]	.013
Physical activity × North America	0.15 [0.10, 0.20]	< .001	0.06 [0.02, 0.11]	.010
Physical activity × South America	0.14 [0.11, 0.17]	< .001	0.07 [0.04, 0.11]	< .001
Europe × Physical activity²	−0.01 [−0.02, 0.00]	.014	0.02 [0.00, 0.03]	.019
Africa × Physical activity²	−0.06 [−0.11, −0.01]	.030	0.02 [−0.02, 0.05]	.358
Asia × Physical activity²	−0.21 [−0.28, −0.13]	< .001	−0.04 [−0.09, 0.00]	.061

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
North America × Physical activity²	−0.04 [−0.06, −0.02]	< .001	−0.02 [−0.05, 0.02]	.316
South America × Physical activity²	−0.03 [−0.04, −0.02]	< .001	−0.03 [−0.06, −0.01]	.002
Sleep onset (z)				
(Intercept)	0.06 [−0.22, 0.34]	.675	0.07 [−0.19, 0.33]	.575
Physical activity	−0.10 [−0.11, −0.08]	< .001	−0.06 [−0.08, −0.04]	< .001
Europe	0.07 [0.02, 0.12]	.010	0.05 [0.00, 0.10]	.067
Africa	0.08 [0.02, 0.14]	.007	0.00 [−0.05, 0.06]	.882
Asia	0.23 [0.17, 0.30]	< .001	0.16 [0.10, 0.22]	< .001
North America	0.07 [0.01, 0.13]	.021	−0.04 [−0.10, 0.01]	.150
South America	0.04 [−0.01, 0.10]	.105	0.08 [0.02, 0.13]	.005
Physical activity²	0.00 [0.00, 0.01]	.162	−0.05 [−0.06, −0.04]	< .001
Physical activity × Europe	−0.02 [−0.04, 0.00]	.084	0.00 [−0.03, 0.02]	.697
Physical activity × Africa	0.07 [0.02, 0.11]	.002	0.02 [−0.02, 0.05]	.455
Physical activity × Asia	0.02 [−0.05, 0.09]	.573	0.01 [−0.03, 0.05]	.565
Physical activity × North America	0.11 [0.07, 0.15]	< .001	0.06 [0.03, 0.10]	< .001
Physical activity × South America	−0.06 [−0.09, −0.04]	< .001	−0.02 [−0.04, 0.01]	.250
Europe × Physical activity²	0.01 [0.01, 0.02]	< .001	0.05 [0.04, 0.07]	< .001
Africa × Physical activity²	−0.02 [−0.07, 0.02]	.249	0.04 [0.01, 0.07]	.002

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Asia × Physical activity²	−0.06 [−0.12, 0.00]	.041	0.02 [−0.01, 0.05]	.249
North America × Physical activity²	−0.02 [−0.04, −0.01]	.006	0.02 [−0.01, 0.04]	.236
South America × Physical activity²	0.00 [−0.01, 0.00]	.467	−0.02 [−0.03, 0.00]	.039
Sleep regularity (z)				
(Intercept)	0.00 [−0.15, 0.15]	.966	0.05 [−0.13, 0.23]	.561
Physical activity	0.32 [0.29, 0.35]	< .001	0.21 [0.18, 0.24]	< .001
Europe	0.27 [0.21, 0.34]	< .001	0.16 [0.09, 0.23]	< .001
Africa	0.43 [0.36, 0.51]	< .001	0.38 [0.30, 0.46]	< .001
Asia	0.46 [0.37, 0.55]	< .001	0.42 [0.34, 0.50]	< .001
North America	0.15 [0.07, 0.22]	< .001	0.03 [−0.04, 0.11]	.421
South America	0.05 [−0.02, 0.11]	.184	−0.05 [−0.12, 0.02]	.169
Physical activity²	−0.05 [−0.06, −0.05]	< .001	−0.12 [−0.14, −0.11]	< .001
Physical activity × Europe	0.01 [−0.02, 0.04]	.391	−0.13 [−0.16, −0.10]	< .001
Physical activity × Africa	−0.34 [−0.40, −0.28]	< .001	0.10 [0.04, 0.16]	< .001
Physical activity × Asia	−0.36 [−0.45, −0.26]	< .001	0.03 [−0.03, 0.08]	.339
Physical activity × North America	0.02 [−0.04, 0.07]	.508	0.02 [−0.04, 0.07]	.551
Physical activity × South America	0.00 [−0.03, 0.03]	.998	0.05 [0.02, 0.09]	.006
Europe × Physical activity²	−0.03 [−0.04, −0.02]	< .001	0.12 [0.10, 0.14]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Africa × Physical activity²	−0.26 [−0.32, −0.20]	< .001	−0.06 [−0.09, −0.02]	.005
Asia × Physical activity²	−0.26 [−0.33, −0.18]	< .001	0.02 [−0.03, 0.07]	.439
North America × Physical activity²	−0.03 [−0.05, −0.01]	.003	−0.02 [−0.05, 0.02]	.350
South America × Physical activity²	0.00 [−0.01, 0.01]	.579	0.01 [−0.01, 0.03]	.399

Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction.

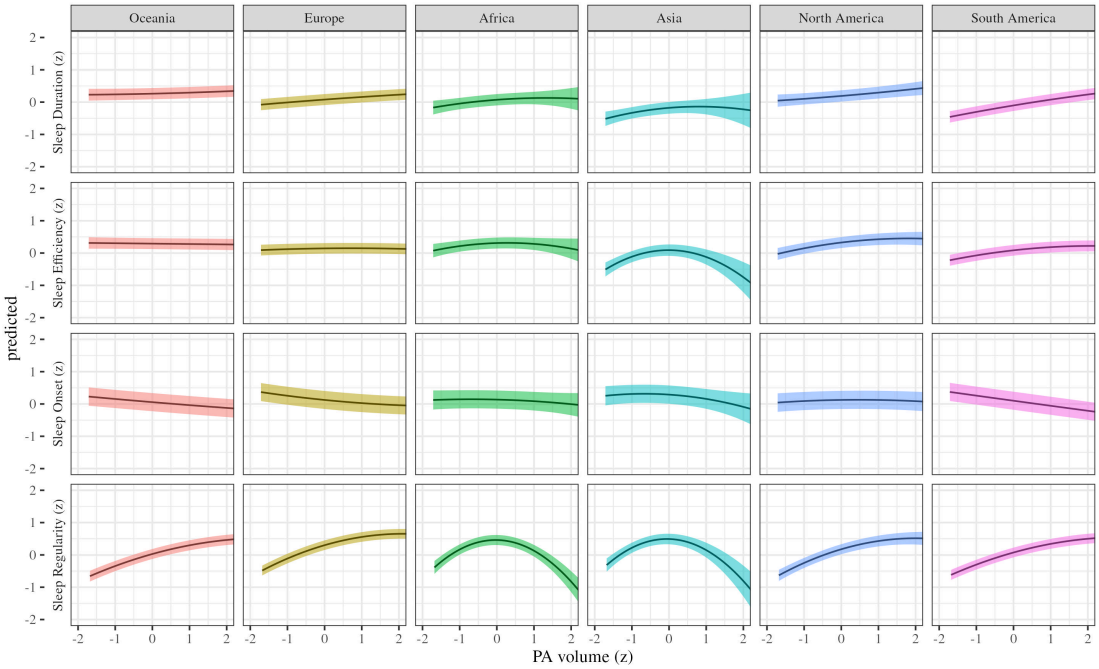

Figure 54: Sleep metrics on Physical activity volume by region

The association between physical activity intensity and sleep by region

We estimated how physical activity intensity is associated with sleep across region. We present the results in Table 22 and Figure 55.

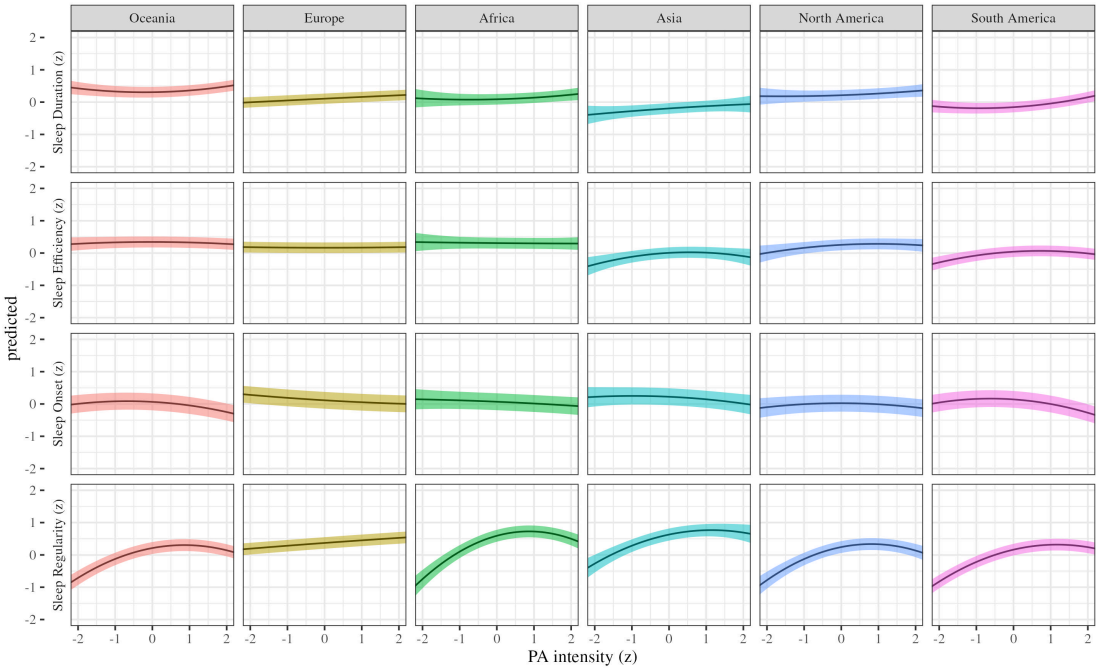

Figure 55: Sleep metrics on Physical activity intensity moderated by region

The association between sleep duration and next-day physical activity

We estimated the association between sleep duration and next-day physical activity by region. Results, controlling for SES, age, sex and BMI z-score are presented in Table 23 and Figure 56.

Table 23: Sleep predicting physical activity controlling for SES, age, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.28 [−0.01, 0.58]	.062	0.08 [−0.05, 0.20]	.224
Sleep duration	0.10 [0.09, 0.11]	< .001	0.06 [0.05, 0.07]	< .001
Europe	−0.15 [−0.20, −0.09]	< .001	−0.15 [−0.20, −0.10]	< .001
Africa	−0.12 [−0.18, −0.06]	< .001	−0.02 [−0.08, 0.04]	.491
Asia	−0.36 [−0.43, −0.30]	< .001	−0.42 [−0.48, −0.36]	< .001
North America	−0.29 [−0.35, −0.22]	< .001	−0.24 [−0.30, −0.19]	< .001
South America	−0.19 [−0.25, −0.14]	< .001	−0.25 [−0.31, −0.20]	< .001
Sleep duration²	−0.04 [−0.04, −0.03]	< .001	−0.01 [−0.02, 0.00]	.002
Sleep duration × Europe	−0.11 [−0.12, −0.09]	< .001	−0.03 [−0.05, −0.02]	< .001
Sleep duration × Africa	−0.06 [−0.10, −0.03]	< .001	−0.06 [−0.09, −0.02]	< .001
Sleep duration × Asia	−0.09 [−0.13, −0.06]	< .001	0.01 [−0.03, 0.05]	.614
Sleep duration × North America	−0.04 [−0.07, −0.01]	.003	−0.02 [−0.05, 0.00]	.104
Sleep duration × South America	−0.08 [−0.10, −0.07]	< .001	−0.07 [−0.09, −0.06]	< .001
Europe × Sleep duration²	0.02 [0.02, 0.03]	< .001	0.00 [0.00, 0.01]	.269
Africa × Sleep duration²	0.02 [0.00, 0.04]	.088	−0.03 [−0.05, −0.01]	.004
Asia × Sleep duration²	0.03 [0.00, 0.06]	.021	−0.05 [−0.08, −0.02]	< .001
North America × Sleep duration²	0.02 [0.01, 0.04]	.009	0.00 [−0.01, 0.02]	.693

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
South America × Sleep duration²	0.00 [−0.01, 0.01]	.452	−0.01 [−0.02, 0.00]	.047
Sleep efficiency (z)				
(Intercept)	0.31 [0.02, 0.61]	.038	0.11 [−0.02, 0.23]	.096
Sleep efficiency	−0.02 [−0.04, 0.00]	.019	0.01 [−0.01, 0.02]	.522
Europe	−0.20 [−0.26, −0.15]	< .001	−0.17 [−0.22, −0.12]	< .001
Africa	−0.17 [−0.23, −0.11]	< .001	−0.12 [−0.18, −0.07]	< .001
Asia	−0.42 [−0.49, −0.36]	< .001	−0.47 [−0.53, −0.42]	< .001
North America	−0.31 [−0.37, −0.25]	< .001	−0.28 [−0.33, −0.23]	< .001
South America	−0.28 [−0.34, −0.22]	< .001	−0.33 [−0.38, −0.28]	< .001
Sleep efficiency²	−0.03 [−0.03, −0.02]	< .001	−0.02 [−0.03, −0.01]	< .001
Sleep efficiency × Europe	0.04 [0.02, 0.06]	< .001	0.00 [−0.01, 0.02]	.587
Sleep efficiency × Africa	0.02 [−0.01, 0.05]	.178	−0.02 [−0.05, 0.01]	.144
Sleep efficiency × Asia	0.04 [0.00, 0.07]	.038	0.00 [−0.03, 0.04]	.843
Sleep efficiency × North America	0.02 [−0.01, 0.05]	.192	0.00 [−0.03, 0.03]	.929
Sleep efficiency × South America	−0.01 [−0.03, 0.01]	.432	−0.01 [−0.03, 0.01]	.448
Europe × Sleep efficiency²	0.02 [0.01, 0.03]	< .001	0.01 [0.00, 0.01]	.253
Africa × Sleep efficiency²	0.02 [0.01, 0.03]	< .001	0.01 [0.00, 0.03]	.031
Asia × Sleep efficiency²	0.03 [0.02, 0.04]	< .001	0.01 [0.00, 0.02]	.187

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
North America × Sleep efficiency²	0.02 [0.01, 0.03]	.001	0.01 [0.00, 0.03]	.014
South America × Sleep efficiency²	0.02 [0.01, 0.02]	.002	0.01 [0.00, 0.02]	.003
Sleep onset (z)				
(Intercept)	0.14 [−0.16, 0.44]	.350	0.01 [−0.12, 0.13]	.925
Sleep onset	−0.16 [−0.18, −0.15]	< .001	−0.10 [−0.11, −0.08]	< .001
Europe	−0.02 [−0.08, 0.04]	.523	−0.06 [−0.12, −0.01]	.020
Africa	0.07 [−0.02, 0.15]	.142	−0.04 [−0.12, 0.05]	.382
Asia	−0.20 [−0.28, −0.12]	< .001	−0.35 [−0.43, −0.28]	< .001
North America	−0.07 [−0.15, 0.00]	.060	−0.17 [−0.24, −0.10]	< .001
South America	0.01 [−0.06, 0.07]	.861	−0.18 [−0.24, −0.13]	< .001
Sleep onset²	0.01 [0.01, 0.02]	< .001	0.01 [0.00, 0.01]	.012
Sleep onset × Europe	0.14 [0.12, 0.15]	< .001	0.09 [0.07, 0.11]	< .001
Sleep onset × Africa	0.16 [0.03, 0.30]	.019	−0.01 [−0.15, 0.13]	.876
Sleep onset × Asia	0.14 [0.03, 0.25]	.015	0.02 [−0.10, 0.13]	.765
Sleep onset × North America	0.21 [0.15, 0.27]	< .001	0.08 [0.03, 0.14]	.004
Sleep onset × South America	0.21 [0.19, 0.23]	< .001	0.11 [0.09, 0.13]	< .001
Europe × Sleep onset²	−0.02 [−0.03, −0.01]	< .001	−0.01 [−0.01, 0.00]	.114
Africa × Sleep onset²	−0.02 [−0.10, 0.05]	.517	−0.05 [−0.13, 0.02]	.169
Asia × Sleep onset²	−0.03 [−0.10, 0.04]	.377	−0.07 [−0.15, 0.00]	.048

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
North America × Sleep onset²	0.01 [−0.02, 0.04]	.511	0.00 [−0.03, 0.03]	.983
South America × Sleep onset²	−0.02 [−0.03, −0.01]	< .001	−0.01 [−0.02, 0.00]	.014
Sleep regularity (z)				
(Intercept)	0.23 [−0.07, 0.53]	.135	0.00 [−0.13, 0.12]	.948
Sleep regularity	0.15 [0.14, 0.17]	< .001	0.07 [0.05, 0.08]	< .001
Europe	−0.24 [−0.29, −0.18]	< .001	−0.18 [−0.23, −0.13]	< .001
Africa	−0.20 [−0.26, −0.14]	< .001	−0.13 [−0.18, −0.07]	< .001
Asia	−0.46 [−0.52, −0.40]	< .001	−0.49 [−0.55, −0.43]	< .001
North America	−0.32 [−0.38, −0.26]	< .001	−0.24 [−0.29, −0.18]	< .001
South America	−0.27 [−0.33, −0.22]	< .001	−0.31 [−0.36, −0.26]	< .001
Sleep regularity²	0.00 [0.00, 0.01]	.139	0.00 [0.00, 0.01]	.182
Sleep regularity × Europe	−0.03 [−0.05, −0.02]	< .001	−0.01 [−0.02, 0.01]	.458
Sleep regularity × Africa	−0.09 [−0.12, −0.07]	< .001	0.03 [0.01, 0.06]	.008
Sleep regularity × Asia	−0.10 [−0.13, −0.07]	< .001	0.06 [0.04, 0.09]	< .001
Sleep regularity × North America	−0.05 [−0.08, −0.02]	< .001	0.03 [0.00, 0.05]	.048
Sleep regularity × South America	−0.01 [−0.03, 0.00]	.077	0.01 [0.00, 0.03]	.156
Europe × Sleep regularity²	−0.01 [−0.02, 0.00]	.016	−0.01 [−0.02, 0.00]	.003
Africa × Sleep regularity²	−0.03 [−0.04, −0.01]	< .001	0.00 [−0.02, 0.01]	.797
Asia × Sleep regularity²	−0.02 [−0.04, 0.00]	.044	−0.01 [−0.03, 0.01]	.349

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
North America × Sleep regularity²	−0.01 [−0.02, 0.00]	.178	−0.02 [−0.03, 0.00]	.030
South America × Sleep regularity²	0.00 [−0.01, 0.01]	.804	0.00 [−0.01, 0.01]	.802

Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction.

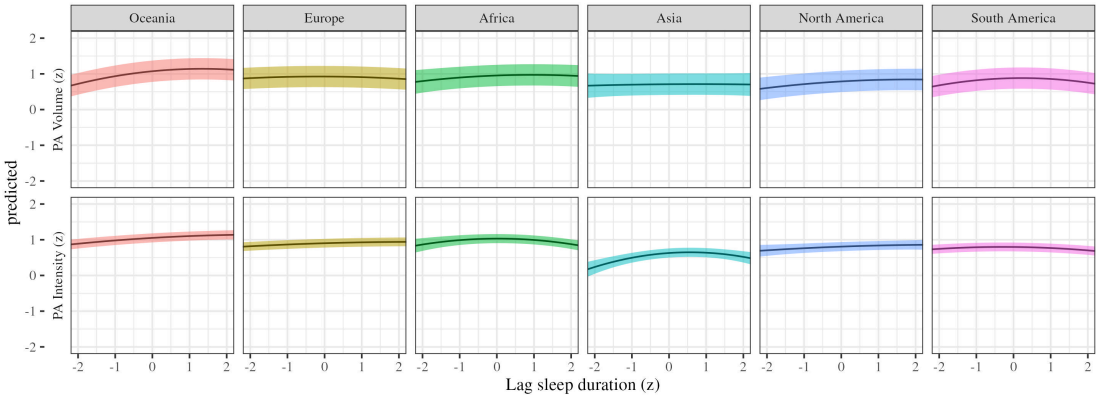

Figure 56: Physical activity by sleep duration moderated by region

The association between sleep efficiency and next-day physical activity

We estimated the association between sleep efficiency and next-day physical activity by region. Results, controlling for SES, age, sex and BMI z-score are presented in Table 23 and Figure 57.

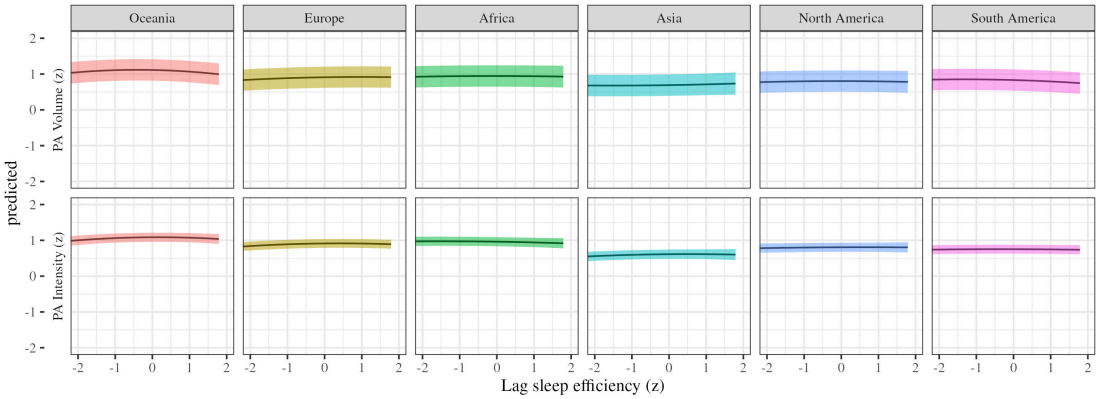

Figure 57: Physical activity by sleep efficiency moderated by region

The association between sleep onset and next-day physical activity

We estimated the association between sleep onset and next-day physical activity by region. Results, controlling for SES, age, sex and BMI z-score are presented in Table 23 and Figure 58.

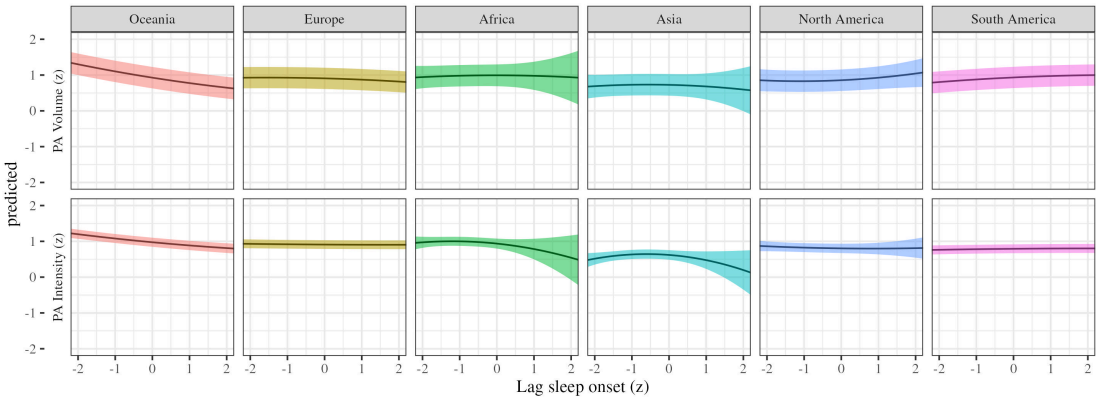

Figure 58: Physical activity by sleep onset moderated by region

The association between sleep regularity and next-day physical activity

We estimated the association between sleep regularity and next-day physical activity by region. Results are presented in Table 23 and Figure 59.

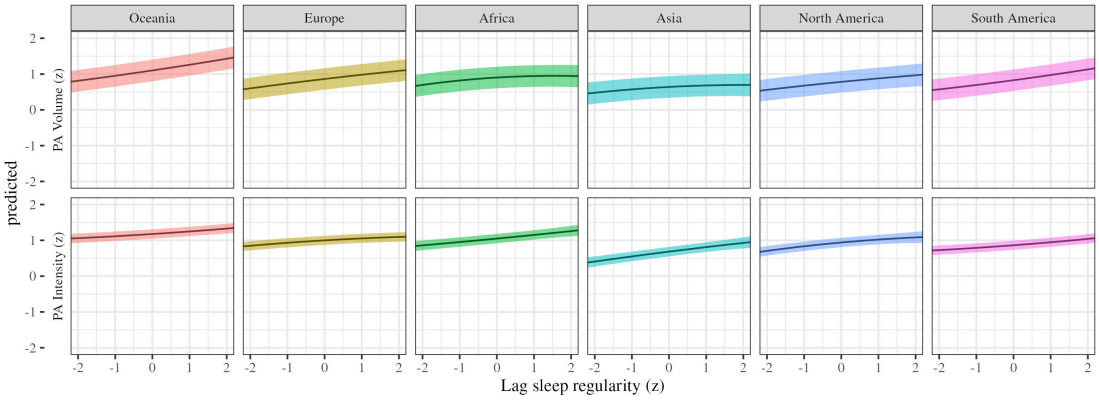

Figure 59: Physical activity by sleep regularity moderated by region

daylight hours

The association between physical activity volume and sleep by daylight hours

We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by daylight hours, and the results are presented in Table 24 and Figure 60

Table 24: Physical activity predicting sleep controlling for SES, age, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.38 [0.22, 0.54]	< .001	0.34 [0.19, 0.49]	< .001
Physical activity	0.20 [0.15, 0.24]	< .001	0.13 [0.10, 0.16]	< .001
Daylight hours	−0.03 [−0.03, −0.03]	< .001	−0.03 [−0.03, −0.02]	< .001
Physical activity²	−0.03 [−0.04, −0.01]	.002	0.01 [−0.01, 0.03]	.238
Physical activity × Daylight hours	−0.01 [−0.01, 0.00]	< .001	−0.01 [−0.01, 0.00]	< .001
Daylight hours × Physical activity²	0.00 [0.00, 0.00]	.009	0.00 [0.00, 0.00]	.698
Sleep efficiency (z)				
(Intercept)	0.27 [0.10, 0.43]	.001	0.24 [0.07, 0.41]	.006
Physical activity	0.13 [0.09, 0.18]	< .001	0.04 [0.01, 0.07]	.006
Daylight hours	−0.01 [−0.01, 0.00]	< .001	−0.01 [−0.01, 0.00]	.004
Physical activity²	−0.03 [−0.05, −0.02]	< .001	−0.01 [−0.03, 0.01]	.369
Physical activity × Daylight hours	−0.01 [−0.01, 0.00]	< .001	0.00 [−0.01, 0.00]	.015
Daylight hours × Physical activity²	0.00 [0.00, 0.00]	.019	0.00 [0.00, 0.00]	.425
Sleep onset (z)				
(Intercept)	−0.06 [−0.33, 0.22]	.693	−0.02 [−0.27, 0.24]	.904
Physical activity	−0.20 [−0.23, −0.17]	< .001	−0.14 [−0.16, −0.12]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Daylight hours	0.01 [0.01, 0.02]	< .001	0.01 [0.01, 0.01]	< .001
Physical activity²	−0.01 [−0.03, 0.00]	.044	−0.04 [−0.06, −0.03]	< .001
Physical activity × Daylight hours	0.01 [0.00, 0.01]	< .001	0.00 [0.00, 0.01]	< .001
Daylight hours × Physical activity²	0.00 [0.00, 0.00]	.012	0.00 [0.00, 0.00]	< .001
Sleep regularity (z)				
(Intercept)	0.04 [−0.11, 0.20]	.571	0.05 [−0.14, 0.23]	.629
Physical activity	0.32 [0.27, 0.37]	< .001	0.19 [0.16, 0.22]	< .001
Daylight hours	0.01 [0.00, 0.01]	< .001	0.01 [0.00, 0.01]	.003
Physical activity²	−0.04 [−0.06, −0.02]	< .001	−0.06 [−0.08, −0.04]	< .001
Physical activity × Daylight hours	0.00 [0.00, 0.00]	.933	−0.01 [−0.01, 0.00]	< .001
Daylight hours × Physical activity²	0.00 [0.00, 0.00]	.047	0.00 [0.00, 0.01]	< .001

Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction.

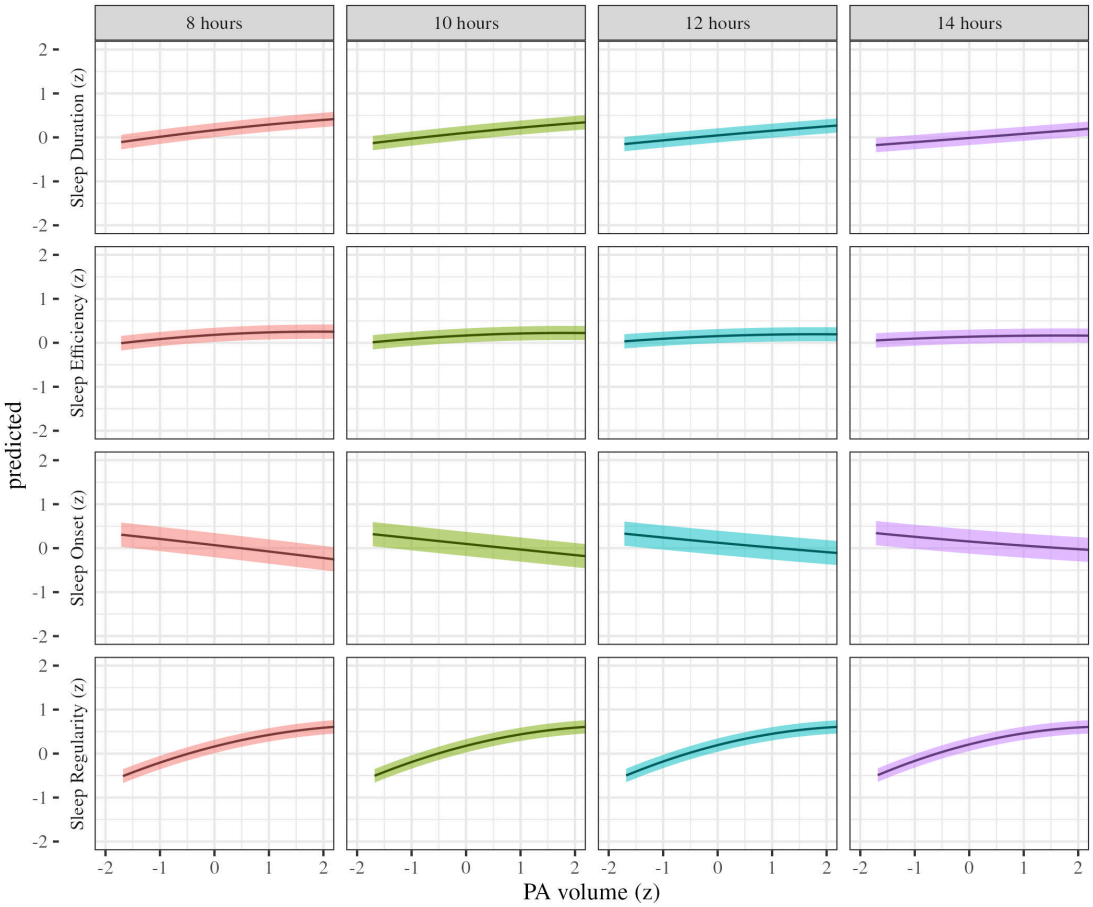

Figure 60: Sleep metrics on Physical activity volume by daylight hours

The association between physical activity intensity and sleep by daylight hours

We estimated how physical activity intensity is associated with sleep across daylight hours. We present the results in Table 24 and Figure 61.

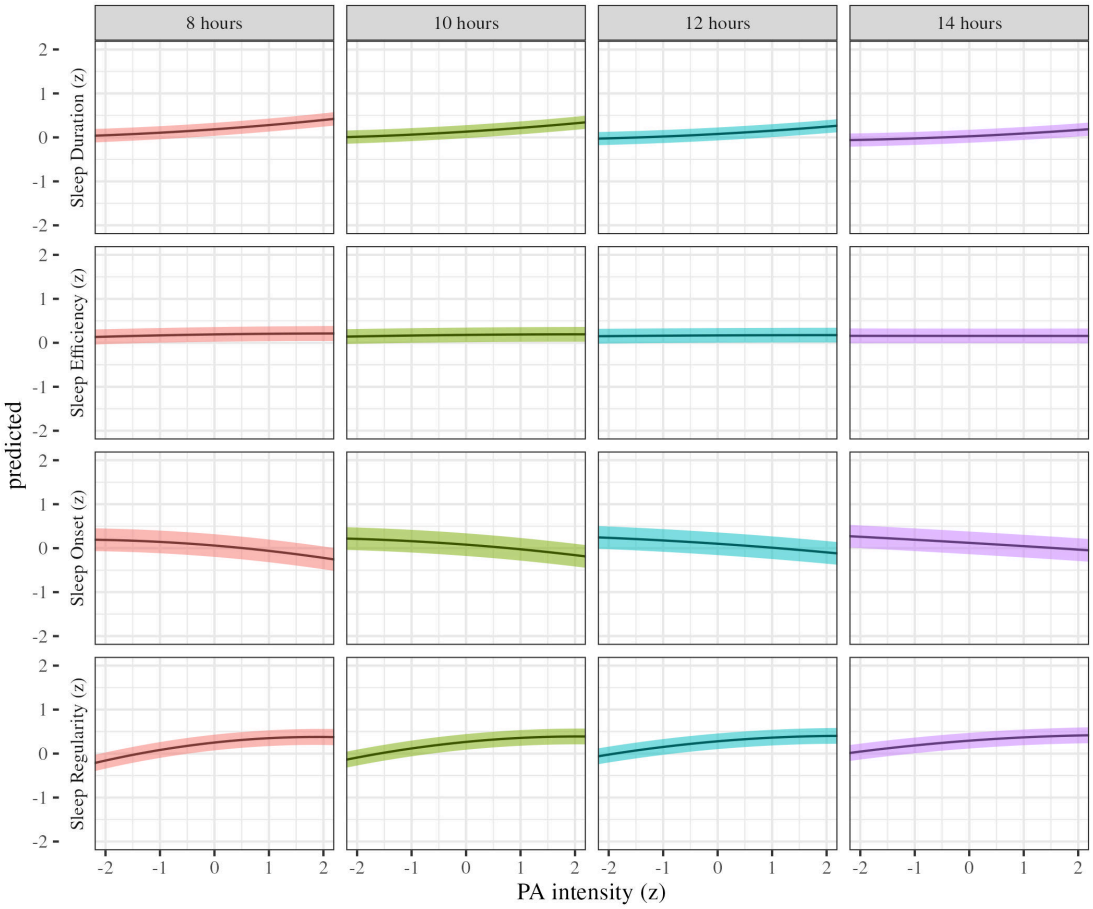

Figure 61: Sleep metrics on Physical activity intensity moderated by daylight hours

The association between sleep duration and next-day physical activity

We estimated the association between sleep duration and next-day physical activity by daylight hours. Results, controlling for SES, age, sex and BMI z-score are presented in Table 25 and Figure 62.

Table 25: Sleep predicting physical activity controlling for SES, age, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	−0.07 [−0.37, 0.24]	.658	−0.10 [−0.23, 0.03]	.116
Sleep duration	0.12 [0.10, 0.14]	< .001	0.09 [0.07, 0.11]	< .001
Daylight hours	0.02 [0.01, 0.02]	< .001	0.00 [0.00, 0.01]	.094
Sleep duration²	−0.01 [−0.02, 0.00]	.160	0.02 [0.00, 0.03]	.011
Sleep duration × Daylight hours	−0.01 [−0.01, −0.01]	< .001	−0.01 [−0.01, 0.00]	< .001
Daylight hours × Sleep duration²	0.00 [0.00, 0.00]	.088	0.00 [0.00, 0.00]	< .001
Sleep efficiency (z)				
(Intercept)	−0.08 [−0.38, 0.23]	.619	−0.06 [−0.19, 0.06]	.320
Sleep efficiency	0.04 [0.01, 0.06]	.003	0.02 [−0.01, 0.04]	.171
Daylight hours	0.02 [0.01, 0.02]	< .001	0.00 [0.00, 0.00]	.586
Sleep efficiency²	−0.02 [−0.03, 0.00]	.006	−0.03 [−0.04, −0.01]	< .001
Sleep efficiency × Daylight hours	0.00 [0.00, 0.00]	.008	0.00 [0.00, 0.00]	.379
Daylight hours × Sleep efficiency²	0.00 [0.00, 0.00]	.086	0.00 [0.00, 0.00]	.002
Sleep onset (z)				
(Intercept)	−0.08 [−0.38, 0.23]	.619	−0.09 [−0.22, 0.04]	.181
Sleep onset	−0.05 [−0.08, −0.02]	< .001	−0.04 [−0.06, −0.01]	.008

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Daylight hours	0.02 [0.01, 0.02]	< .001	0.00 [0.00, 0.00]	.662
Sleep onset²	−0.01 [−0.02, 0.01]	.466	0.00 [−0.01, 0.02]	.556
Sleep onset × Daylight hours	0.00 [0.00, 0.00]	.078	0.00 [0.00, 0.00]	.105
Daylight hours × Sleep onset²	0.00 [0.00, 0.00]	.775	0.00 [0.00, 0.00]	.806
Sleep regularity (z)				
(Intercept)	−0.21 [−0.51, 0.10]	.191	−0.19 [−0.32, −0.05]	.006
Sleep regularity	0.17 [0.15, 0.20]	< .001	0.15 [0.13, 0.18]	< .001
Daylight hours	0.02 [0.02, 0.02]	< .001	0.00 [0.00, 0.00]	.649
Sleep regularity²	0.01 [−0.01, 0.02]	.243	0.01 [0.00, 0.03]	.156
Sleep regularity × Daylight hours	0.00 [−0.01, 0.00]	< .001	−0.01 [−0.01, 0.00]	< .001
Daylight hours × Sleep regularity²	0.00 [0.00, 0.00]	.165	0.00 [0.00, 0.00]	.092

Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction.

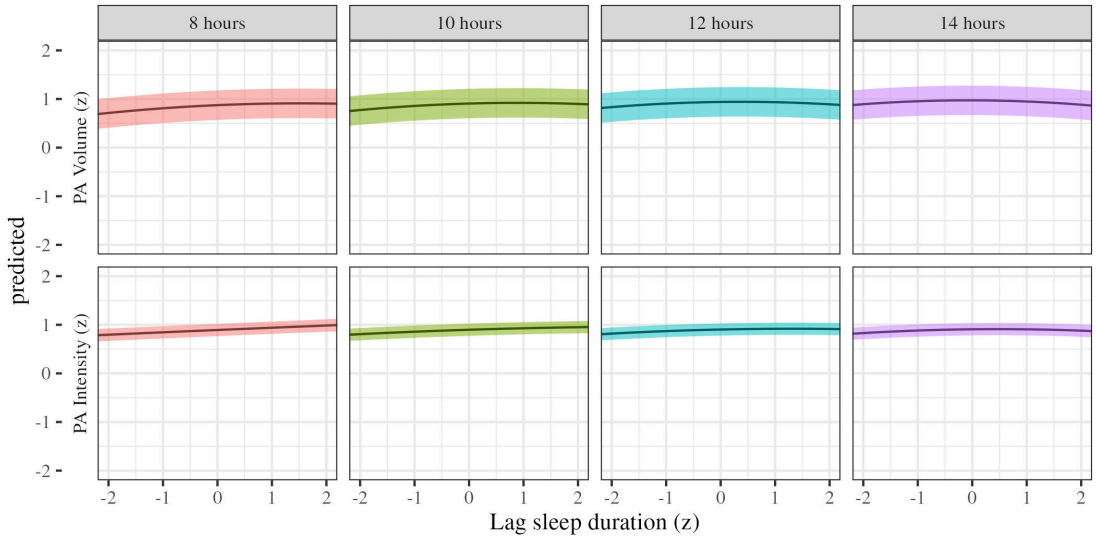

Figure 62: Physical activity by sleep duration moderated by daylight hours

The association between sleep efficiency and next-day physical activity

We estimated the association between sleep efficiency and next-day physical activity by daylight hours. Results, controlling for SES, age, sex and BMI z-score are presented in Table 25 and Figure 63.

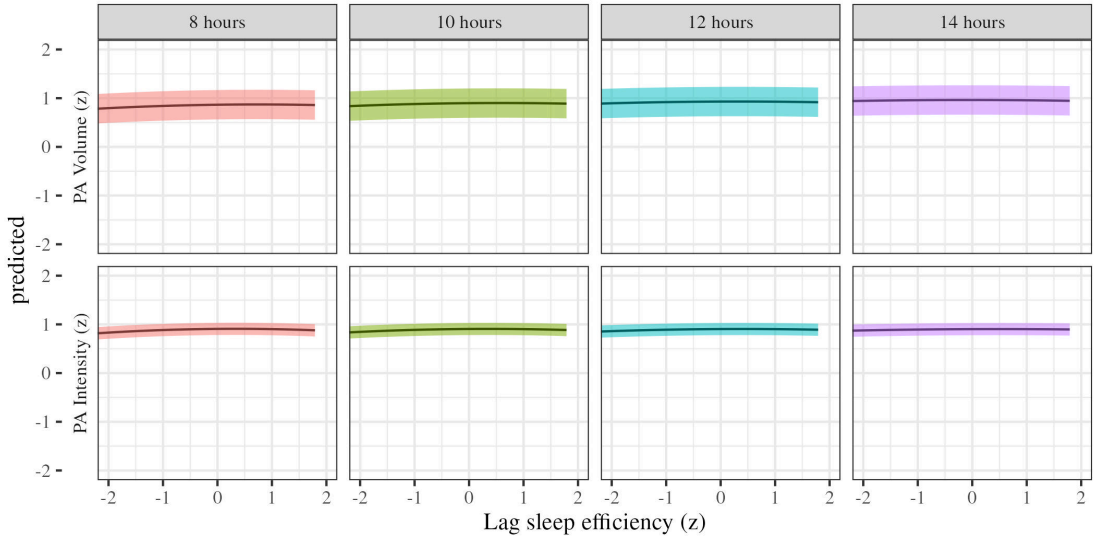

Figure 63: Physical activity by sleep efficiency moderated by daylight hours

The association between sleep onset and next-day physical activity

We estimated the association between sleep onset and next-day physical activity by daylight hours. Results, controlling for SES, age, sex and BMI z-score are presented in Table 25 and Figure 64.

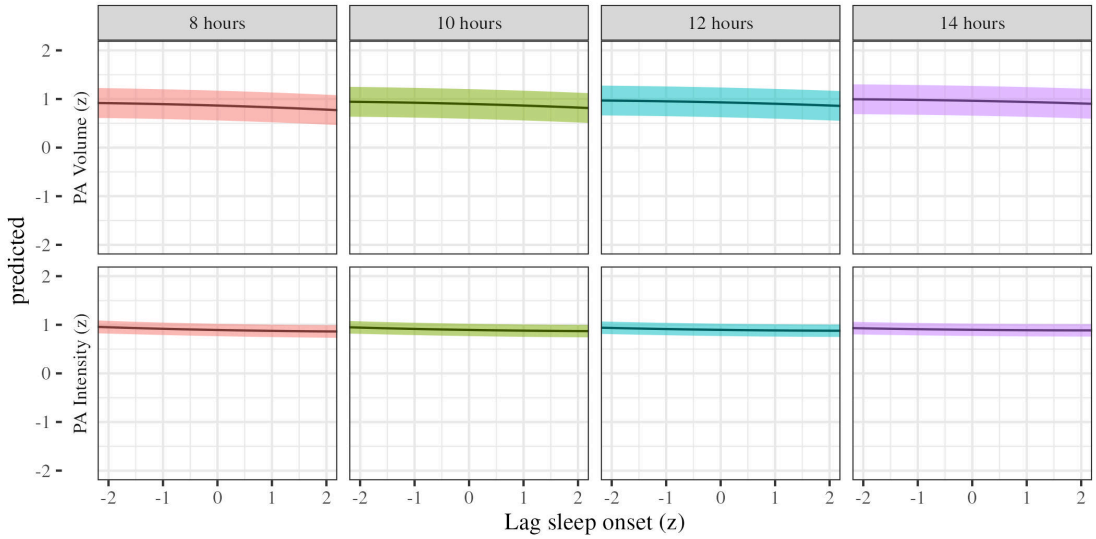

Figure 64: Physical activity by sleep onset moderated by daylight hours

The association between sleep regularity and next-day physical activity

We estimated the association between sleep regularity and next-day physical activity by daylight hours. Results are presented in Table 25 and Figure 65.

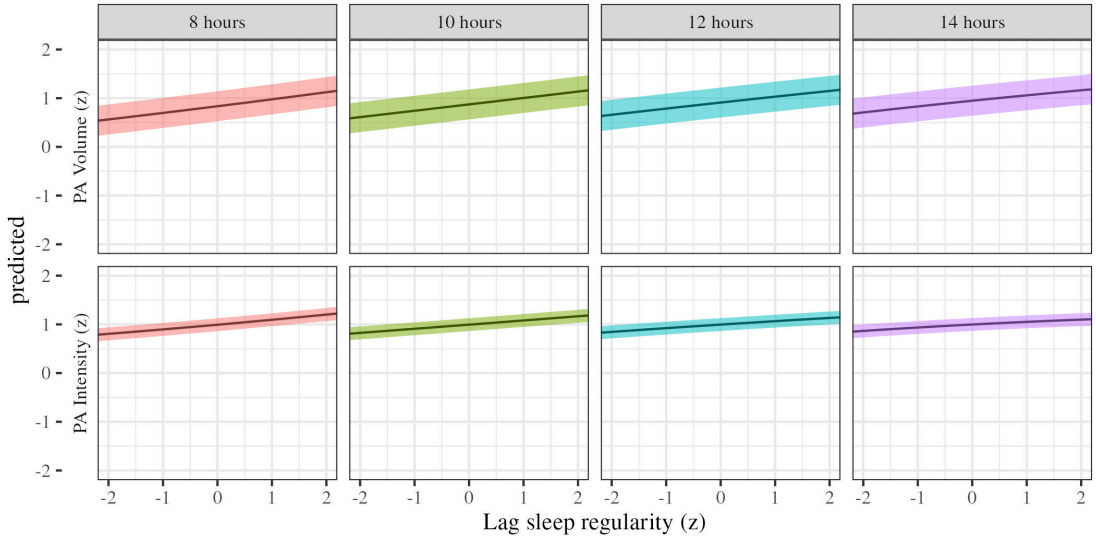

Figure 65: Physical activity by sleep regularity moderated by daylight hours

wear location

The association between physical activity volume and sleep by wear location

We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by wear location, and the results are presented in Table 26 and Figure 66

Table 26: Physical activity predicting sleep controlling for SES, age, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.56 [0.33, 0.80]	< .001	0.46 [0.22, 0.70]	< .001
Physical activity	0.11 [0.09, 0.13]	< .001	0.08 [0.06, 0.09]	< .001
Wrist	−0.69 [−0.95, −0.43]	< .001	−0.56 [−0.82, −0.29]	< .001
Physical activity²	−0.03 [−0.05, −0.02]	< .001	0.00 [−0.01, 0.01]	.594
Physical activity × Wrist	−0.01 [−0.03, 0.01]	.332	−0.01 [−0.03, 0.00]	.081
Wrist × Physical activity²	0.03 [0.02, 0.05]	< .001	0.01 [−0.01, 0.02]	.312
Sleep efficiency (z)				
(Intercept)	−0.35 [−0.55, −0.15]	< .001	−0.47 [−0.67, −0.27]	< .001
Physical activity	0.12 [0.10, 0.13]	< .001	0.09 [0.07, 0.10]	< .001
Wrist	0.68 [0.46, 0.90]	< .001	0.79 [0.57, 1.01]	< .001
Physical activity²	−0.12 [−0.14, −0.10]	< .001	−0.04 [−0.06, −0.03]	< .001
Physical activity × Wrist	−0.14 [−0.16, −0.12]	< .001	−0.11 [−0.12, −0.09]	< .001
Wrist × Physical activity²	0.12 [0.11, 0.14]	< .001	0.04 [0.03, 0.05]	< .001
Sleep onset (z)				
(Intercept)	−0.85 [−1.21, −0.49]	< .001	−0.81 [−1.18, −0.45]	< .001
Physical activity	−0.02 [−0.03, −0.01]	.001	−0.03 [−0.04, −0.02]	< .001
Wrist	1.22 [0.81, 1.62]	< .001	1.15 [0.75, 1.56]	< .001
Physical activity²	−0.03 [−0.04, −0.02]	< .001	−0.02 [−0.03, −0.01]	< .001
Physical activity × Wrist	−0.15 [−0.16, −0.13]	< .001	−0.08 [−0.09, −0.06]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Wrist × Physical activity²	0.05 [0.03, 0.06]	< .001	0.01 [0.00, 0.02]	.057
Sleep regularity (z)				
(Intercept)	−0.01 [−0.32, 0.29]	.936	−0.37 [−0.69, −0.06]	.020
Physical activity	0.19 [0.17, 0.21]	< .001	0.30 [0.28, 0.31]	< .001
Wrist	0.20 [−0.14, 0.54]	.243	0.62 [0.26, 0.97]	< .001
Physical activity²	−0.25 [−0.27, −0.23]	< .001	−0.14 [−0.15, −0.13]	< .001
Physical activity × Wrist	0.13 [0.11, 0.15]	< .001	−0.25 [−0.27, −0.24]	< .001
Wrist × Physical activity²	0.19 [0.17, 0.21]	< .001	0.13 [0.11, 0.14]	< .001

Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction.

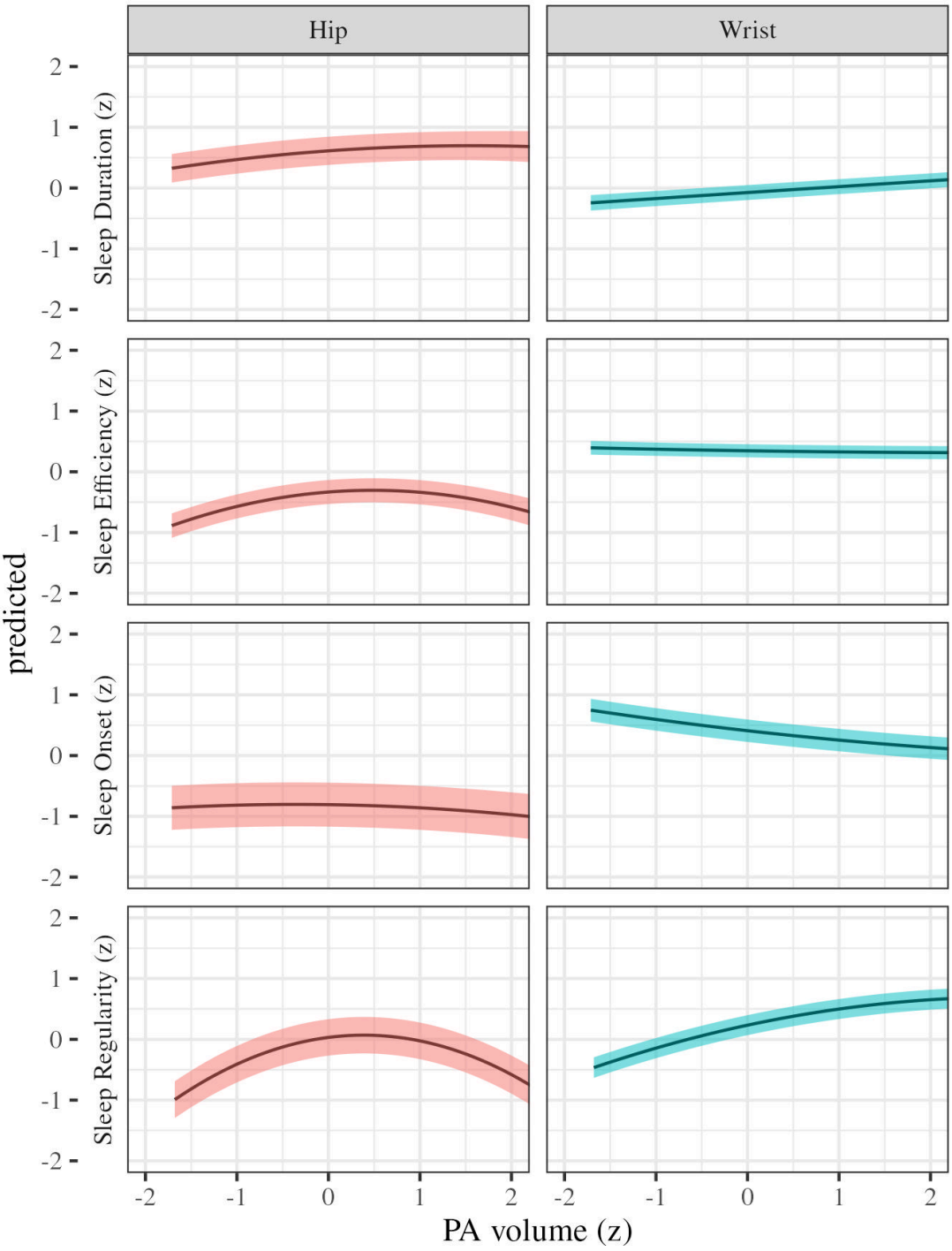

Figure 66: Sleep metrics on Physical activity volume by wear location

The association between physical activity intensity and sleep by wear location

We estimated how physical activity intensity is associated with sleep across wear location. We present the results in Table 26 and Figure 67.

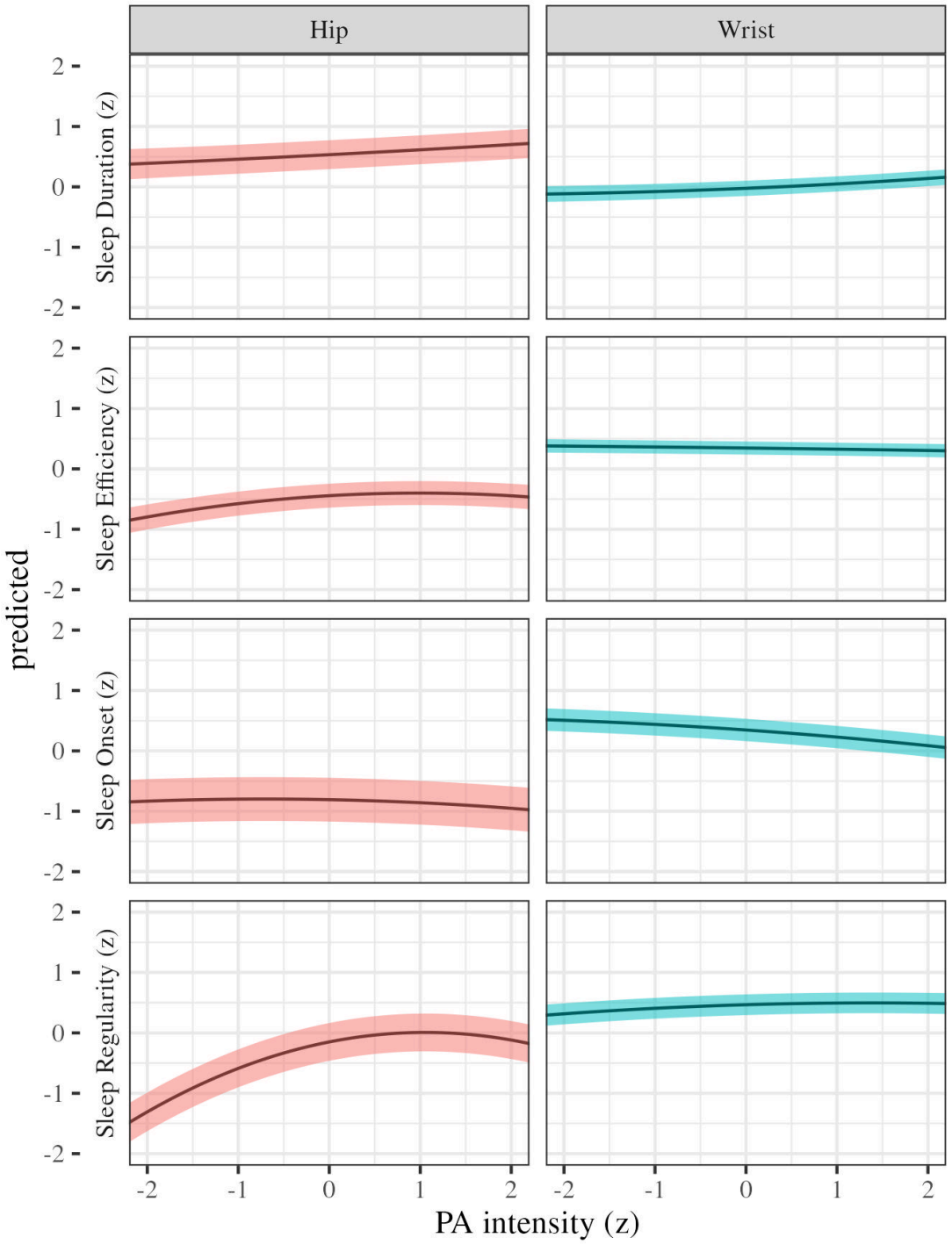

Figure 67: Sleep metrics on Physical activity intensity moderated by wear location

The association between sleep duration and next-day physical activity

We estimated the association between sleep duration and next-day physical activity by wear location. Results, controlling for SES, age, sex and BMI z-score are presented in Table 27 and Figure 68.

Table 27: Sleep predicting physical activity controlling for SES, age, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	−0.98 [−1.38, −0.59]	< .001	−0.40 [−0.63, −0.17]	< .001
Sleep duration	0.06 [0.05, 0.07]	< .001	0.07 [0.06, 0.08]	< .001
Wrist	1.40 [0.96, 1.84]	< .001	0.41 [0.15, 0.66]	.002
Sleep duration²	−0.02 [−0.03, −0.01]	< .001	−0.03 [−0.03, −0.02]	< .001
Sleep duration × Wrist	−0.06 [−0.07, −0.05]	< .001	−0.05 [−0.06, −0.04]	< .001
Wrist × Sleep duration²	0.00 [−0.01, 0.01]	.699	0.02 [0.01, 0.03]	< .001
Sleep efficiency (z)				
(Intercept)	−0.95 [−1.35, −0.56]	< .001	−0.37 [−0.60, −0.14]	.002
Sleep efficiency	0.02 [0.01, 0.03]	< .001	0.01 [0.01, 0.02]	.001
Wrist	1.35 [0.90, 1.79]	< .001	0.37 [0.11, 0.63]	.005
Sleep efficiency²	−0.01 [−0.01, 0.00]	< .001	−0.01 [−0.01, −0.01]	< .001
Sleep efficiency × Wrist	−0.02 [−0.03, −0.01]	< .001	−0.01 [−0.02, 0.00]	.010
Wrist × Sleep efficiency²	0.00 [0.00, 0.01]	.039	0.01 [0.00, 0.01]	.015
Sleep onset (z)				
(Intercept)	−0.97 [−1.37, −0.58]	< .001	−0.41 [−0.64, −0.18]	< .001
Sleep onset	−0.02 [−0.05, 0.01]	.188	−0.06 [−0.09, −0.03]	< .001
Wrist	1.38 [0.93, 1.82]	< .001	0.40 [0.15, 0.66]	.002
Sleep onset²	−0.01 [−0.02, 0.01]	.298	−0.02 [−0.04, −0.01]	.003
Sleep onset × Wrist	−0.01 [−0.04, 0.02]	.418	0.05 [0.02, 0.07]	.001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Wrist × Sleep onset²	0.01 [−0.01, 0.02]	.381	0.03 [0.01, 0.04]	.001
Sleep regularity (z)				
(Intercept)	−1.03 [−1.41, −0.65]	< .001	−0.53 [−0.75, −0.31]	< .001
Sleep regularity	0.11 [0.11, 0.12]	< .001	0.13 [0.13, 0.14]	< .001
Wrist	1.34 [0.91, 1.76]	< .001	0.44 [0.20, 0.69]	< .001
Sleep regularity²	−0.01 [−0.01, 0.00]	.019	0.00 [0.00, 0.01]	.474
Sleep regularity × Wrist	0.01 [0.00, 0.02]	.020	−0.10 [−0.11, −0.09]	< .001
Wrist × Sleep regularity²	0.01 [0.00, 0.01]	.053	0.00 [0.00, 0.01]	.579

Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction.

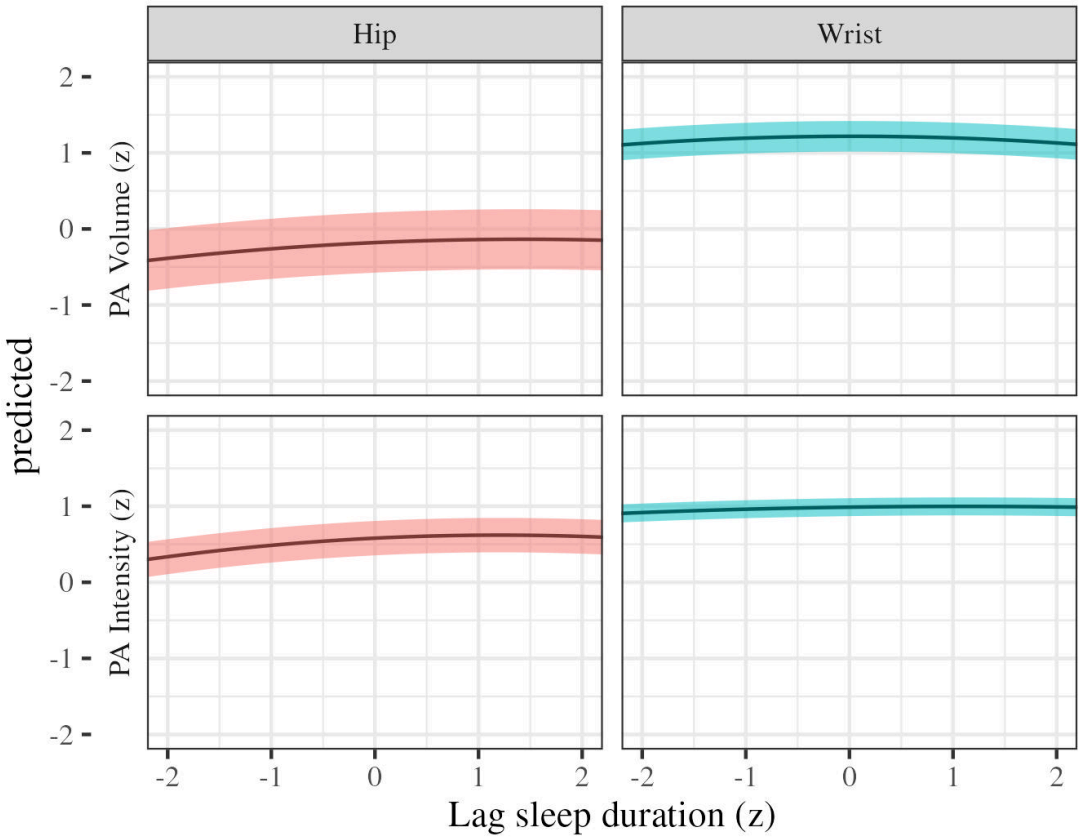

Figure 68: Physical activity by sleep duration moderated by wear location

The association between sleep efficiency and next-day physical activity

We estimated the association between sleep efficiency and next-day physical activity by wear location. Results, controlling for SES, age, sex and BMI z-score are presented in Table 27 and Figure 69.

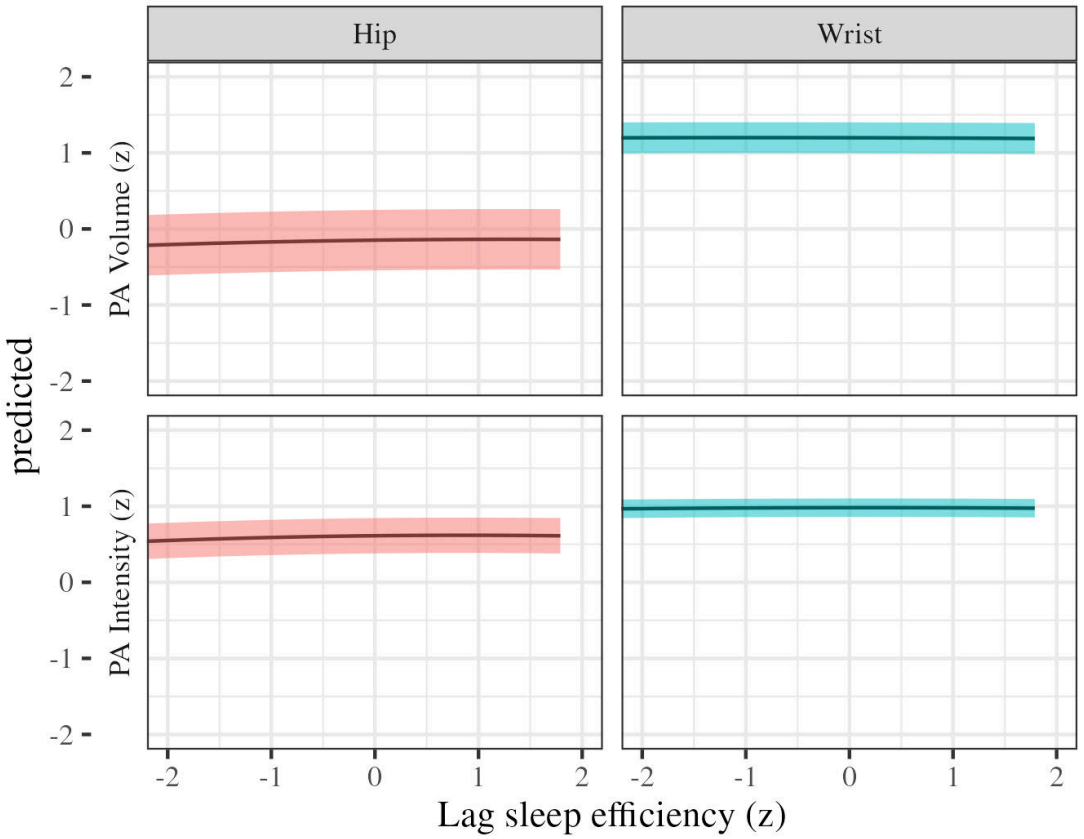

Figure 69: Physical activity by sleep efficiency moderated by wear location

The association between sleep onset and next-day physical activity

We estimated the association between sleep onset and next-day physical activity by wear location. Results, controlling for SES, age, sex and BMI z-score are presented in Table 27 and Figure 70.

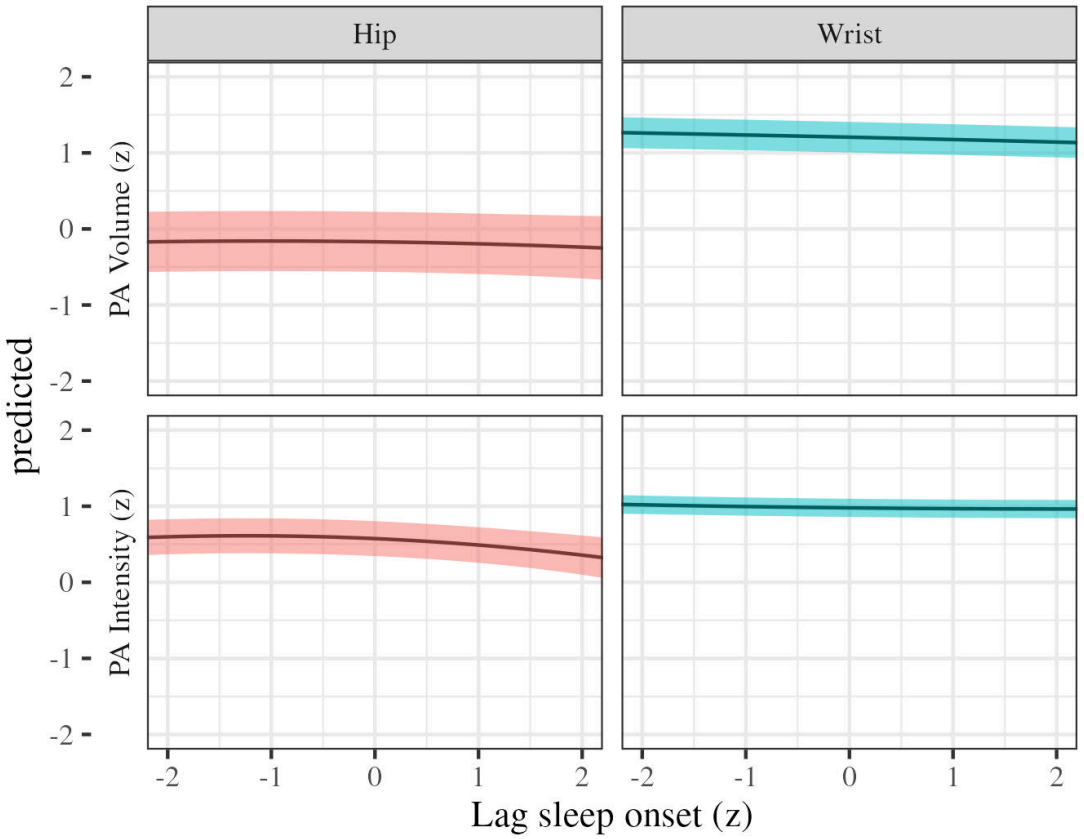

Figure 70: Physical activity by sleep onset moderated by wear location

The association between sleep regularity and next-day physical activity

We estimated the association between sleep regularity and next-day physical activity by wear location. Results are presented in Table 27 and Figure 71.

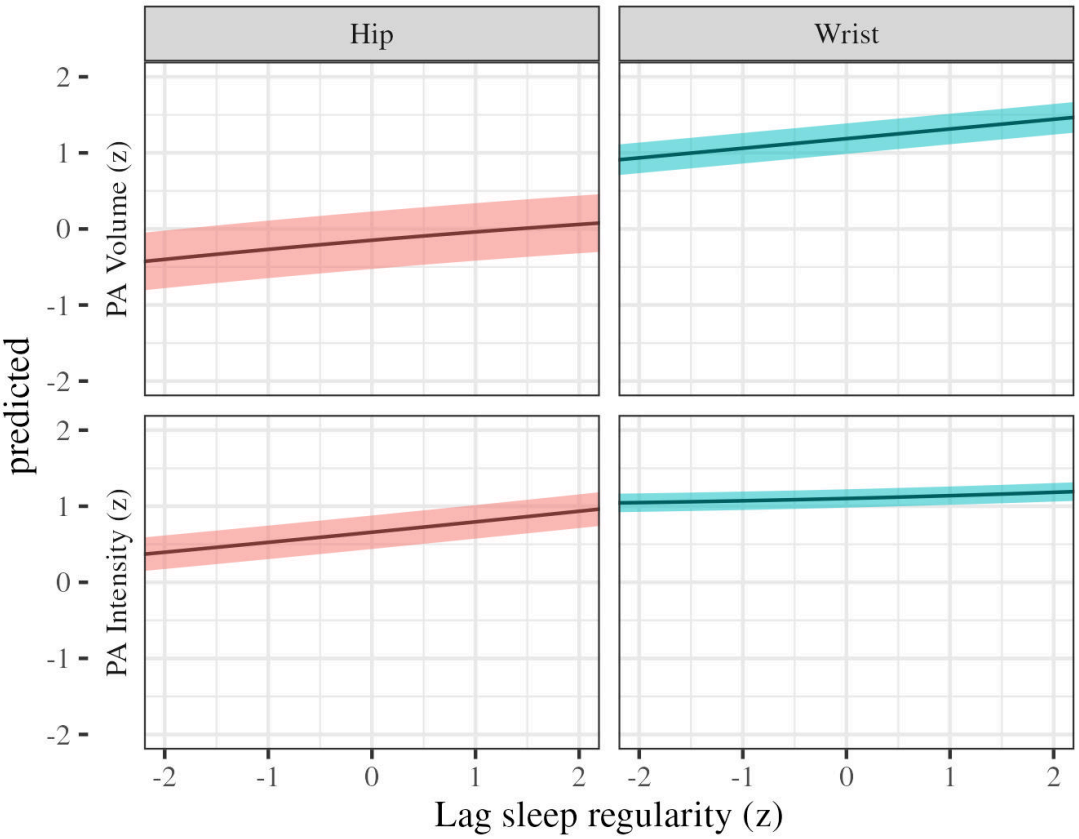

Figure 71: Physical activity by sleep regularity moderated by wear location

most active hour

The association between physical activity volume and sleep by most active hour

We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by most active hour, and the results are presented in Table 28 and Figure 72

Table 28: Physical activity predicting sleep controlling for SES, age, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.09 [−0.07, 0.25]	.261	0.11 [−0.03, 0.26]	.130
Physical activity	0.07 [0.05, 0.09]	< .001	0.06 [0.05, 0.08]	< .001
Most active hour	−0.01 [−0.01, 0.00]	< .001	−0.01 [−0.01, −0.01]	< .001
Physical activity²	0.00 [−0.01, 0.01]	.832	−0.01 [−0.02, 0.00]	.046
Physical activity × Most active hour	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	.818
Most active hour × Physical activity²	0.00 [0.00, 0.00]	.264	0.00 [0.00, 0.00]	< .001
Sleep efficiency (z)				
(Intercept)	0.18 [0.02, 0.34]	.025	0.20 [0.04, 0.37]	.016
Physical activity	−0.04 [−0.06, −0.02]	< .001	−0.02 [−0.04, −0.01]	.009
Most active hour	0.00 [0.00, 0.00]	.447	0.00 [0.00, 0.00]	< .001
Physical activity²	0.00 [−0.01, 0.01]	.852	−0.02 [−0.03, 0.00]	.010
Physical activity × Most active hour	0.01 [0.00, 0.01]	< .001	0.00 [0.00, 0.00]	< .001
Most active hour × Physical activity²	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	.014
Sleep onset (z)				
(Intercept)	−0.06 [−0.34, 0.21]	.647	−0.06 [−0.31, 0.20]	.651
Physical activity	−0.20 [−0.21, −0.18]	< .001	−0.11 [−0.12, −0.09]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Most active hour	0.01 [0.01, 0.01]	< .001	0.01 [0.01, 0.01]	< .001
Physical activity²	0.01 [0.00, 0.02]	.007	0.00 [−0.01, 0.01]	.477
Physical activity × Most active hour	0.01 [0.00, 0.01]	< .001	0.00 [0.00, 0.00]	< .001
Most active hour × Physical activity²	0.00 [0.00, 0.00]	.009	0.00 [0.00, 0.00]	.006
Sleep regularity (z)				
(Intercept)	0.23 [0.08, 0.37]	.002	0.22 [0.04, 0.39]	.014
Physical activity	0.27 [0.24, 0.29]	< .001	0.06 [0.04, 0.08]	< .001
Most active hour	−0.01 [−0.01, 0.00]	< .001	−0.01 [−0.01, 0.00]	< .001
Physical activity²	−0.06 [−0.07, −0.05]	< .001	−0.04 [−0.05, −0.02]	< .001
Physical activity × Most active hour	0.00 [0.00, 0.01]	< .001	0.00 [0.00, 0.00]	< .001
Most active hour × Physical activity²	0.00 [0.00, 0.00]	.528	0.00 [0.00, 0.00]	.051

Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction.

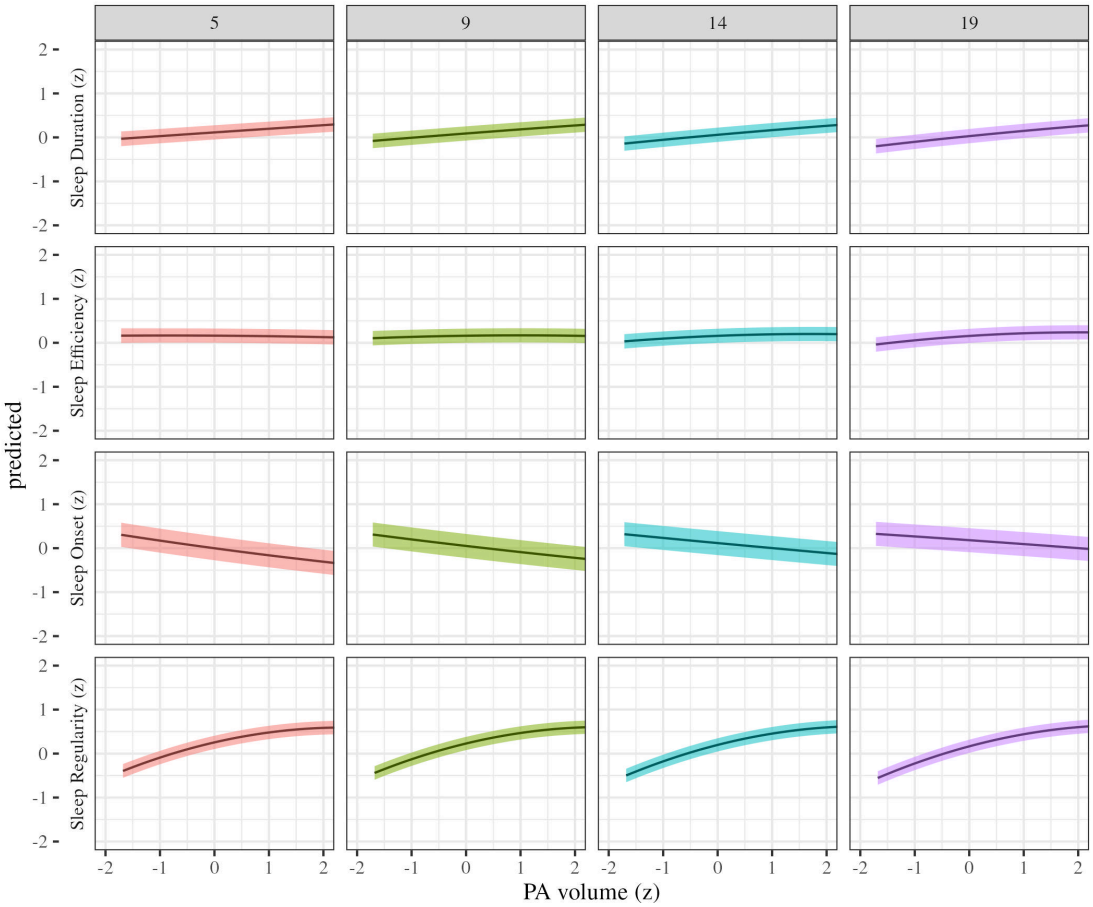

Figure 72: Sleep metrics on Physical activity volume by most active hour

The association between physical activity intensity and sleep by most active hour

We estimated how physical activity intensity is associated with sleep across most active hour. We present the results in Table 28 and Figure 73.

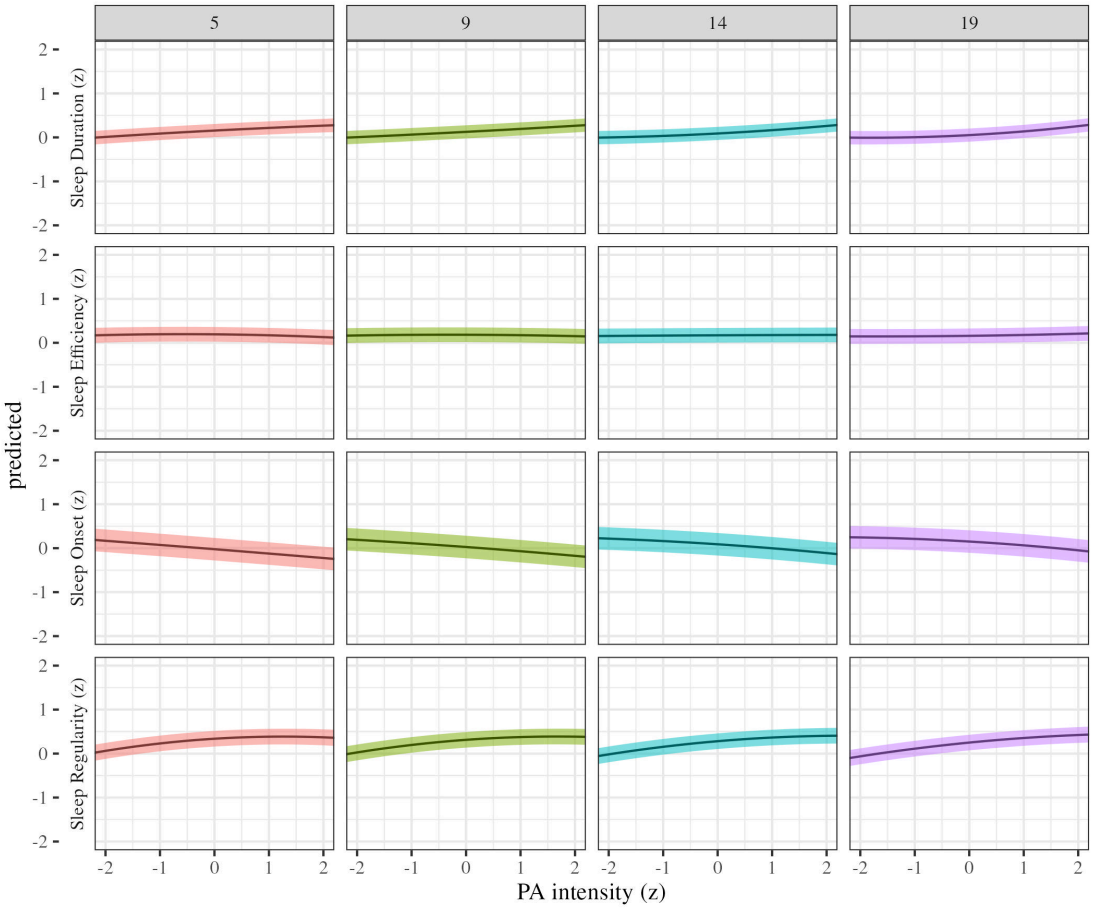

Figure 73: Sleep metrics on Physical activity intensity moderated by most active hour

The association between sleep duration and next-day physical activity

We estimated the association between sleep duration and next-day physical activity by most active hour. Results, controlling for SES, age, sex and BMI z-score are presented in Table 29 and Figure 74.

Table 29: Sleep predicting physical activity controlling for SES, age, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.02 [−0.27, 0.32]	.871	−0.09 [−0.22, 0.03]	.135
Sleep duration	0.00 [−0.01, 0.01]	.641	0.06 [0.05, 0.07]	< .001
Most active hour	0.01 [0.01, 0.01]	< .001	0.00 [0.00, 0.00]	< .001
Sleep duration²	0.00 [−0.01, 0.01]	.726	0.00 [−0.01, 0.01]	.942
Sleep duration × Most active hour	0.00 [0.00, 0.00]	.005	0.00 [0.00, 0.00]	< .001
Most active hour × Sleep duration²	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	.019
Sleep efficiency (z)				
(Intercept)	0.03 [−0.27, 0.32]	.866	−0.10 [−0.22, 0.03]	.120
Sleep efficiency	−0.01 [−0.02, 0.00]	.092	−0.04 [−0.05, −0.03]	< .001
Most active hour	0.01 [0.01, 0.01]	< .001	0.00 [0.00, 0.00]	< .001
Sleep efficiency²	0.00 [−0.01, 0.01]	.718	−0.01 [−0.01, 0.00]	.110
Sleep efficiency × Most active hour	0.00 [0.00, 0.00]	.010	0.00 [0.00, 0.00]	< .001
Most active hour × Sleep efficiency²	0.00 [0.00, 0.00]	.022	0.00 [0.00, 0.00]	.654
Sleep onset (z)				
(Intercept)	0.02 [−0.28, 0.33]	.877	−0.12 [−0.24, 0.01]	.068
Sleep onset	−0.04 [−0.05, −0.03]	< .001	−0.06 [−0.08, −0.05]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Most active hour	0.01 [0.01, 0.01]	< .001	0.00 [0.00, 0.00]	< .001
Sleep onset²	0.00 [0.00, 0.01]	.161	0.02 [0.01, 0.02]	< .001
Sleep onset × Most active hour	0.00 [0.00, 0.00]	.009	0.00 [0.00, 0.00]	< .001
Most active hour × Sleep onset²	0.00 [0.00, 0.00]	.004	0.00 [0.00, 0.00]	< .001
Sleep regularity (z)				
(Intercept)	−0.06 [−0.36, 0.24]	.703	−0.19 [−0.32, −0.06]	.005
Sleep regularity	0.06 [0.05, 0.07]	< .001	0.03 [0.01, 0.04]	< .001
Most active hour	0.01 [0.01, 0.01]	< .001	0.00 [0.00, 0.00]	.341
Sleep regularity²	0.00 [−0.01, 0.01]	.630	0.00 [−0.01, 0.01]	.771
Sleep regularity × Most active hour	0.00 [0.00, 0.01]	< .001	0.00 [0.00, 0.00]	< .001
Most active hour × Sleep regularity²	0.00 [0.00, 0.00]	.467	0.00 [0.00, 0.00]	.899

Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction.

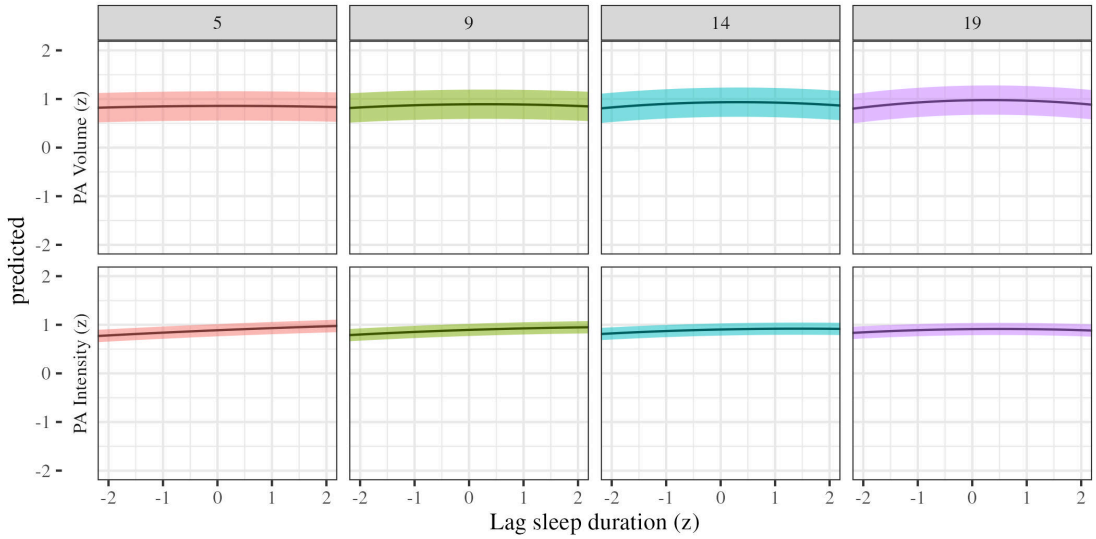

Figure 74: Physical activity by sleep duration moderated by most active hour

The association between sleep efficiency and next-day physical activity

We estimated the association between sleep efficiency and next-day physical activity by most active hour. Results, controlling for SES, age, sex and BMI z-score are presented in Table 29 and Figure 75.

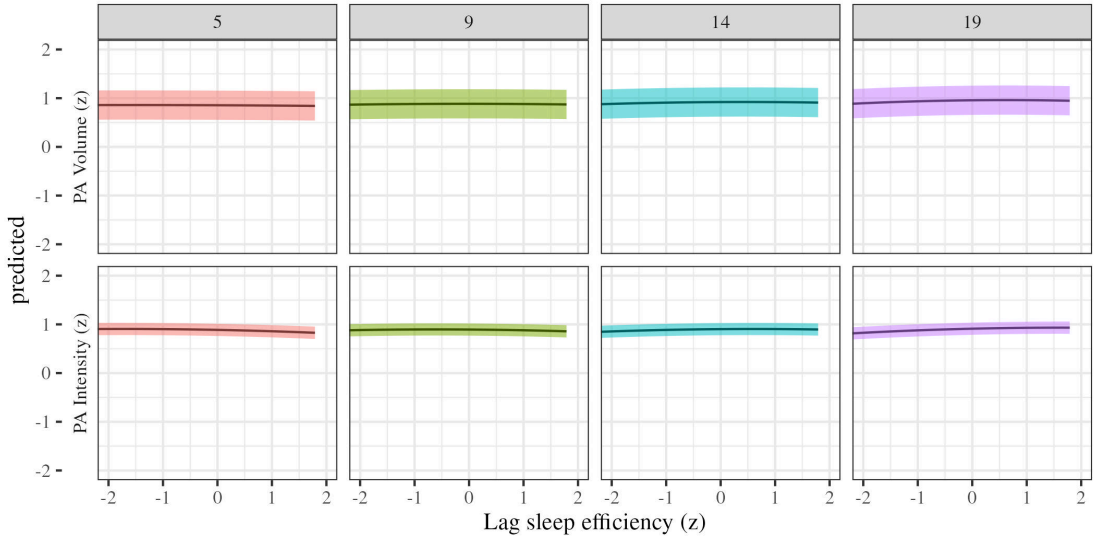

Figure 75: Physical activity by sleep efficiency moderated by most active hour

The association between sleep onset and next-day physical activity

We estimated the association between sleep onset and next-day physical activity by most active hour. Results, controlling for SES, age, sex and BMI z-score are presented in Table 29 and Figure 76.

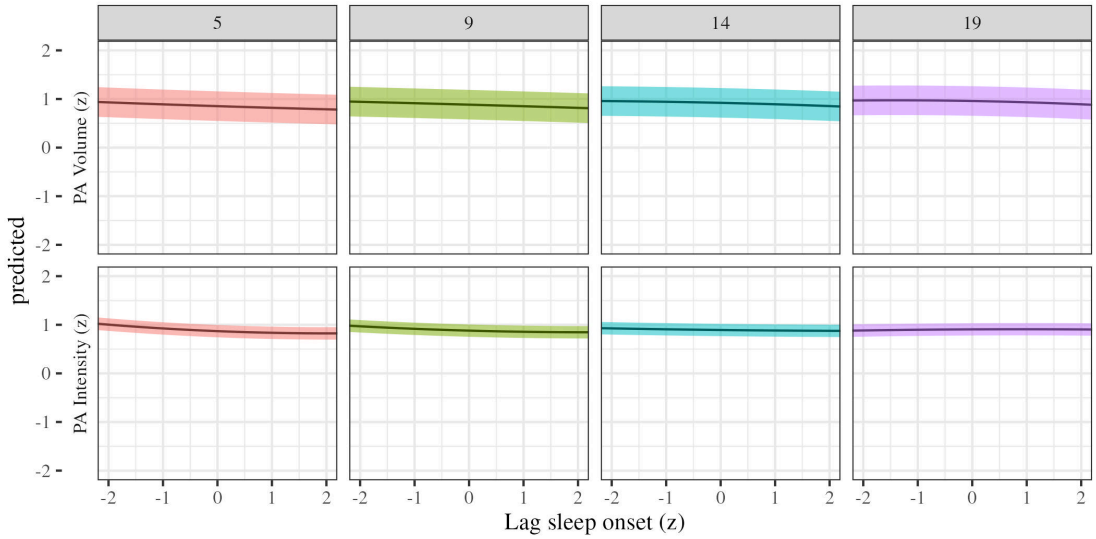

Figure 76: Physical activity by sleep onset moderated by most active hour

The association between sleep regularity and next-day physical activity

We estimated the association between sleep regularity and next-day physical activity by most active hour. Results are presented in Table 29 and Figure 77.

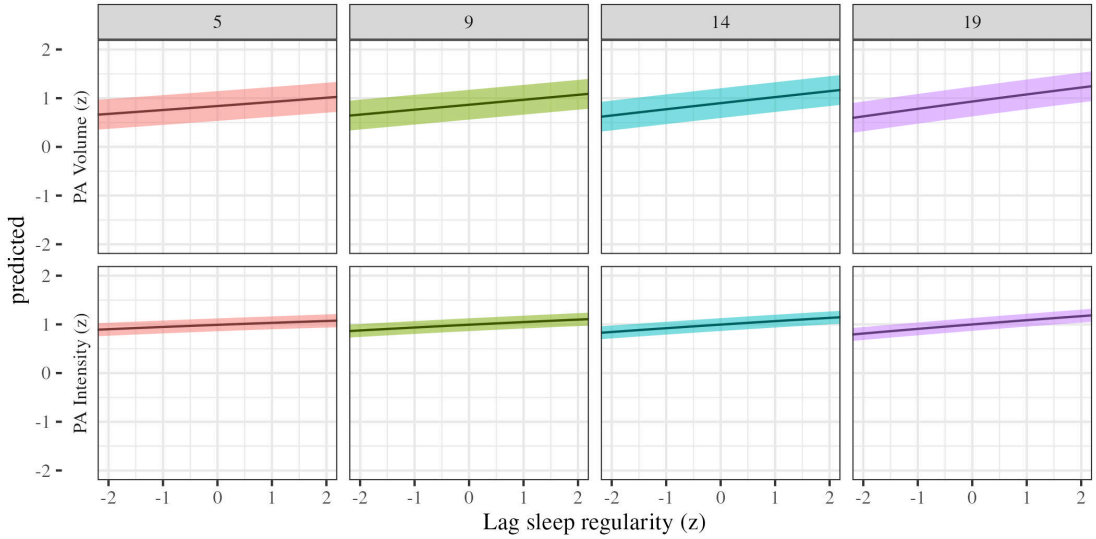

Figure 77: Physical activity by sleep regularity moderated by most active hour

ethnicity

The association between physical activity volume and sleep by ethnicity

We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by ethnicity, and the results are presented in Table 30 and Figure 78

Table 30: Physical activity predicting sleep controlling for SES, age, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.08 [−0.09, 0.24]	.365	0.08 [−0.07, 0.23]	.316
Physical activity	0.11 [0.09, 0.12]	< .001	0.07 [0.06, 0.08]	< .001
Non-white	−0.14 [−0.17, −0.11]	< .001	−0.16 [−0.19, −0.14]	< .001
Unclear	−0.11 [−0.14, −0.08]	< .001	−0.10 [−0.13, −0.07]	< .001
Physical activity²	0.00 [0.00, 0.00]	.993	0.00 [0.00, 0.01]	.118
Physical activity × Non-white	0.06 [0.03, 0.08]	< .001	0.03 [0.01, 0.05]	.002
Physical activity × Unclear	−0.02 [−0.04, 0.00]	.045	−0.02 [−0.04, −0.01]	.002
Non-white × Physical activity²	0.00 [−0.01, 0.01]	.512	0.01 [−0.01, 0.02]	.374
Unclear × Physical activity²	0.00 [−0.01, 0.00]	.162	0.01 [0.00, 0.02]	.060
Sleep efficiency (z)				
(Intercept)	0.19 [0.03, 0.35]	.019	0.18 [0.02, 0.34]	.031
Physical activity	0.04 [0.03, 0.05]	< .001	0.01 [0.00, 0.01]	.246
Non-white	−0.07 [−0.10, −0.04]	< .001	−0.11 [−0.14, −0.08]	< .001
Unclear	−0.01 [−0.04, 0.02]	.604	0.00 [−0.03, 0.03]	.938
Physical activity²	−0.01 [−0.01, −0.01]	< .001	0.00 [−0.01, 0.00]	.408
Physical activity × Non-white	0.07 [0.05, 0.10]	< .001	0.06 [0.04, 0.08]	< .001
Physical activity × Unclear	0.00 [−0.02, 0.02]	.802	−0.02 [−0.03, 0.00]	.025
Non-white × Physical activity²	−0.02 [−0.03, −0.01]	< .001	−0.01 [−0.03, 0.00]	.149

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Unclear × Physical activity²	0.00 [−0.01, 0.01]	.871	0.00 [0.00, 0.01]	.429
Sleep onset (z)				
(Intercept)	0.09 [−0.19, 0.37]	.520	0.09 [−0.17, 0.34]	.505
Physical activity	−0.13 [−0.14, −0.12]	< .001	−0.08 [−0.09, −0.07]	< .001
Non-white	0.07 [0.05, 0.09]	< .001	0.06 [0.03, 0.08]	< .001
Unclear	0.04 [0.02, 0.07]	< .001	0.04 [0.02, 0.07]	< .001
Physical activity²	0.00 [0.00, 0.01]	.078	−0.01 [−0.01, 0.00]	.006
Physical activity × Non-white	0.03 [0.01, 0.04]	.007	0.00 [−0.02, 0.02]	.999
Physical activity × Unclear	0.04 [0.02, 0.05]	< .001	−0.01 [−0.02, 0.00]	.276
Non-white × Physical activity²	−0.02 [−0.02, −0.01]	< .001	−0.01 [−0.02, 0.00]	.205
Unclear × Physical activity²	0.00 [0.00, 0.01]	.746	0.00 [−0.01, 0.00]	.262
Sleep regularity (z)				
(Intercept)	0.21 [0.05, 0.36]	.008	0.17 [−0.01, 0.35]	.062
Physical activity	0.33 [0.32, 0.35]	< .001	0.11 [0.10, 0.12]	< .001
Non-white	0.00 [−0.03, 0.03]	.851	0.05 [0.02, 0.09]	.001
Unclear	−0.12 [−0.15, −0.09]	< .001	−0.07 [−0.10, −0.03]	< .001
Physical activity²	−0.06 [−0.06, −0.06]	< .001	−0.01 [−0.01, 0.00]	.082
Physical activity × Non-white	−0.09 [−0.11, −0.06]	< .001	0.08 [0.06, 0.11]	< .001
Physical activity × Unclear	−0.02 [−0.04, 0.00]	.091	−0.04 [−0.06, −0.03]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Non-white × Physical activity²	0.02 [0.01, 0.03]	< .001	−0.08 [−0.09, −0.06]	< .001
Unclear × Physical activity²	0.00 [−0.01, 0.01]	.770	−0.03 [−0.03, −0.02]	< .001

Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction.

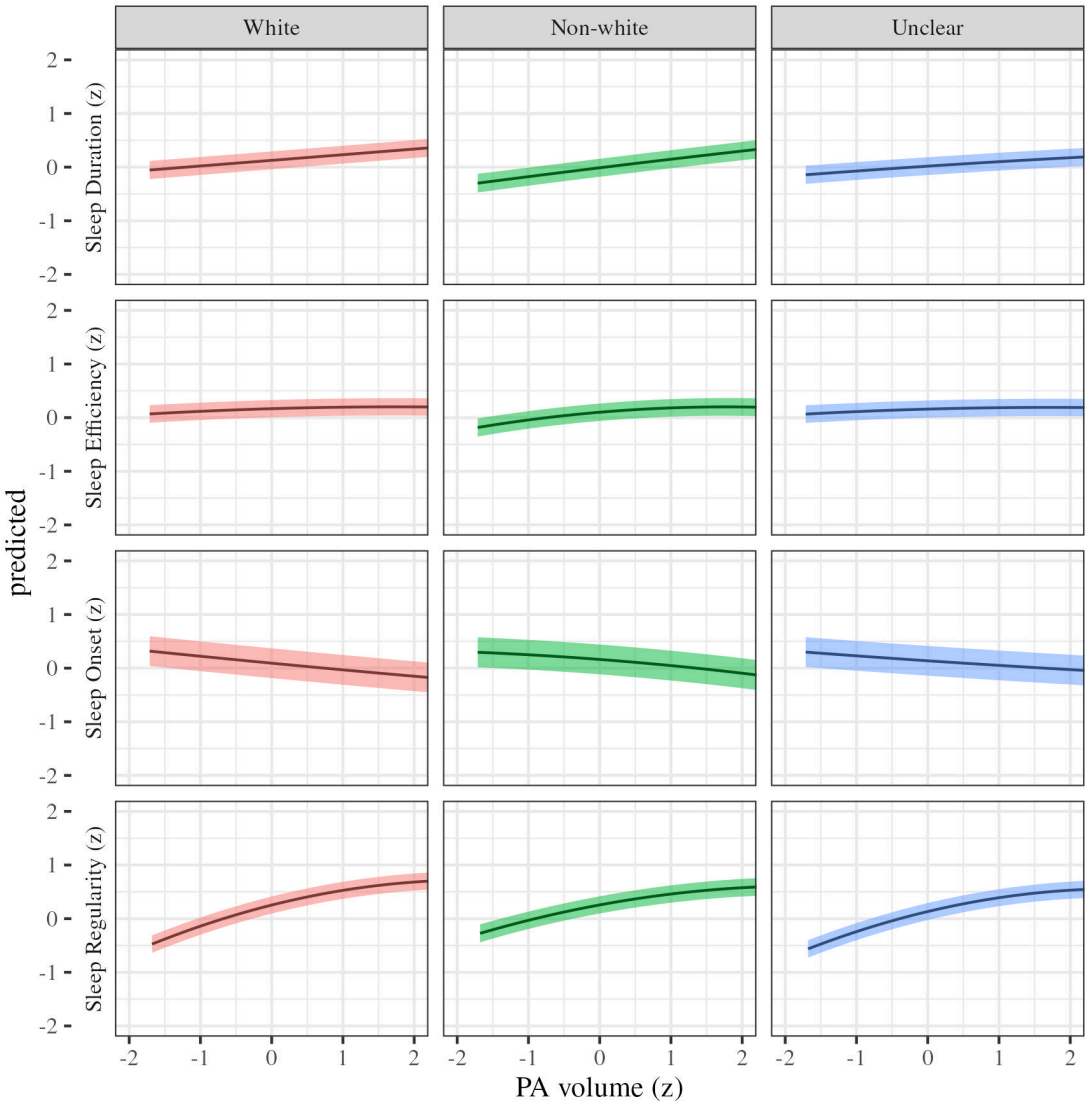

Figure 78: Sleep metrics on Physical activity volume by ethnicity

The association between physical activity intensity and sleep by ethnicity

We estimated how physical activity intensity is associated with sleep across ethnicity. We present the results in Table 30 and Figure 79.

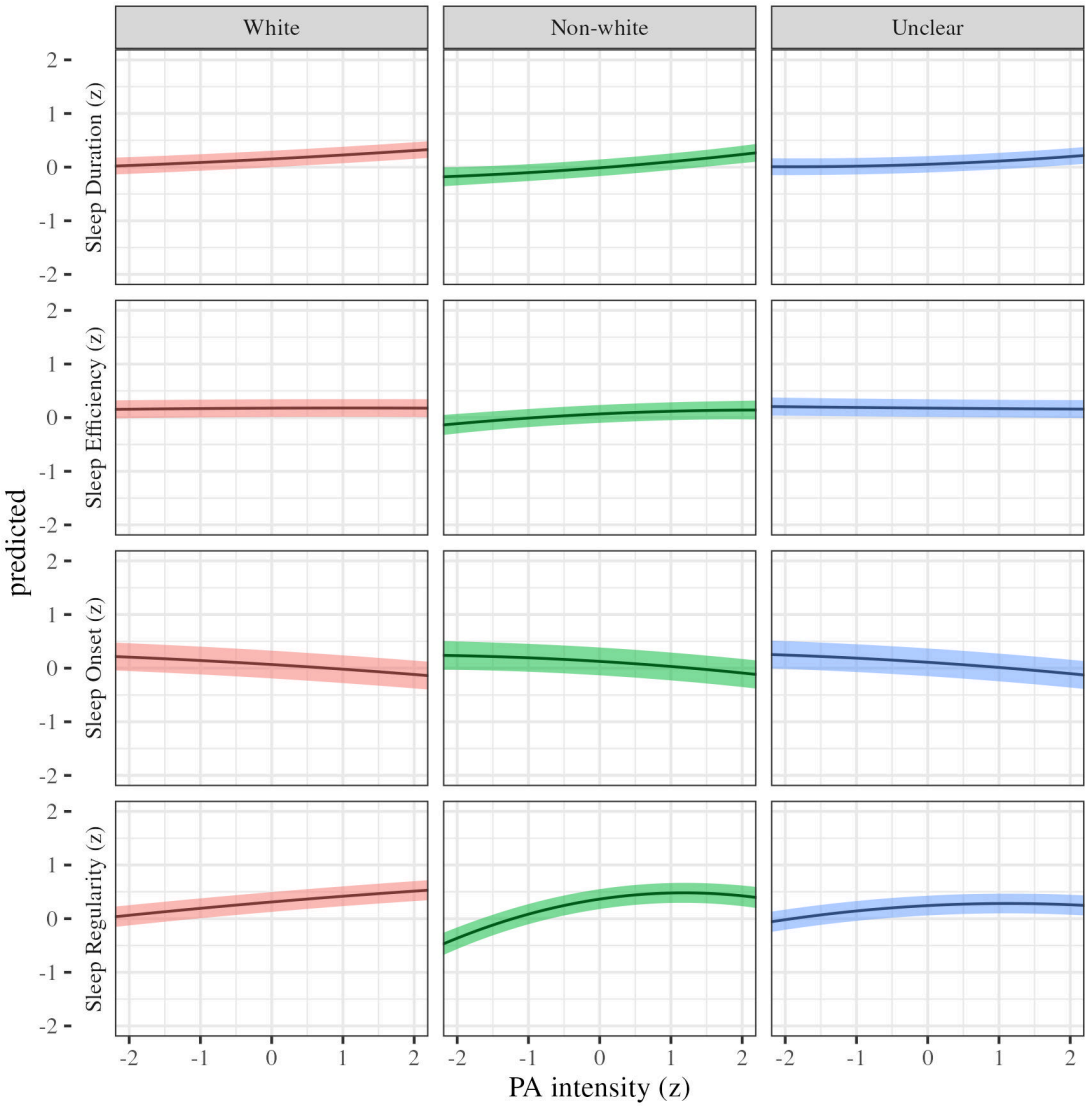

Figure 79: Sleep metrics on Physical activity intensity moderated by ethnicity

The association between sleep duration and next-day physical activity

We estimated the association between sleep duration and next-day physical activity by ethnicity. Results, controlling for SES, age, sex and BMI z-score are presented in Table 31 and Figure 80.

Table 31: Sleep predicting physical activity controlling for SES, age, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.13 [−0.18, 0.43]	.414	−0.05 [−0.18, 0.07]	.381
Sleep duration	0.02 [0.01, 0.02]	< .001	0.04 [0.03, 0.04]	< .001
Non-white	−0.03 [−0.05, 0.00]	.025	−0.03 [−0.05, −0.01]	.012
Unclear	0.03 [0.01, 0.06]	.016	−0.03 [−0.05, −0.01]	.013
Sleep duration²	−0.01 [−0.02, −0.01]	< .001	0.00 [−0.01, 0.00]	.095
Sleep duration × Non-white	−0.01 [−0.02, 0.00]	.150	−0.04 [−0.05, −0.02]	< .001
Sleep duration × Unclear	−0.01 [−0.02, 0.00]	.002	−0.02 [−0.03, −0.01]	< .001
Non-white × Sleep duration²	−0.01 [−0.02, 0.00]	.071	−0.01 [−0.02, −0.01]	< .001
Unclear × Sleep duration²	−0.01 [−0.01, 0.00]	< .001	−0.01 [−0.02, −0.01]	< .001
Sleep efficiency (z)				
(Intercept)	0.12 [−0.18, 0.42]	.427	−0.04 [−0.16, 0.08]	.506
Sleep efficiency	0.01 [0.00, 0.01]	.017	0.01 [0.01, 0.02]	< .001
Non-white	−0.05 [−0.07, −0.02]	< .001	−0.06 [−0.08, −0.04]	< .001
Unclear	0.01 [−0.01, 0.04]	.266	−0.06 [−0.08, −0.03]	< .001
Sleep efficiency²	−0.01 [−0.01, −0.01]	< .001	−0.01 [−0.01, −0.01]	< .001
Sleep efficiency × Non-white	−0.02 [−0.03, 0.00]	.045	−0.02 [−0.03, 0.00]	.030
Sleep efficiency × Unclear	0.00 [−0.01, 0.01]	.529	−0.01 [−0.02, 0.00]	.015
Non-white × Sleep efficiency²	0.00 [0.00, 0.01]	.270	0.00 [0.00, 0.01]	.312
Unclear × Sleep efficiency²	0.00 [0.00, 0.01]	.490	0.00 [0.00, 0.01]	.045

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep onset (z)				
(Intercept)	0.13 [−0.17, 0.43]	.399	−0.05 [−0.17, 0.07]	.413
Sleep onset	−0.02 [−0.03, −0.01]	< .001	−0.01 [−0.02, −0.01]	< .001
Non-white	−0.03 [−0.06, 0.00]	.020	−0.04 [−0.06, −0.01]	.002
Unclear	−0.01 [−0.03, 0.02]	.683	−0.05 [−0.08, −0.03]	< .001
Sleep onset²	0.00 [−0.01, 0.00]	.009	0.01 [0.00, 0.01]	< .001
Sleep onset × Non-white	0.05 [0.04, 0.07]	< .001	0.03 [0.01, 0.04]	< .001
Sleep onset × Unclear	−0.03 [−0.04, −0.01]	< .001	−0.02 [−0.03, −0.01]	.004
Non-white × Sleep onset²	0.01 [0.00, 0.02]	.011	0.00 [−0.01, 0.01]	.446
Unclear × Sleep onset²	0.00 [0.00, 0.01]	.345	−0.01 [−0.01, 0.00]	.017
Sleep regularity (z)				
(Intercept)	0.01 [−0.29, 0.32]	.934	−0.15 [−0.28, −0.02]	.022
Sleep regularity	0.13 [0.13, 0.14]	< .001	0.08 [0.07, 0.08]	< .001
Non-white	−0.03 [−0.06, −0.01]	.005	−0.05 [−0.07, −0.02]	< .001
Unclear	0.04 [0.01, 0.06]	.009	−0.05 [−0.07, −0.02]	< .001
Sleep regularity²	0.00 [−0.01, 0.00]	.223	0.00 [−0.01, 0.00]	.199
Sleep regularity × Non-white	−0.04 [−0.06, −0.03]	< .001	0.01 [0.00, 0.03]	.130
Sleep regularity × Unclear	−0.01 [−0.02, 0.00]	.079	−0.02 [−0.03, −0.01]	.001
Non-white × Sleep regularity²	−0.01 [−0.02, 0.00]	.035	−0.01 [−0.01, 0.00]	.211
Unclear × Sleep regularity²	0.00 [0.00, 0.01]	.161	0.00 [0.00, 0.01]	.365

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p

Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction.

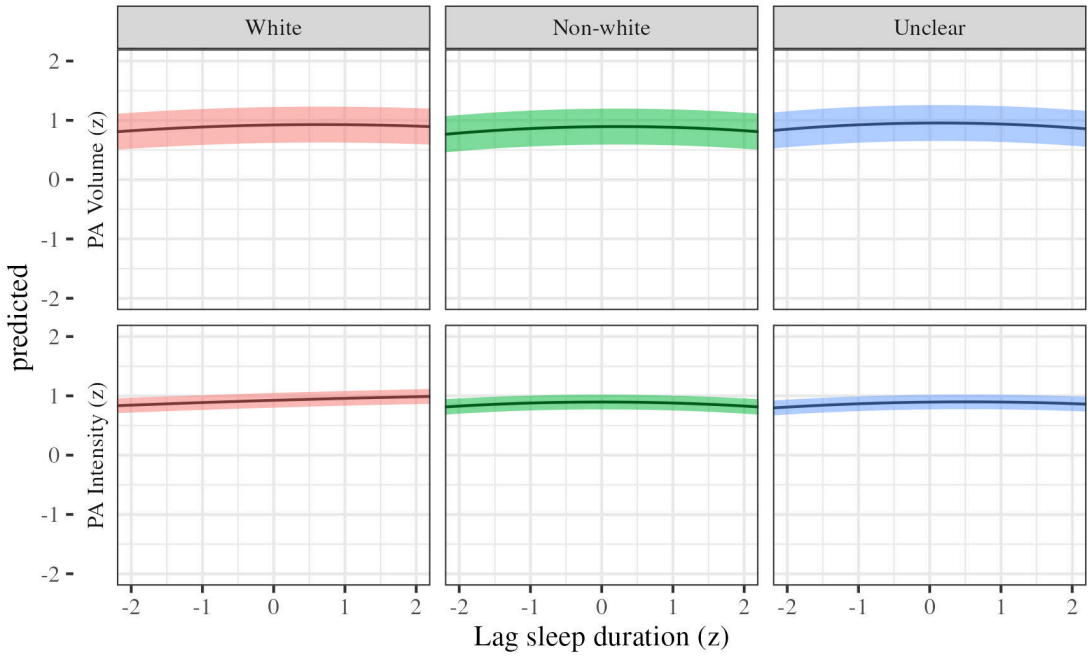

Figure 80: Physical activity by sleep duration moderated by ethnicity

The association between sleep efficiency and next-day physical activity

We estimated the association between sleep efficiency and next-day physical activity by ethnicity. Results, controlling for SES, age, sex and BMI z-score are presented in Table 31 and Figure 81.

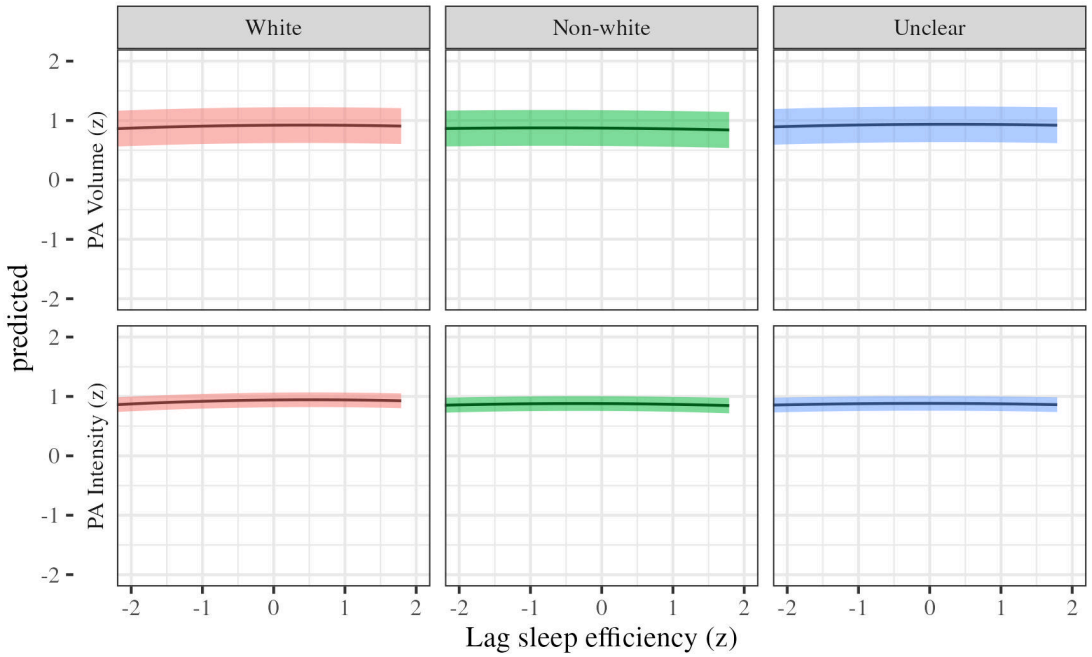

Figure 81: Physical activity by sleep efficiency moderated by ethnicity

The association between sleep onset and next-day physical activity

We estimated the association between sleep onset and next-day physical activity by ethnicity. Results, controlling for SES, age, sex and BMI z-score are presented in Table 31 and Figure 82.

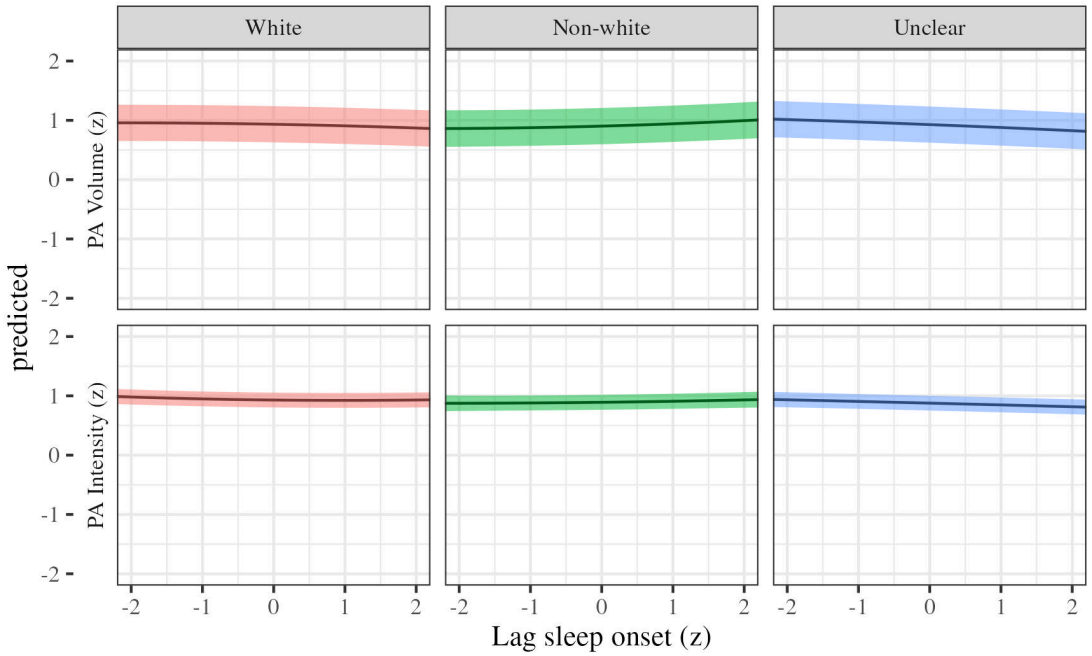

Figure 82: Physical activity by sleep onset moderated by ethnicity

The association between sleep regularity and next-day physical activity

We estimated the association between sleep regularity and next-day physical activity by ethnicity. Results are presented in Table 31 and Figure 83.

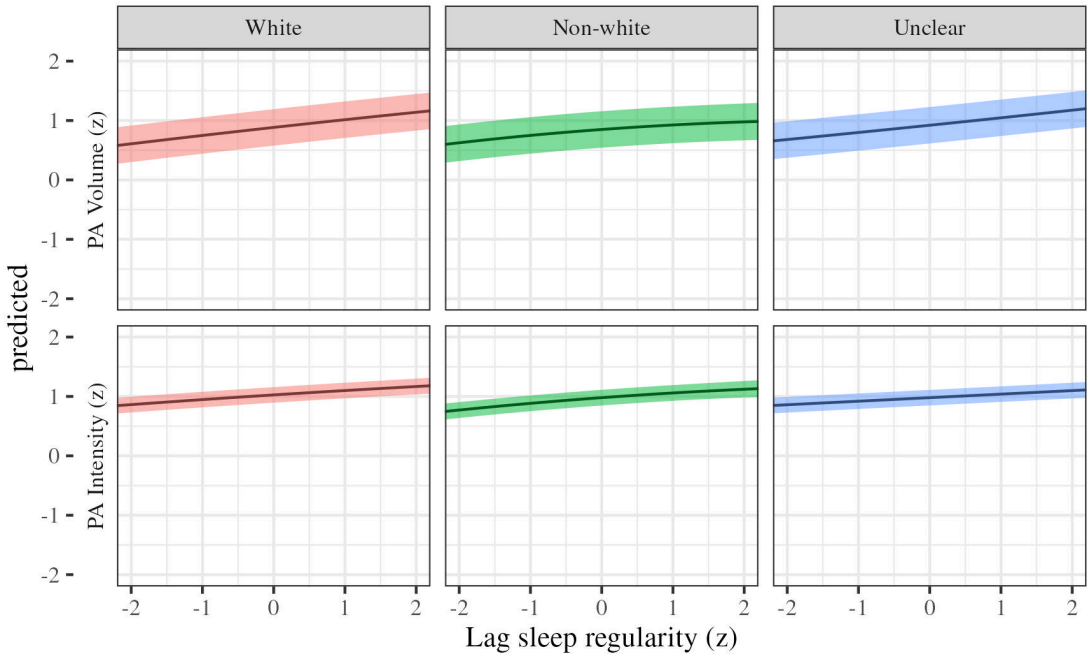

Figure 83: Physical activity by sleep regularity moderated by ethnicity